PERFECTOID PURE AND SIGNATURE PAIRS

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ABSTRACT. We extend perfectoid purity and related invariants to divisor pairs in mixed characteristic. The resulting purity condition agrees with F-purity of pairs in characteristic p as defined by Hara–Watanabe. For normal (respectively, demi-normal) log $\mathbb{Q}$-Gorenstein pairs, we show that perfectoid pure implies log canonical (respectively, semi-log canonical). We similarly establish adjunction for perfectoid pure pairs and inversion of adjunction for principal pairs in complete intersections along a Cartier divisor. We also define perfectoid pure thresholds and prove localization and base change results, including a formula for local extensions with regular closed fiber. We establish upper bounds in terms of analytic spread, dimension, and log canonical thresholds. Finally, we also introduce perfectoid signatures for divisor pairs in mixed characteristic whose positivity characterizes the BCM-regularity of log $\mathbb{Q}$-Gorenstein pairs. We prove a transformation rule under finite extensions and show continuity and convexity for scalings of a Cartier divisor.

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1. Introduction

Within birational algebraic geometry and characteristic $p > 0$ commutative algebra, instead of simply studying the singularities of a ring R, one often studies the singularities of a pair (R, f^t), (R, Δ) or $(R, \mathfrak{a}^t)$. Here $f \in R$ and $t \in \mathbb{Q}_{\geq 0}$, $\Delta \geq 0$ is a $\mathbb{Q}$-divisor on $\operatorname{Spec} R$, or $\mathfrak{a} \subseteq R$ is an ideal and $t \in \mathbb{Q}_{\geq 0}$. Pairs arise naturally as correction terms when changing rings or varieties and they have become an important tool for proving facts even about regular rings, see for instance [ELS01].

Recently, by applying methods originating in p-adic Hodge theory, [Sch12, And18, BS22, Bha20], a theory of singularities in mixed characteristic has been developed analogous to the theories coming out of birational algebraic geometry and characteristic $p > 0$ commutative algebra, see for instance [MS21, MST+22, TY23, BMP+23, CLM+26, BMP+24, Yos25, CPQG+25]. Within those works, there is a suitable theory of singularities of pairs analogous to the theory of log terminal or strongly F-regular pairs (namely, BCM-regular pairs). However, there is no general theory analogous to log canonical and F-pure pairs ([Kol13, HW02, Tak04]). Furthermore, we also lack a theory analogous to log canonical (or F-pure) thresholds and a theory analogous to the theory of F-signature of pairs. Indeed, there are aspects of the BCM-regular definitions that make some of these generalizations difficult to even formulate, and we discuss these below in Section 1.2.

In this paper we develop these missing theories in mixed characteristic. In particular, we introduce and develop the following notions.

(a) A definition of perfectoid purity for pairs (R, Δ) (or (R, f^t)) agreeing with the definition of F-purity for pairs from [HW02], see Section 3 and Section 4.

(b) An analog of the F-pure threshold of [TW04], denoted $\operatorname{ppt}(\mathfrak{a}, \Delta)$, see Section 7.

(c) An analog of the F-signature for pairs from [BST12, PT18], see Section 8.

1.1. Main results.

We prove the following results about these definitions. First we show that perfectoid pure pairs are log canonical.

Theorem A (Theorem 5.6)**.** *Let R be an excellent normal ring admitting a dualizing complex, with p in its Jacobson radical. Let Δ be an effective $\mathbb{Q}$-divisor on R such that $K_R + \Delta$ is $\mathbb{Q}$-Cartier. If (R, Δ) is perfectoid pure, then it is log canonical.*

We in fact prove the non-normal semi-log canonical version as well where, instead of assuming that R is normal, we assume that R is demi-normal ([Kol13]) and that Δ is a Mumford $\mathbb{Q}$-divisor ([Kol18]), see Corollary 5.8.

Indeed, the above result is new even when $\Delta = 0$ where the result had only been shown when K_R was $\mathbb{Q}$-Cartier with *index not divisible by* p, see [BMP+24, Corollary 5.11]. In fact, it is the flexibility that pairs provides that lets us handle the case where the index is divisible by p.

We also obtain the following adjunction-type result.

Theorem B (Theorem 6.5)**.** *Let R be an excellent equidimensional demi-normal ring with p in its Jacobson radical and which admits a dualizing complex. Let D be a reduced effective Mumford divisor on R and let $i\colon D = V(I_D) \hookrightarrow \operatorname{Spec} R$. Assume that $S \coloneqq R/I_D$ is deminormal. Let Δ be an effective Mumford $\mathbb{Q}$-divisor on $\operatorname{Spec}(R)$ having no prime component in common with D.*

Assume that $K_R + D + \Delta$ is $\mathbb{Q}$-Cartier at every height 1-prime of S and that $(R, D + \Delta)$ is perfectoid pure. Then (S, Δ_D) is perfectoid pure where Δ_D is the different.

Under moderate geometric hypotheses, we can weaken the hypothesis that S is demi-normal to require instead that S is $\mathbf{S}_2$ (and we can even remove that if D is $\mathbb{Q}$-Cartier), see Theorem 6.7. Indeed, if all the divisors are Cartier, then we can ignore these demi-normality and even equidimensionality assumptions completely and obtain a quite general adjunction statement in Proposition 3.13. If additionally we assume that the ambient ring is a complete intersection, we also obtain inversion of adjunction Theorem 3.14.

We now move to our analogs of F-pure thresholds. We prove the following, which answers [BMP+24, Conjecture 4.22(a)] and even generalizes the statement to the context of pairs.

Theorem C (Theorem 7.28)**.** *Let $(R, \mathfrak{m}) \to (S, \mathfrak{n})$ be a faithfully flat local map of Noetherian local rings of residue characteristic p such that $S/\mathfrak{m}S$ is regular. Let $f_1, \ldots, f_n \in S$ be elements whose images form a regular system of parameters on $S/\mathfrak{m}S$.*

Let $\mathfrak{a}$ be an ideal of R and Δ be an effective Mumford $\mathbb{Q}$-divisor on R. If $\Delta \neq 0$, further assume that R and S are excellent reduced $\mathbf{S}_2$ rings and that the flat pullback Δ_S of Δ is a Mumford $\mathbb{Q}$-divisor. Then

$$\operatorname{ppt}((\mathfrak{a}S + (f_1, \ldots, f_n))S; \Delta_S) = \operatorname{ppt}(\mathfrak{a}; \Delta) + \dim(S/\mathfrak{m}S).$$

In particular, if (R, Δ) is perfectoid pure, then so is (S, Δ_S).

Using this, in Corollary 7.30 we show that $\operatorname{ppt}(\mathfrak{a}) \leq \ell(\mathfrak{a})$, where $\ell(\mathfrak{a})$ denotes the analytic spread of $\mathfrak{a}$. Moreover, we prove that

$$\operatorname{ppt}(\mathfrak{a}) \leq \min\{\dim(R_Q) \mid (p, \mathfrak{a}) \subseteq Q \in \operatorname{Spec}(R)\}.$$

We also compare the perfectoid pure threshold to the log canonical threshold.

Theorem D (Proposition 7.40)**.** *Let $(R, \mathfrak{m})$ be an excellent normal local ring of residual characteristic $p > 0$ admitting a dualizing complex and let $\mathfrak{a} \subseteq R$ be an ideal. Let Δ be an effective $\mathbb{Q}$-divisor on R such that $K_R + \Delta$ is $\mathbb{Q}$-Cartier. Then $\operatorname{ppt}(\mathfrak{a}; \Delta) \leq \operatorname{lct}(\mathfrak{a}; \Delta)$.*

We also compute some interesting examples of perfectoid pure thresholds. Suppose for instance that $R = \mathbb{Z}_p[\![x,y,z]\!]/(x^3+y^3+z^3)$. Then we prove that

$$\operatorname{ppt}(\mathfrak{m}) = 1 \quad \text{if } p \equiv 1 (\operatorname{mod} 3)$$

If $p \equiv 2 (\operatorname{mod} 3)$ then we know that R is perfectoid pure by [Yos25] and in fact also that $\operatorname{ppt}(p) = 1 - p/(p^2-1)$. Building upon this, and the theory we develop, we also show that

$$\operatorname{ppt}(\mathfrak{m}) = 1 - p/(p^2-1) \quad \text{if } p \equiv 2 (\operatorname{mod} 3).$$

We now move on to the perfectoid signature of a pair (R,Δ), which we denote by $s^{\underline{x}}_{\text{perfd}}(R,\Delta)$. As in [CLM+26], R is a complete Noetherian normal domain and $\underline{x}$ is used to identify a certain choice of system of parameters from a Noether-Cohen normalization $A \subseteq R$. We show a general transformation rule in the context of perfectoid signature, proving the following analog of [CRST18, Theorem 4.4] and [XZ21, Theorem 1.3].

Theorem E (Theorem 8.9)**.** *Let $\phi\colon (R,\mathfrak{m},k) \to (R',\mathfrak{n},l)$ be a finite generically étale extension of complete normal local domains of mixed characteristic $(0,p)$ with perfect residue fields and with fraction fields $K = K(R)$ and $L = K(R')$. Let $\tau\colon \operatorname{Spec}(R') \to \operatorname{Spec}(R)$ be the corresponding morphism of schemes. Suppose $\Delta_R \geq 0$ is a $\mathbb{Q}$-divisor such that $\Delta_{R'} \coloneqq \tau^*\Delta_R - K_{R'/R} \geq 0$. Let t be the maximal number of direct* Tr*-summands*[1] *of R' and $r = [L:K]$. Then we have*

$$t \cdot s^{\underline{x}}_{\text{perfd}}(R',\Delta_{R'}) = r \cdot s^{\underline{x}}_{\text{perfd}}(R,\Delta_R).$$

Just as in the non-pair case, we show that perfectoid signature of pairs measures BCM-regularity. Indeed, in Theorem 8.16 we prove the following. Suppose $\Delta \geq 0$ is a $\mathbb{Q}$-divisor such that $K_R + \Delta$ is $\mathbb{Q}$-Cartier. Then (R,Δ) is BCM-regular if and only if

$$s^{\underline{x}}_{\text{perfd}}(R,\Delta) > 0.$$

Similar to [BST13], we also study the function $t \mapsto s^{\underline{x}}_{\text{perfd}}(R,tD)$ for $D = \operatorname{div} f$. We prove the following.

Theorem F (Theorem 8.17, Theorem 8.18)**.** *Suppose $(R,\mathfrak{m},k)$ is a complete Noetherian normal local domain of mixed characteristic $(0,p)$ with perfect residue field and $f \in \mathfrak{m}$ is non-zero. Let $D = \operatorname{div}(f)$ on $\operatorname{Spec}(R)$. The function $\mathbb{Q}_{\geq 0} \to \mathbb{R}$ given by*

$$t \mapsto s^{\underline{x}}_{\text{perfd}}(R,tD)$$

is continuous. In particular, it extends naturally to a function $\mathbb{R}_{\geq 0} \to \mathbb{R}$. Furthermore, that function is convex.

1.2. **On the definitions.** Perhaps the most challenging part of the work above was developing each definition. We describe the idea behind each one below.

[1] That is, a rank-1 free R-module summand of R' such that if $\rho : R' \to R$ is projection on that summand, then $\rho = \operatorname{Tr} \cdot s$ for some $s \in R'$, see also [CRST18, Definition 2.12].

Perfectoid purity for pairs (R, Δ). In order to mimic the idea of [HW02], we want some map

$$R \to B((1-\varepsilon)\Delta)$$

to be pure for all $1 \gg \varepsilon > 0$ and where B is some perfectoid algebra. The problem is interpreting twisting by a $\mathbb{Q}$-divisor on B. Unlike [MS18], we cannot assume that B is an R^+-algebra, as if it was then we would also have a splitting $R \to R^+ \to R^+((1-\varepsilon)\Delta)$ which would imply that R itself was a splinter (and hence is KLT if $\mathbb{Q}$-Gorenstein). While we may (and do) assume that B is absolutely integrally closed, this means that in principle there can be multiple different ways to pullback a $\mathbb{Q}$-divisor. In this generality, nth roots of elements of R are *not* unique up to units. Nevertheless, the ε perturbation saves us as long as we pass all the way to B.

We sketch the idea which is developed carefully at the start of Section 4. Suppose we have $\varepsilon_1 > \varepsilon_2 > 0$. One can always pick a finite flat extension $R \subseteq S_1$ and a map $S_1 \to B$ and an integer $n_1 > 0$ such that $n_1(1-\varepsilon_1)\Delta$ is an integral divisor that pulls back to $n_1\Delta_{S_1}$, where Δ_{S_1} is an effective Weil generalized divisor. Such an S_1, n_1 and Δ_{S_1} exist: simply formally adjoin an appropriate root to R and use that B is absolutely integrally closed (Construction 2.18). Note even if R is normal, we cannot assume S_1 is normal while keeping $R \subseteq S_1$ flat, and normalizing leads to problems similar to those with R^+.

We likewise construct $R \subseteq S_2 \to B$ and Δ_{S_2} an effective $\mathbb{Q}$-divisor such that $n_2((1-\varepsilon_2)\Delta) = n_2\Delta_{S_2}$. Set $\mathscr{M}_i$ to be the sheaf on R associated to $B \otimes_{S_i} S_i(\Delta_{S_i})$. Then in Lemma 4.6, we show that there is a canonical pointed[2] map

$$\mathscr{M}_1|_U \to \mathscr{M}_2|_U$$

for some open subset $U \subseteq \operatorname{Spec} R$ whose complement has codimension ≥ 2 (a *big open set*). The key observation that lets us construct this map is that B is seminormal, since it is perfectoid, see Lemma 2.5. We then show we can take a filtered colimit over all $\varepsilon > 0$ and over all big open sets to obtain a pointed[3] object $M_B(<\Delta)$ (one can take other colimits too as we show in Lemma 4.16). With that in mind, one has that (R, Δ) is perfectoid pure if and only if the structure map

$$R \to M_B(<\Delta)$$

is pure, see Proposition 4.20. This is equivalent to asking that for every $\varepsilon > 0$, there exists a finite flat R-algebra S and Δ_S as above with $n(1-\varepsilon)\Delta$ an integral divisor that pulls back to $n\Delta_S$ for some $n > 0$ as well as some map $S \to B_\varepsilon$ for some perfectoid S-algebra, such that $R \to B_\varepsilon \otimes_S S(\Delta_S)$ is pure. In fact, this latter characterization is our original definition Definition 4.8.

Perfectoid pure thresholds. In this section we introduce analogs of the F-pure threshold of an ideal $\mathfrak{a} \subseteq R$ (or even for a pair (R, Δ) as above), which we denote by $\operatorname{ppt}(R; \mathfrak{a})$ or $\operatorname{ppt}(R, \Delta; \mathfrak{a})$ and call it *perfectoid pure threshold* of $\mathfrak{a}$.

We explain the definition, set $\mathcal{I}(B, <\Delta)$ to be elements $b \in B$ such that $R \xrightarrow{1 \mapsto b} B \to M_B(<\Delta)$ is not pure, see Definition 2.23 and Definition 4.22. We then define

$$\operatorname{ppt}(\mathfrak{a}; \Delta) = \sup_B \sup\{r \in \mathbb{Q}_{\geq 0} \mid \overline{(\mathfrak{a}B)^r} \not\subseteq \mathcal{I}(B, <\Delta)\}.$$

where $\overline{(\mathfrak{a}B)^r}$ is the fractional integral closure (see also [HS06, Section 10.5] and [RS25]) and B runs over perfectoid R-algebras where certain key elements have compatible systems of roots, (see

[2] sending $1 \mapsto 1$

[3] meaning there is a given map $R \to M_B(<\Delta)$

Definition 7.10). It turns out one can compute this from concretely defined perfectoid algebras $\widehat{R}^{A_\infty}_{\mathrm{perfd}}$ (Proposition 7.17).

Thanks to the Briançon–Skoda theorem for perfectoid algebras ([MMRS25, Corollary 3.20]) one can reduce this computation to the principal case similar to the summation formula for multiplier ideals: supposing R is a local normal domain, there are minimal generators $0 \neq \mathfrak{a} = (x_1, \ldots, x_m) \subsetneq R$ with

$$\mathrm{ppt}(\mathfrak{a}; \Delta) = \sup\left\{ \frac{|a|}{p^e} \mid e \in \mathbb{Z}_{\geq 0}, a \in \mathbb{Z}^m_{\geq 0}, (R, \Delta + \frac{1}{p^e}\operatorname{div}(x_1^{a_1} \cdots x_m^{a_m})) \text{ is perfectoid pure} \right\},$$

see Corollary 7.18. This is one key result that underlies many other results in Section 7.

Perfectoid signature pairs. Our strategy here is slightly different. We are working with perfectoid signature which measures BCM-regularity and not perfectoid purity. Because of this, we can use finite (normal) extensions of integral domains $R \subseteq S$ instead of the finite flat extensions mentioned above.

Suppose we are given a normal pair $(R, \Delta \geq 0)$ where R is a complete local Noetherian normal local domain of mixed characteristic. We fix $A \subseteq R$ to be a Noether-Cohen normalization (R is a finite A-module) with a chosen regular system of parameters $\underline{x}$. We set $S \supseteq R$ to be a finite extension such that Δ pulls back to a divisor with integer coefficients. In Definition 8.2, we then define

$$\mathcal{I}^R(S^{A_\infty}_{\mathrm{perfd}}, \Delta_R) = \left\{ x \in S^{A_\infty}_{\mathrm{perfd}} \mid R \xrightarrow{1 \mapsto x \otimes 1} S^{A_\infty}_{\mathrm{perfd}} \otimes_S S(\Delta_R) \text{ is not pure} \right\}.$$

Then we define the *perfectoid signature* of (R, Δ_R) to be

$$s^{\underline{x}}_{\mathrm{perfd}}(R, \Delta_R) = \frac{1}{r} \cdot \lambda_\infty(S^{A_\infty}_{\mathrm{perfd}} / \mathcal{I}^R(S^{A_\infty}_{\mathrm{perfd}}, \Delta_R))$$

where $r = \operatorname{rank}_R S$.

We show that this is independent of the choice of S. This computation is essentially also the proof of the transformation rule under finite maps for perfectoid signature. Note that, like the non-pair case ([CLM+26]), we do not know whether perfectoid signature is independent of the choice of A or $\underline{x}$.

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adjunction arguments), and subsequently to edit it for readability. We then rewrote the whole section ourselves. More generally, we have also made use of LLMs when finalizing this document. They have been used to help identify typos and errors, and to suggest improvements in exposition. We take full responsibility for the mathematical content and final form of the article.

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2. Preliminaries

All rings considered are commutative with unity. If R is a ring, I is an ideal of R, and M is an R-module then $M^{\wedge I}$ is the classical I-adic completion of M. We write $\widetilde{M}$ for the associated quasicoherent sheaf on $X = \operatorname{Spec} R$. Let p be a prime and suppose that it is in the Jacobson radical of R. We denote by $I^- = \bigcap_{n\geq 1}(I + p^n R)$ the p-adic closure of I in R.

2.1. Perfectoid rings.

We first recall the definition and various properties of a perfectoid ring as in [BMS18, BIM19, Definitions 3.5] (see also [MT25] for an introduction).

Definition 2.1. A ring S is perfectoid if it is ϖ-adically complete for some element $\varpi \in S$ such that $\varpi^p u = p$ for some unit u, the Frobenius map $\phi\colon S/pS \to S/pS$ is surjective and the kernel of Fontaine's map $\theta\colon W(S^\flat) \to S$ is principal for $S^\flat = \varprojlim_\phi S/p$.

Proposition 2.2. *Let R be a perfectoid ring, $f_1, \ldots, f_n \in R$, and $I = (f_1, \ldots, f_n)$.*

(a) R is reduced [ČS24, Section 2.1.3] *(see also* [MT25, Corollary 2.4]*);*

(b) R is absolutely weakly normal, and hence seminormal [BMP+24, Lemma 4.28]*;*

(c) for some unit u, up has a compatible system of p-power roots [BIM19, Lemma 3.6]*;*

(d) for every $f \in R$ there exists $g \in R$ admitting a compatible system of p-power roots such that $f - g \in pR$ [IY25, Lemma 2.2]*;*

(e) if $p \in I$, then there are generators $g_1, \ldots, g_n$ of I, each admitting a compatible system of p-power roots [IY25, Lemma 2.3]*;*

(f) if $p \in I$ then $R^{\wedge I}$ is perfectoid and the completion agrees with derived I-completion [ČS24, 2.1.11(e)].

Theorem 2.3 ([BS22, Theorem 7.14, Remark 7.15], cf. [And18]). *Let R be a perfectoid ring. Then there exists a p-completely faithfully flat map $R \to S$ of perfectoid rings such that S is absolutely integrally closed (i.e., every monic polynomial has a solution). In particular, every element of S admits a compatible system of p-power roots.*

Next, let S be a ring with p in its Jacobson radical and denote its perfectoidization by S_{perfd}, as in [BS22, Definition 8.2] (see also [BMP+24, Definition 3.10]). When S admits a finite map from a perfectoid ring, S_{perfd} is a perfectoid ring by [BS22, Theorem 10.11], and the natural map $S \to S_{\mathrm{perfd}}$ is universal among maps from S to perfectoid rings by [BS22, Corollary 8.14] (see also [BMP+24, Proposition 3.14]). By [BS22, Theorem 7.4], if S is perfectoid and $I \subset S$ is an ideal, then $S \to (S/I)_{\mathrm{perfd}}$ is surjective. We denote its kernel by I_{perfd}, so $(S/I)_{\mathrm{perfd}} = S/I_{\mathrm{perfd}}$. If I is finitely generated, then I_{perfd} gives a theory of almost mathematics for derived p-complete S-modules; see [BS22, Section 10]. More precisely, for every $n \geq 1$, the pair $(S/p^nS, I_{\mathrm{perfd}}S/p^nS)$ is a basic setup for almost mathematics in the sense of [GR03, Section 2.1.1].

Corollary 2.4. *Let S be a perfectoid ring and $I \subset S$ an ideal. Then I_{perfd} is a p-adically closed radical ideal and $I_{\mathrm{perfd}} = (\sqrt{I})_{\mathrm{perfd}} = (I^-)_{\mathrm{perfd}}$. Moreover, if $I = (f_1, \ldots, f_n)$ and each f_i admits a compatible system of p-power roots f_i^{1/p^∞} in S, then $I_{\mathrm{perfd}} = (\sqrt{I})^- = \left(f_1^{1/p^\infty}, \ldots, f_n^{1/p^\infty}\right)^-$.*

Proof. The first assertion follows because I_{perfd} is the kernel of a map to a perfectoid ring. Moreover, every map from S/I to a perfectoid ring kills both $\sqrt{I}$ and I^-, since perfectoid rings are reduced and p-adically separated. The first two equalities therefore follow from the universal property of perfectoidization. The final equalities follow from [Fay25, Corollary 2.8], building on the principal case from [CLM+26, Lemma 2.19]. □

Lemma 2.5. *Let B be a seminormal ring. Then the following hold.*

(a) If $g, x \in B$ and $g^m \in x^n B$ for some $m < n \in \mathbb{Z}_{>0}$, then there is a unique $c \in \sqrt{xB}$ such that $g = cx$.

(b) Suppose that $f \in B$ and $\{f^{1/p^e}\}_{e \in \mathbb{Z}_{>0}}$ is a compatible system of p-power roots of f. Then

$$\sqrt{fB} = (f^{1/p^\infty})B \cong \varinjlim \left(B \xrightarrow{\cdot f^{1-1/p}} B \xrightarrow{\cdot f^{1/p-1/p^2}} B \xrightarrow{\cdot f^{1/p^2-1/p^3}} \cdots \right).$$

For any map $B \to B'$ of seminormal rings, the natural map $\sqrt{fB} \otimes_B B' \to \sqrt{fB'}$ is an isomorphism. Moreover, $(f^{1/p^\infty})B = (f'^{1/p^\infty})B$ for any other compatible system $\{f'^{1/p^e}\}_{e \in \mathbb{Z}_{>0}}$ of p-power roots of f.

(c) Suppose that $\pi, y, z \in B$ satisfy $y^N = u\pi^a$ and $z^N = v\pi^b$ for some $N \in \mathbb{Z}_{>0}$, units $u, v \in B^\times$, and integers $0 \leq a < b$. Then there is a unique $w \in \sqrt{yB}$ such that $z = wy$, and $w^N = vu^{-1}\pi^{b-a}$.

In particular, all these assertions apply when the rings involved are localizations of perfectoid rings.

Proof. For (a), write $g^m = x^n h$ for some $h \in B$, and set

$$C = \left(B[T]/(xT - g, T^m - x^{n-m}h)\right)_{\mathrm{red}}.$$

The map $B \to C$ is finite and subintegral. Indeed, away from $V(x)$ the relation $xT = g$ determines T, while over a prime containing x (and hence g), every prime above it contains T and has the same

residue field. Its kernel is contained in every prime of B, and hence is zero because B is reduced. Since C is reduced and B is seminormal, the map is an isomorphism by [Sta26, Tag 0H3G].

The image c of T in B satisfies $g = cx$ and $c^m = x^{n-m}h \in xB$, so $c \in \sqrt{xB}$. If $c' \in \sqrt{xB}$ also satisfies $g = c'x$, then $x(c-c') = 0$ and $(c-c')^q \in xB$ for some $q > 0$. Thus $(c-c')^{q+1} = 0$, and $c = c'$ because B is reduced.

For (b), certainly $(f^{1/p^\infty})B \subseteq \sqrt{fB}$. Conversely, this ideal is radical. Indeed, if $g^m \in (f^{1/p^\infty})B$, we may choose e such that $g^m \in f^{1/p^e}B$ and k such that $p^k > m$. Applying (a) with $x = f^{1/p^{e+k}}$ and $n = p^k$ gives $g \in f^{1/p^{e+k}}B$. This proves the first equality and the independence of the chosen root system.

The colimit description of $(f^{1/p^\infty})B$ in the statement follows from [AH97] and requires only that B is reduced (see also Observation (a) in the proof of [MT25, Theorem 1.2]). Tensor products commute with filtered colimits, giving $(f^{1/p^\infty})B \otimes_B B' \cong (f^{1/p^\infty})B'$ for any map $B \to B'$ of seminormal rings. Applying the first equality to B' identifies the right side with $\sqrt{fB'}$.

For (c), note that $z^{(a+1)N} = y^{(b+1)N}v^{a+1}u^{-(b+1)}\pi^{b-a}$. Since $(a+1)N < (b+1)N$, applying (a) with $(g, x, m, n) = (z, y, (a+1)N, (b+1)N)$ gives a unique $w \in \sqrt{yB}$ such that $z = wy$. Put $d = w^N - vu^{-1}\pi^{b-a}$; then $y^N d = z^N - v\pi^b = 0$. If $a = 0$, then y is a unit, so $d = 0$. If $a > 0$, then $\pi^a = u^{-1}y^N \in yB$, so $\pi, d \in \sqrt{yB}$. Choose $\ell > 0$ with $d^\ell \in yB$. Then $d^{\ell N+1} = 0$, and hence $d = 0$ because B is reduced.

The final assertion follows because every localization of a perfectoid ring is seminormal by Proposition 2.2(b) and [Sta26, Tag 0EUM]. □

Corollary 2.6. *Let $R \to B$ be a ring map with B perfectoid, and let $f \in R$ be such that its image in B admits a compatible system of p-power roots. For every p^∞-torsion R-module M, the natural map*

$$M \otimes_R (f^{1/p^\infty})B \longrightarrow M \otimes_R (fB)_{\mathrm{perfd}}$$

is an isomorphism.

Proof. The ring $B/\sqrt{fB}$ is reduced, so its p^∞-torsion is killed by p: if $p^n\bar{b} = 0$, then $(p\bar{b})^n = 0$. By [CLM+26, Proposition 2.13], the derived p-completion of $B/\sqrt{fB}$ is therefore discrete and agrees with $B/(\sqrt{fB})^- = B/(fB)_{\mathrm{perfd}}$ by Corollary 2.4.

A complex and its derived p-completion become equivalent after derived tensoring with a p^∞-torsion module. This is immediate for a module killed by p^N, and the general case follows from $M = \operatorname{colim}_N M[p^N]$. Applying this to $B/\sqrt{fB}$ and using Lemma 2.5 gives $M \otimes^{\mathbb{L}}_R (f^{1/p^\infty})B \simeq M \otimes^{\mathbb{L}}_R (fB)_{\mathrm{perfd}}$. Taking zeroth cohomology proves the result. □

We will often work with a special perfectoid ring, constructed as follows.

Notation 2.7. Let $(R, \mathfrak{m}, k)$ be a complete Noetherian local ring of residue characteristic $p > 0$. Fix a Cohen ring C_k for k and a local map $C_k \to W(k^{1/p^\infty})$ lifting the inclusion $k \subseteq k^{1/p^\infty}$. Let

$$\varphi\colon A = C_k[\![x_1, \dots, x_n]\!] \longrightarrow R$$

be a local homomorphism inducing the identity on residue fields such that R is module-finite over $\varphi(A)$. The map φ need not be injective or surjective. When it is surjective, we call it a *Cohen*

presentation of R; when it is injective, we call it a *Noether–Cohen normalization* of R. Set

$$A_\infty := \left(A \underset{C_k}{\widehat{\otimes}} W(k^{1/p^\infty}) \right) \left[p^{1/p^\infty}, x_1^{1/p^\infty}, \dots, x_n^{1/p^\infty} \right]^{\wedge p}.$$

Here both completions are p-adic. Then A_∞ is perfectoid by [BIM19, Example 3.8(4)]. Set $R^{A_\infty}_{\mathrm{perfd}} := (R \otimes_A A_\infty)_{\mathrm{perfd}}$, which is a perfectoid R-algebra by [BS22, Theorem 10.11]. Both A_∞ and $R^{A_\infty}_{\mathrm{perfd}}$ depend on the chosen regular system of parameters $p, x_1, \dots, x_n$ of A, and $R^{A_\infty}_{\mathrm{perfd}}$ also depends on φ.

Lemma 2.8. *Let $(R, \mathfrak{m})$ be a Noetherian complete local domain of mixed characteristic and let $A \to R$ be as in Notation 2.7. Then every nonzero element of R is a nonzerodivisor on $R^{A_\infty}_{\mathrm{perfd}}$. Moreover, for every $0 \neq g \in R$, there exists an integer $N \geq 0$ such that $(p^n R^{A_\infty}_{\mathrm{perfd}} :_{R^{A_\infty}_{\mathrm{perfd}}} g) \subseteq p^{n-N} R^{A_\infty}_{\mathrm{perfd}}$ for every $n > N$.*

Proof. Let $0 \neq g \in R$ and we will first show that g is a nonzerodivisor on $R^{A_\infty}_{\mathrm{perfd}}$. If necessary, we may adjoin variables to A to obtain a new ring S with $S \twoheadrightarrow R$. We may further assume that g admits a compatible system of p-power roots in $R^{S_\infty}_{\mathrm{perfd}}$ (by replacing S by $S[\![x]\!]$ and considering $S[\![x]\!] \to R$ where $x \mapsto g$). Because $R^{A_\infty}_{\mathrm{perfd}} \to R^{S_\infty}_{\mathrm{perfd}}$ is p-completely faithfully flat and in particular injective by André's flatness lemma, it suffices to show that g is not a zerodivisor on $R^{S_\infty}_{\mathrm{perfd}}$. Writing $R = S/Q$, we also have that $S_\infty/Q \twoheadrightarrow (S_\infty/Q)_{\mathrm{perfd}} \cong R^{S_\infty}_{\mathrm{perfd}}$ by [BS22, Theorem 7.4] (essentially also André's flatness lemma).

In what follows, for a p-torsion free ring T, we denote T' the p-root closure of T in $T[1/p]$. That is,

$$T' = \{\eta \in T[1/p] \mid \eta^{p^e} \in T \text{ for some } e\}.$$

We consider the following commutative diagram

$$\begin{array}{ccccc} S_\infty/Q & \longrightarrow & (S_\infty/Q)' & \longrightarrow & (S_\infty/Q)'^{\wedge p} \\ & \searrow & & & \downarrow \cong \\ & & & & (S_\infty/Q)_{\mathrm{perfd}} \end{array}$$

where the isomorphism follows from [Ish24, Corollary 5.9]. Note that all rings in the above diagrams are p-torsion free (see [MST+22, Lemma A.2]).

Suppose $gz = 0$ in $(S_\infty/Q)_{\mathrm{perfd}}$. By the surjection $S_\infty/Q \twoheadrightarrow (S_\infty/Q)_{\mathrm{perfd}}$, we will abuse notation and also use g and z as elements in S_∞/Q. Since $(S_\infty/Q)_{\mathrm{perfd}}$ is reduced, we know that $g^{1/p^\infty} z = 0$ in $(S_\infty/Q)_{\mathrm{perfd}}$ and thus

$$g^{1/p^\infty} z \in \ker(S_\infty/Q \to (S_\infty/Q)'^{\wedge p})$$

Unwinding definitions, this means

$$g^{1/p^\infty} z \in \cap_n p^n (S_\infty/Q)' \cap (S_\infty/Q).$$

By the definition of p-root closure, this means

$$\forall m, n, \exists e_0 \text{ such that } g^{p^{e-m}} z^{p^e} \in p^{np^e}(S_\infty/Q) \text{ for all } e \geq e_0.$$

It follows that for all $e \gg 0$ (depending on m, n),

$$z^{p^e} \in p^{np^e}(S_\infty/Q) : g^{p^{e-m}} = (p^{np^e} :_{S/Q} g^{p^{e-m}})(S_\infty/Q) \tag{2.8.1}$$

by faithful flatness of $S/Q \to S_\infty/Q$. Let $v_1, \dots, v_t$ be the Rees valuations of $(p) \subseteq R$. For any $x \in p^{np^e} :_R g^{p^{e-m}}$, we have $v_i(x) + p^{e-m}v_i(g) \geq np^e v_i(p)$. Thus

$$\frac{v_i(x)}{p^e} \geq nv_i(p) - \frac{v_i(g)}{p^m}.$$

Choose $m \gg 0$ so that $v_i(g)/p^m \leq v_i(p)$ for all i. We have that

$$v_i(x) \geq (n-1)p^e v_i(p)$$

for all i and thus $x \in \overline{p^{(n-1)p^e}R} \subseteq p^{(n-1)p^e - C}R$ by valuative criterion of integral closure and the uniform Briançon–Skoda theorem (C is a constant that does not depend on n and e) [Hun92, MMRS25]. Now (2.8.1) says that for any fixed n, for all $e \gg 0$, we have

$$z^{p^e} \in p^{(n-1)p^e - C}(S_\infty/Q) \subseteq p^{(n-2)p^e}(S_\infty/Q).$$

It follows that $\frac{z}{p^{n-2}} \in (S_\infty/Q)'$ and thus $\frac{z}{p^{n-2}} \in (S_\infty/Q)_{\mathrm{perfd}}$ (note that we are abusing notations, it should be interpreted as the image of $\frac{z}{p^{n-2}}$). Since this is true for all n, we obtain that

$$z \in \cap_n p^{n-2}(S_\infty/Q)_{\mathrm{perfd}} = 0$$

since $(S_\infty/Q)_{\mathrm{perfd}}$ is p-adically complete.

For the final statement, choose $N \geq 1$ such that $v_i(g) \leq (N-1)v_i(p)$ for every i. If $n > N$, $z \in (S_\infty/Q)'$, and $gz \in p^n(S_\infty/Q)'$, the calculation above applied to

$$g^{p^e} z^{p^e} \in p^{np^e}(S_\infty/Q)$$

for $e \gg 0$ gives $z^{p^e} \in p^{(n-N)p^e}(S_\infty/Q)$. Thus $z/p^{n-N} \in (S_\infty/Q)'$. Since $(S_\infty/Q)' \to (S_\infty/Q)_{\mathrm{perfd}}$ is surjective (the kernel is exactly $\cap_n p^n(S_\infty/Q)'$), passing to p-adic completion we obtain that

$$(p^n(S_\infty/Q)_{\mathrm{perfd}} :_{(S_\infty/Q)_{\mathrm{perfd}}} g) \subseteq p^{n-N}(S_\infty/Q)_{\mathrm{perfd}}.$$

Since $R^{A_\infty}_{\mathrm{perfd}} \to (S_\infty/Q)_{\mathrm{perfd}}$ is faithfully flat modulo p^n,

$$\begin{aligned}(0 :_{R^{A_\infty}_{\mathrm{perfd}}/p^n} g) \otimes_{R^{A_\infty}_{\mathrm{perfd}}} (S_\infty/Q)_{\mathrm{perfd}} &\cong 0 :_{(S_\infty/Q)_{\mathrm{perfd}}/p^n} g \\ &\subseteq p^{n-N}(S_\infty/Q)_{\mathrm{perfd}}/p^n(S_\infty/Q)_{\mathrm{perfd}} \\ &\cong \left(p^{n-N}R^{A_\infty}_{\mathrm{perfd}}/p^n R^{A_\infty}_{\mathrm{perfd}}\right) \otimes_{R^{A_\infty}_{\mathrm{perfd}}} (S_\infty/Q)_{\mathrm{perfd}},\end{aligned}$$

which implies that $0 :_{R^{A_\infty}_{\mathrm{perfd}}/p^n} g \subseteq p^{n-N}R^{A_\infty}_{\mathrm{perfd}}/p^n R^{A_\infty}_{\mathrm{perfd}}$ as wanted. □

Proposition 2.9. *In the setting of Notation 2.7, suppose also that $0 \neq g \in A$ is such that $A[g^{-1}] \to R[g^{-1}]$ is finite étale. Let N be a p^∞-torsion R-module and $\zeta \in N$. For each minimal prime $\mathfrak{p}_i$ of R, set $R_i = R/\mathfrak{p}_i$ and let ζ_i be the image of ζ in $R_i \otimes_R N$. Suppose that, whenever $g \notin \mathfrak{p}_i$, the element ζ_i maps to zero in $B_i \otimes_R N$ for some perfectoid balanced big Cohen–Macaulay R_i^+-algebra B_i. Then $(gA_\infty)_{\mathrm{perfd}}(1 \otimes \zeta) = 0$ in $R^{A_\infty}_{\mathrm{perfd}} \otimes_R N$.*

Proof. We first show that $(gA_\infty)_{\mathrm{perfd}}(1 \otimes \zeta) = 0$ in $(R_i)^{A_\infty}_{\mathrm{perfd}} \otimes_R N$ for every i. If $g \in \mathfrak{p}_i$, then g is in the kernel of $A_\infty \to (R_i)^{A_\infty}_{\mathrm{perfd}}$. Since the kernel is a perfectoid ideal, it must contain $(gA_\infty)_{\mathrm{perfd}}$, so the claim follows. If $g \notin \mathfrak{p}_i$, the ideal $\mathfrak{p}_i R[g^{-1}]$ is a minimal prime of the finite étale $A[g^{-1}]$-algebra $R[g^{-1}]$ and contracts to (0) in $A[g^{-1}]$. Since $R[g^{-1}]$ is finite étale over the normal domain $A[g^{-1}]$, it is normal. Thus $R[g^{-1}]$ is the product of the $R_i[g^{-1}]$ ranging over all i with $g \notin \mathfrak{p}_i$ and applying [Sta26, Tag 00U2] gives that $A[g^{-1}] \to R_i[g^{-1}]$ is finite étale. Then, $A \subset R_i$ is a Noether-Cohen

normalization so the argument of [CLM+26, Lemma 5.6], which applies to the present choice of A_∞ by [BMP+25, Lemma 7.9 and the proof of Claim 7.11], gives $(gA_\infty)_{\mathrm{perfd}}(1\otimes\zeta)=0$ in $(R_i)^{A_\infty}_{\mathrm{perfd}}\otimes_R N$, proving the componentwise claim.

Since perfectoidization commutes with finite products, the target of

$$R^{A_\infty}_{\mathrm{perfd}}\longrightarrow\prod_i (R_i)^{A_\infty}_{\mathrm{perfd}}$$

is the perfectoidization of the finite $R^{A_\infty}_{\mathrm{perfd}}$-algebra $(\prod_i R_i)\otimes_R R^{A_\infty}_{\mathrm{perfd}}$, which agrees with $R^{A_\infty}_{\mathrm{perfd}}$ after inverting g. Thus the displayed map is p-completely $(gR^{A_\infty}_{\mathrm{perfd}})_{\mathrm{perfd}}$-almost pure by [CLM+26, Proposition 2.25]. Since N is p^∞-torsion, it follows that

$$\ker\left(R^{A_\infty}_{\mathrm{perfd}}\otimes_R N\to\prod_i (R_i)^{A_\infty}_{\mathrm{perfd}}\otimes_R N\right)$$

is annihilated by $(gR^{A_\infty}_{\mathrm{perfd}})_{\mathrm{perfd}}$. Now the componentwise vanishing shows that $(gA_\infty)_{\mathrm{perfd}}(1\otimes\zeta)$ is in the above kernel. Thus we obtain that

$$(gA_\infty)_{\mathrm{perfd}}(1\otimes\zeta)=(gA_\infty)_{\mathrm{perfd}}\left((gA_\infty)_{\mathrm{perfd}}(1\otimes\zeta)\right)\subseteq(gR^{A_\infty}_{\mathrm{perfd}})_{\mathrm{perfd}}\left((gA_\infty)_{\mathrm{perfd}}(1\otimes\zeta)\right)=0$$

where we used that $(gA_\infty)_{\mathrm{perfd}}$ is idempotent (after tensoring with N) and that $(gA_\infty)_{\mathrm{perfd}}R^{A_\infty}_{\mathrm{perfd}}\subseteq(gR^{A_\infty}_{\mathrm{perfd}})_{\mathrm{perfd}}$ (which follows from universality as $A_\infty\to(R^{A_\infty}_{\mathrm{perfd}}/gR^{A_\infty}_{\mathrm{perfd}})_{\mathrm{perfd}}$ kills g). □

2.2. **Generalized and Mumford divisors.** We use Hartshorne's theory of generalized divisors in a form based on codimension-two extension; see [Har94, Har07]. Let X be a Noetherian scheme. For a coherent $\mathscr{O}_X$-module $\mathscr{F}$, we say that $\mathscr{F}$ is $\mathbf{S}_k$ if $\operatorname{depth}_{\mathscr{O}_{X,x}}\mathscr{F}_x\geq\min\{k,\dim(\operatorname{Supp}(\mathscr{F}_x))\}$ for every $x\in X$. We say that X is $\mathbf{S}_k$ if $\mathscr{O}_X$ is $\mathbf{S}_k$ [Sta26, Tag 0341]. If every local ring of X is catenary and X is $\mathbf{S}_2$, then X is locally equidimensional [Sta26, Tag 0FIW]. This applies, in particular, when X is excellent and $\mathbf{S}_2$. We say that a closed subset $Z\subseteq X$ has codimension at least two if $\dim\mathscr{O}_{X,x}\geq 2$ for every $x\in Z$. For a coherent $\mathscr{O}_X$-module $\mathscr{F}$, set $\mathscr{F}^{**}:=\mathscr{H}\mathrm{om}_X(\mathscr{H}\mathrm{om}_X(\mathscr{F},\mathscr{O}_X),\mathscr{O}_X)$, and call $\mathscr{F}$ *reflexive* if the natural map $\mathscr{F}\to\mathscr{F}^{**}$ is an isomorphism. If X is a Noetherian scheme, recall that a *big open subset* $U\subseteq X$ is an open subset whose complement has codimension ≥ 2.

Definition 2.10. Let X be a Noetherian scheme and $\mathscr{G}$ an $\mathscr{O}_X$-module. We say that $\mathscr{G}$ has the *codimension-two extension property* if for every open immersion $j\colon U\hookrightarrow X$ from a big open, the canonical map $\mathscr{G}\longrightarrow j_*(\mathscr{G}|_U)$ is an isomorphism. Equivalently, for every $\mathscr{O}_X$-module $\mathscr{F}$, restriction induces an isomorphism

$$\mathrm{Hom}_X(\mathscr{F},\mathscr{G})\xrightarrow{\sim}\mathrm{Hom}_U(\mathscr{F}|_U,\mathscr{G}|_U).$$

Let $\mathcal{S}_X\subseteq\mathscr{O}_X$ be the subsheaf of local nonzerodivisors, and set $\mathscr{K}_X:=\mathcal{S}_X^{-1}\mathscr{O}_X$, the sheaf of total quotient rings of X. For every affine open $V=\operatorname{Spec}A\subseteq X$, one has $\Gamma(V,\mathscr{K}_X)=K(A)$, the total quotient ring of A [Sta26, Tag 02OV]. A *fractional ideal* on X is a coherent $\mathscr{O}_X$-submodule $\mathscr{I}\subseteq\mathscr{K}_X$; it is *nondegenerate* if $\mathscr{I}_\eta=(\mathscr{K}_X)_\eta$ at every generic point η of an irreducible component of X. It is *principal* if $\mathscr{I}=g\mathscr{O}_X$ for some $g\in\Gamma(X,\mathscr{K}_X)^\times$, *locally principal* if it is locally of this form, and *locally principal in codimension one* if it is locally principal on a big open subset. A nondegenerate $\mathbf{S}_2$ fractional ideal has the codimension-two extension property [Sta26, Tag 0E9I].

Consequently, these fractional ideals and their morphisms are determined by restriction to any big open. This motivates the following construction.

Definition 2.11. For an $\mathscr{O}_X$-module $\mathscr{F}$, set

$$\mathscr{F}^{\mathrm{c1}} \coloneqq \varinjlim_{U} (j_U)_* \big(\mathscr{F}|_U\big),$$

where the filtered colimit is over big open subsets $U \subseteq X$, ordered by reverse inclusion, and $j_U \colon U \hookrightarrow X$ is the open immersion. There is a natural map $\mathscr{F} \to \mathscr{F}^{\mathrm{c1}}$, which is an isomorphism at every point of codimension at most one. We call $\mathscr{F}^{\mathrm{c1}}$ the *codimension-one envelope* of $\mathscr{F}$. If $X = \operatorname{Spec} R$ and N is an R-module, we use the same notation for

$$N^{\mathrm{c1}} \coloneqq \Gamma\big(X, (\widetilde{N})^{\mathrm{c1}}\big) = \varinjlim_{U} \Gamma(U, \widetilde{N}),$$

further writing $N^{\mathrm{c1},R}$ to specify the base as necessary. In particular, $\widetilde{N}^{\mathrm{c1}} \cong \widetilde{N^{\mathrm{c1}}}$ (which can be quickly deduced from [Sta26, Tag 01LC] and [Sta26, Tag 01BH]). In either setting, we observe that $(-)^{\mathrm{c1}}$ is a functor and $M \to M^{\mathrm{c1}}$ is a natural transformation.

Proposition 2.12. *Fix R to be a Noetherian ring and suppose that M and N are R-modules. Then:*

(a) The sheaf $\widetilde{N^{\mathrm{c1}}}$ has the codimension-2 extension property.

(b) $N^{\mathrm{c1}} \to (N^{\mathrm{c1}})^{\mathrm{c1}}$ is an isomorphism.

(c) For every height ≤ 1 prime P of R, the map $N_P \to (N^{\mathrm{c1}})_P$ is an isomorphism.

(d) Suppose $M \to N$ is a map of R-modules, then $M^{\mathrm{c1}} \to N^{\mathrm{c1}}$ is an isomorphism if and only if $M_P \to N_P$ is an isomorphism for all height ≤ 1 primes $P \subseteq R$.

(e) The natural map $(M \otimes N)^{\mathrm{c1}} \to (M^{\mathrm{c1}} \otimes N)^{\mathrm{c1}}$ is an isomorphism.

(f) The formation of $(-)^{\mathrm{c1}}$ commutes with filtered colimits. In other words, for a filtered system $\{M_i\}$ with $\varinjlim_i M_i = M$, we have a natural isomorphism $\varinjlim_i (M_i^{\mathrm{c1}}) \cong M^{\mathrm{c1}}$.

Proof. Write $X = \operatorname{Spec} R$. For (a), pick a big open subset $V \subseteq X$. We need to show that

$$N^{\mathrm{c1}} = \varinjlim_{U \subseteq X} \Gamma(U, \widetilde{N}) \to \Gamma(V, \varinjlim_{U \subseteq X} (j_U)_* \widetilde{N}|_U)$$

is an isomorphism where the filtered colimits run over big open subsets. But the right side is equal to $\varinjlim_{U \subseteq V} \Gamma(U, \widetilde{N})$ which clearly agrees with N^{c1}. This proves the first statement. But then (b) follows immediately.

For (c), suppose we have two big open subsets $P \in U \subseteq V \subseteq X$. Then by [Sta26, Tag 0EHM], [Sta26, Tag 01LC], and the correspondence between modules and quasi-coherent sheaves, all the restriction maps $N_P = \Gamma(X, \widetilde{N})_P \to \Gamma(V, \widetilde{N})_P \to \Gamma(U, \widetilde{N})_P$ are isomorphisms. It follows that $N_P \to (N^{\mathrm{c1}})_P$ is also an isomorphism.

The ($\Rightarrow$) direction of (d) is immediate from (c). For the converse, suppose that K and C are the kernel and cokernel of $M^{\mathrm{c1}} \to N^{\mathrm{c1}}$ so that $0 \to K \to M^{\mathrm{c1}} \to N^{\mathrm{c1}} \to C \to 0$ is an exact sequence. Note that by (a), M^{c1} has codimension-2 extension property. In particular, this implies that $H^0_I(M^{\mathrm{c1}}) = H^1_I(M^{\mathrm{c1}}) = 0$ for all ideals $I \subseteq R$ of height ≥ 2 (and likewise for N^{c1}). Now for every finitely generated $K' \subseteq K$, $K'_P = 0$ for all height ≤ 1 primes P implies that $K' = H^0_J(K')$

for some $J \subseteq R$ of height ≥ 2. But then $K' = 0$ as $H^0_J(M^{\mathrm{c1}}) = 0$. It follows that $K = 0$ and we have $0 \to M^{\mathrm{c1}} \to N^{\mathrm{c1}} \to C \to 0$. By examining the long exact sequence of local cohomology we obtain that $H^0_I(C) = 0$ for all $I \subseteq R$ of height ≥ 2. But for every finitely generated $C' \subseteq C$, we know that $C' = H^0_J(C')$ for some $J \subseteq R$ of height ≥ 2 because $C'_P \subseteq C_P = 0$ for all height ≤ 1 primes P. Thus $C' = 0$ as $H^0_J(C) = 0$. Therefore $C = 0$ and we have $M^{\mathrm{c1}} \to N^{\mathrm{c1}}$ is an isomorphism.

For (e), notice that $M \otimes N \to M^{\mathrm{c1}} \otimes N$ becomes an isomorphism after localizing at a height ≤ 1 prime ideal P. Thus, the result follows from (d).

Finally, (f) follows as $\Gamma(U, -)$ commutes with filtered colimits and since colimits commute with colimits. □

Lemma 2.13. *Let X be a Noetherian, locally equidimensional $\mathbf{S}_2$-scheme. Let $\mathscr{I}$ be a nondegenerate fractional ideal. Then $\mathscr{I}^{\mathrm{c1}}$ is a nondegenerate $\mathbf{S}_2$ fractional ideal and is the unique such fractional ideal agreeing with $\mathscr{I}$ at every point of codimension at most one. In particular, the natural map $\mathscr{I} \longrightarrow \mathscr{I}^{\mathrm{c1}}$ is an isomorphism if and only if $\mathscr{I}$ is $\mathbf{S}_2$.*

If $\mathscr{I}$ is locally principal in codimension one, we have a canonical identification $\mathscr{I}^{\mathrm{c1}} \cong \mathscr{I}^{**}$. In particular, $\mathscr{I}^{\mathrm{c1}}$ is reflexive, and, under the same hypothesis, $\mathscr{I}$ is $\mathbf{S}_2$ if and only if it is reflexive.

Proof. The assertion is local, so we may clear denominators and assume $\mathscr{I} \subseteq \mathscr{O}_X$. Applied to $\mathscr{O}_X$, the codimension-two extension property gives $(j_U)_*\mathscr{O}_U = \mathscr{O}_X$. Thus $(j_U)_*(\mathscr{I}|_U)$ is a quasi-coherent ideal of $\mathscr{O}_X$ [Sta26, Tag 01LC]. Their filtered union $\mathscr{I}^{\mathrm{c1}}$ is therefore a quasi-coherent ideal; since X is Noetherian, it is coherent and the system stabilizes. As a submodule of $\mathscr{K}_X$, it is $\mathbf{S}_1$. A stabilizing U may be shrunk to omit any prescribed point of codimension at least two, so [Sta26, Tag 0DX4] gives depth at least two there. Thus $\mathscr{I}^{\mathrm{c1}}$ is $\mathbf{S}_2$. Nondegeneracy and agreement with $\mathscr{I}$ in codimension at most one are immediate, while the codimension-two extension property gives uniqueness and the final assertion.

For the additional statement, choose an open immersion $j\colon U \hookrightarrow X$ whose complement has codimension at least two and such that $\mathscr{I}|_U$ is invertible. The sheaf $\mathscr{I}^{**}$ is a nondegenerate fractional ideal and is $\mathbf{S}_2$ by [Sta26, Tag 0AV6]; its restriction to U is canonically $\mathscr{I}|_U$. The codimension-two extension property therefore identifies $\mathscr{I}^{**}$ with $j_*(\mathscr{I}|_U)$, and hence with $\mathscr{I}^{\mathrm{c1}}$. Applying this identification to $\mathscr{I}^{\mathrm{c1}}$ shows that the latter is reflexive. The final equivalence now follows from the first part of the lemma. □

Definition 2.14. Let X be a Noetherian, locally equidimensional $\mathbf{S}_2$-scheme.

(1) A *generalized divisor* D is a nondegenerate $\mathbf{S}_2$ fractional ideal $\mathscr{I}_D$.

(2) We call a generalized divisor D *effective* if $\mathscr{I}_D \subseteq \mathscr{O}_X$, *principal* if $\mathscr{I}_D$ is principal, and *Cartier* if $\mathscr{I}_D$ is locally principal (which agrees with the usual notion of a Cartier divisor). If $\mathscr{I}_D = g\mathscr{O}_X$, we write $D = \mathrm{div}_X(g)$. The zero divisor is given by $\mathscr{I}_0 = \mathscr{O}_X$.

(3) A generalized divisor is a *Weil generalized divisor* if $\mathscr{I}_D$ is locally principal in codimension one. Hartshorne calls these almost Cartier divisors [Har07, Section 2]; we use "Weil" because "almost" has a different meaning here.

For generalized divisors D and E, define

$$\mathscr{I}_{D+E} \coloneqq (\mathscr{I}_D\mathscr{I}_E)^{\mathrm{c1}}, \qquad \mathscr{I}_{-D} \coloneqq \mathscr{H}\mathrm{om}_X(\mathscr{I}_D, \mathscr{O}_X), \tag{2.14.1}$$

where $\mathscr{I}_D\mathscr{I}_E$ is the image of $\mathscr{I}_D \otimes_{\mathscr{O}_X} \mathscr{I}_E \to \mathscr{K}_X$. The product $\mathscr{I}_D\mathscr{I}_E$ is a coherent nondegenerate fractional ideal, so Lemma 2.13 shows that $\mathscr{I}_{D+E}$ is a generalized divisor. On an open set $V \subset X$, the sheaf $\mathscr{I}_{-D}$ is described by $\{q \in \mathscr{K}_X(V) \mid q\mathscr{I}_D(V) \subseteq \mathscr{O}_X(V)\}$; it is coherent and nondegenerate, and is $\mathbf{S}_2$ by [Sta26, Tag 0AV6]. Thus $-D$ is also a generalized divisor. For any generalized divisor D, set $\mathscr{O}_X(D) \coloneqq \mathscr{I}_{-D}$. If D is effective, the inclusion $\mathscr{O}_X \subseteq \mathscr{O}_X(D)$ determines a distinguished global section 1 of $\mathscr{O}_X(D)$. When $X = \operatorname{Spec} R$, set $I_D = \Gamma(X, \mathscr{I}_D)$ and write

$$R(D) = \Gamma(X, \mathscr{O}_X(D)) = \operatorname{Hom}_R(I_D, R) = \{q \in K(R) \mid qI_D \subseteq R\}.$$

Now suppose that D and E are Weil generalized divisors, and choose a common Cartier locus $j\colon U \hookrightarrow X$ whose complement has codimension at least two. By Lemma 2.13, $\mathscr{I}_D$ and $\mathscr{I}_E$ are reflexive. In particular, $\mathscr{O}_X(-D) = \mathscr{I}_D$. That lemma and the codimension-two extension property give

$$\mathscr{I}_{D+E} \cong (\mathscr{I}_D\mathscr{I}_E)^{**} \cong j_*\big((\mathscr{I}_D\mathscr{I}_E)|_U\big), \qquad \mathscr{I}_{-D} \cong j_*\big((\mathscr{I}_D|_U)^{-1}\big).$$

In particular, $D + E$ and $-D$ are Weil generalized divisors. These descriptions are independent of U and show that the Weil generalized divisors form an abelian group. For $n \in \mathbb{Z}$, write nD for the corresponding multiple. Define $\operatorname{Supp}(D)$ to be the closure of the support of $D|_U$; this is independent of the choice of U. If D is effective, then $\operatorname{Supp}(D) = \operatorname{Supp}(\mathscr{O}_X/\mathscr{I}_D)$. Whenever K_X appears, we assume that X admits a canonical sheaf ω_X. If X is $\mathbf{S}_2$ and $\mathbf{G}_1$, after choosing a generic trivialization of ω_X, we write K_X for the corresponding Weil generalized divisor, so that $\mathscr{O}_X(K_X) \cong \omega_X$.

We record the following compatibility with flat pullback.

Proposition 2.15. *Let $R \to R'$ be a flat homomorphism of Noetherian, locally equidimensional $\mathbf{S}_2$ rings, and let D be a Weil generalized divisor on $\operatorname{Spec} R$. Then $R' \otimes_R I_D \subseteq K(R')$ defines a Weil generalized divisor D' on $\operatorname{Spec} R'$, called the flat pullback of D, and there are canonical identifications*

$$I_{D'} = R' \otimes_R I_D \qquad \text{and} \qquad R'(D') \cong R' \otimes_R R(D).$$

Flat pullback is functorial, defines a homomorphism of groups of Weil generalized divisors, and preserves effective, Cartier, and principal divisors. If D is effective, the second identification carries 1 to $1 \otimes 1$.

Proof. Let W be the set of nonzerodivisors of R. Flatness gives $R' \otimes_R I_D \hookrightarrow R' \otimes_R K(R) \cong W^{-1}R' \hookrightarrow K(R')$, since every element of W remains a nonzerodivisor on R'. Thus $R' \otimes_R I_D$ is a fractional ideal.

Choose an open immersion $j\colon U \hookrightarrow \operatorname{Spec} R$ whose complement has codimension at least two and on which D is Cartier, and let $j'\colon V \hookrightarrow \operatorname{Spec} R'$ be its inverse image. Flat going-down [Sta26, Tag 00HS] shows that $\operatorname{Spec} R' \setminus V$ has codimension at least two and that $R' \otimes_R I_D$ is nondegenerate: its restriction to V is invertible. By Lemma 2.13, I_D is reflexive, so flat base change for Hom shows that $R' \otimes_R I_D$ is reflexive. It is therefore $\mathbf{S}_2$ by [Sta26, Tag 0AV6] and defines the Weil generalized divisor D'. The same base change gives the second displayed identification. Functoriality and the remaining compatibilities follow from the double-dual description of addition and the definitions. □

We conclude by relating Weil generalized divisors to Mumford divisors.

Definition 2.16. Let X be a Noetherian, locally equidimensional $\mathbf{S}_2$-scheme. A *prime Mumford divisor* on X is an integral closed subscheme $Z \subseteq X$ of codimension one such that X is regular at its generic point η_Z. A *Mumford divisor* (respectively, a *Mumford* $\mathbb{Q}$*-divisor*) is a finite formal sum $D = \sum_i a_i Z_i$ of prime Mumford divisors with $a_i \in \mathbb{Z}$ (respectively, $a_i \in \mathbb{Q}$) [Kol18, Definition 1]. Its support is $\bigcup_{a_i \neq 0} Z_i$, and it is *effective* if every $a_i \geq 0$. A Mumford $\mathbb{Q}$-divisor D is $\mathbb{Q}$-*Cartier* if mD is Cartier for some integer $m > 0$. If $D = \sum_i a_i Z_i$ is a Mumford divisor then we write D_{red} for the Mumford divisor $\sum_i b_i Z_i$ where $b_i = 0$ if $a_i = 0$ and $b_i = 1$ if $a_i \neq 0$.

If Z is a prime Mumford divisor, its ideal sequence combined with [Sta26, Tag 00LX] shows that $\mathscr{I}_Z$ is $\mathbf{S}_2$, while regularity at η_Z gives local principality in codimension one. Thus $\mathscr{I}_Z$ defines a Weil generalized divisor, which we again denote by Z. The group law then associates a Weil generalized divisor to every Mumford divisor. For a Mumford divisor D, choose an open immersion $j\colon U \hookrightarrow X$ whose complement has codimension at least two and such that $D|_U$ is Cartier. Then $\mathscr{I}_D = \mathscr{O}_X(-D) = j_*\mathscr{O}_U(-D|_U) \subseteq \mathscr{K}_X$. This association preserves support and effectivity. For Mumford $\mathbb{Q}$-divisors, inequalities and convergence are coefficientwise, while support and reduction have their usual meanings. For effective Mumford $\mathbb{Q}$-divisors D' and D'' on X, write

$$D' \prec D'' \quad \Longleftrightarrow \quad D' \leq (1-\varepsilon)D'' \text{ for some } 0 < \varepsilon < 1. \tag{2.16.1}$$

Remark 2.17. In general, the flat pullback of a Mumford divisor need not be a Mumford divisor. However, flat pullback preserves Mumford divisors – and hence extends $\mathbb{Q}$-linearly to Mumford $\mathbb{Q}$-divisors – when the fibers are R_0 (e.g., when the morphism is regular).

More precisely, let $f\colon R \to S$ be a flat map between excellent reduced $\mathbf{S}_2$ local rings and let Δ be a Mumford $\mathbb{Q}$-divisor on R. Then Δ_S is a Mumford $\mathbb{Q}$-divisor if and only if S_P is regular for every minimal prime P of $\mathfrak{p}S$ and every height-one prime $\mathfrak{p} \in \operatorname{Supp}(\Delta)$. In that case, we often abuse notation and write $\Delta = \Delta_S$ as divisors on S since there is no potential for confusion. We also call $\Delta = \Delta_S$ the *flat pullback* of Δ to S.

Moreover, if Δ is an effective Mumford $\mathbb{Q}$-divisor on R then one can always construct a finite flat extension $f\colon R \to S$ such that Δ admits a flat pullback to an effective Mumford divisor on S as the following construction shows.

Construction 2.18. Let R be an excellent $\mathbf{S}_2$ ring and let Δ be an effective Mumford $\mathbb{Q}$-divisor on R. Choose $m > 0$ such that $D_m \coloneqq m\Delta$ is integral. Using prime avoidance, choose a nonzerodivisor $f \in R$ that vanishes to order one along every prime component of D_m and does not vanish along any codimension-one component of the singular locus. Then $\operatorname{div}_R(f) = (D_m)_{\mathrm{red}} + G$ for an effective Mumford divisor G having no common component with D_m. Set $S_{f,m} \coloneqq R[x]/(x^m - f)$, which is finite free, hence flat, over R. Note that x is a nonzerodivisor since $x^m = f$.

We claim that there is a unique effective Mumford divisor $\Delta_{f,m}$ on $S_{f,m}$ whose m-fold multiple is the flat pullback of D_m. Indeed, write $D_m = \sum_i d_i P_i$ where we identify P_i with the associated height-1 prime of R. For each P_i, the choice of f gives $f = u\pi$ in the DVR R_{P_i}, where $u \in R_{P_i}^{\times}$ and π is a uniformizer. Thus $x^m - u\pi$ is Eisenstein, so the quotient is a domain. In fact, $R_{P_i}[x]/(x^m - u\pi)$ is a DVR with uniformizer x, and there is a unique height one prime Q_i of $S_{f,m}$ over P_i. One can then define $\Delta_{f,m} \coloneqq \sum_i d_i Q_i$, which is Mumford. Note that its m-fold multiple is indeed the flat pullback of D_m since at Q_i, the pullback of a local equation π^{d_i} is x^{md_i} up to a unit. To see uniqueness, note that at Q_i, this calculation forces the coefficient to be d_i while at every point of height 1 not in the pullback of the support of Δ, effectivity and the vanishing of

its m-fold multiple force a coefficient of 0. We also note that $S_{f,m}$ need not be a domain or even reduced. However, if every irreducible component of $\operatorname{Spec} R$ contains a prime component of D_m then $S_{f,m}$ will be reduced.

When X is normal, Mumford divisors and Mumford $\mathbb{Q}$-divisors are precisely the usual Weil divisors and Weil $\mathbb{Q}$-divisors, respectively. Moreover, $D \mapsto \mathscr{I}_D = \mathscr{O}_X(-D)$ identifies the usual Weil divisors with the Weil generalized divisors, compatibly with addition, effectivity, Cartier and principal divisors, support, and the sheaves $\mathscr{O}_X(D)$. Under this identification, the codimension-one envelope of a fractional ideal is its usual reflexive hull (by Lemma 2.13); on affine opens, it is equivalently the intersection of the height-one localizations [Sta26, Tag 0AVB].

Lemma 2.19. *Let $(R, \mathfrak{m})$ be a complete normal local domain of dimension d, fix an effective canonical divisor K_R, giving an embedding $R \subset R(K_R) =: \omega_R$. Let C be an R-algebra, let $q \in C$, and let $\widetilde{q} \in (C \otimes_R R(-K_R))^{\mathrm{c1}}$ have the same image as q in C^{c1}. Then*

$$\operatorname{im}\Big(\operatorname{Hom}_R\big((C \otimes_R R(-K_R))^{\mathrm{c1}}, R\big) \xrightarrow{\mathrm{ev}_{\widetilde{q}}} R\Big) = \operatorname{Ann}_{\omega_R} \ker\Big(H^d_{\mathfrak{m}}(R) \xrightarrow{\cdot q} H^d_{\mathfrak{m}}(C)\Big) \subseteq R \subseteq \omega_R.$$

Proof. The codimension-two extension property and Hom-tensor adjunction give

$$\operatorname{Hom}_R\big((C \otimes_R R(-K_R))^{\mathrm{c1}}, R\big) \cong \operatorname{Hom}_R(C, \omega_R).$$

Suppose that ψ corresponds to φ. At a height-one prime P, choose a generator s of $R(-K_R)_P$ and write $\widetilde{q}_P = t \otimes s$. Since $\widetilde{q}$ maps to q, one has $q = st$ in C_P, and adjunction gives $\psi(\widetilde{q})_P = s\varphi(t)_P = \varphi(q)_P$. Since both sides are sections of the reflexive module ω_R, this height-one equality holds globally. Thus evaluation at $\widetilde{q}$ corresponds to evaluation at q, which under local duality is the Matlis dual of $H^d_{\mathfrak{m}}(R) \xrightarrow{\cdot q} H^d_{\mathfrak{m}}(C)$. Exactness of Matlis duality gives the asserted equality. □

2.3. **Pure and split maps.**

Lemma 2.20. *Let R be a Noetherian $\mathbf{S}_2$ ring. Let M be an R-module and $U \subset \operatorname{Spec} R$ an open set whose complement has codimension ≥ 2. Then, a fixed map $R \to M$ is pure if and only if the induced $R \to \Gamma(U, \widetilde{M})$ is pure.*

Proof. Since U is quasi-compact and quasi-separated, taking sections over U is additive and commutes with filtered colimits. By [Sta26, Tag 058K], pure exact sequences of modules are filtered colimits of split exact sequences, so taking sections over U preserves pure morphisms. By the $\mathbf{S}_2$ property and the codimension-two extension principle, $\Gamma(U, \mathscr{O}_U) = R$. Thus, if $R \to M$ is pure, then so is $R \to \Gamma(U, \widetilde{M})$. Conversely, the latter map factors through $R \to M$. If it is pure, then $R \to M$ is pure: after tensoring with any R-module, the map induced by $R \to M$ is the first map in an injective composite. □

Lemma 2.21. *Let R be a Noetherian $\mathbf{S}_2$ ring. Let $\eta\colon R \to N$ be a map of R-modules. Then η is pure if and only if the composite*

$$R \xrightarrow{\eta} N \longrightarrow N^{\mathrm{c1}}$$

is pure.

Proof. If the composite is pure, then η is pure because the composite factors through η. Conversely, suppose that η is pure. By Lemma 2.20, the induced map

$$R \longrightarrow \Gamma(U, \widetilde{N})$$

is pure for every U occurring in the definition of $N^{\mathrm{c}1}$. Filtered colimits of pure maps are pure, which proves the result. □

Definition 2.22. Let A be a ring. A *pointed* A-module is a pair consisting of an A-module M and a distinguished element $\mathbf{1}_M \in M$, or, equivalently, a homomorphism $A \to M$. If B is a ring and N is a pointed B-module then a homomorphism $(A, M) \to (B, N)$ is a homomorphism of rings $f\colon A \to B$ and a homomorphism of A-modules $\phi\colon M \to f_*N$ such that $\phi(\mathbf{1}_M) = \mathbf{1}_N$.

Definition 2.23. Let R be a ring, A be an R-algebra, and M be a pointed A-module. We use $\mathcal{I}^R(A, M)$ to denote the set $\{x \in A \mid R \xrightarrow{1\mapsto x\cdot \mathbf{1}_M} M \text{ is not pure}\}$.

When R is clear in the context, we will use the shorthand $\mathcal{I}(A, M)$. If $M = A$ we will further shorten to $\mathcal{I}^R(A)$ or $\mathcal{I}(A)$. Note that these loci need not be ideals in general. However, when R is Noetherian local, they are ideals by Lemma 2.24 below and will be referred to as *non-pure ideals*.

Lemma 2.24. *Let R be a Noetherian ring and M be an R-module. For any $x \in M$, the map $R \to M, 1 \mapsto x$, is pure if and only if the map*

$$E_R(R/\mathfrak{m}) \to E_R(R/\mathfrak{m}) \otimes_R M, \quad u \mapsto u \otimes x,$$

is injective for every maximal ideal $\mathfrak{m}$ of R.

Consequently, if $(R, \mathfrak{m})$ is local, A is an R-algebra, M is a pointed A-module, and u is a nonzero socle element of $E(R/\mathfrak{m})$, then

$$\mathcal{I}^R(A, M) = \ker\{A \to E(R/\mathfrak{m}) \otimes_R M, x \mapsto u \otimes x\mathbf{1}_M\} = \operatorname{Ann}_A\left(u \otimes \mathbf{1}_M \text{ in } E(R/\mathfrak{m}) \otimes_R M\right).$$

In particular, $\mathcal{I}^R(A, M)$ is an ideal of A containing $\mathfrak{m}A$.

Proof. Purity may be checked locally, and the first assertion follows from [HR74, Proposition 6.11]. In the local case, every nonzero submodule of E meets its socle, so injectivity may be tested on u. This gives the kernel description, from which the final assertions follow. □

We will use a generalization to our setting of the characterization of purity for approximately Gorenstein rings originating from [Hoc77]. Note that every Noetherian local ring whose completion is reduced is approximately Gorenstein by [Hoc77].

Lemma 2.25. *Let $(R, \mathfrak{m})$ be a Noetherian local ring that is approximately Gorenstein. Let A be an R-algebra and M be a pointed A-module. Then $\mathcal{I}^R(A, M) = \bigcup_t (I_t M :_A u_t \mathbf{1}_M)$, where I_t is a system of Gorenstein ideals cofinal with $\mathfrak{m}^t$ and u_t is a socle representative of R/I_t.*

Proof. Since R is approximately Gorenstein, the injective hull E of the residue field $k = R/\mathfrak{m}$ can be computed as:

$$E = \bigcup_t \operatorname{Ann}_E I_t \cong \varinjlim_t E_{R/I_t} \cong \varinjlim_t R/I_t.$$

It follows from Lemma 2.24 that $R \to M$ is pure if and only if $R/I_t \to M/I_tM$ is injective for all t, which happens if and only if the image of u_t in M/I_tM is nonzero for all t. Thus $x \in \mathcal{I}^R(A, M)$ if and only if $u_t x \cdot \mathbf{1}_M \in I_tM$ for some t, i.e., $x \in \bigcup_t (I_tM :_A u_t\mathbf{1}_M)$. □

Lemma 2.26. *Let R be a Noetherian ring with Jacobson radical J. Suppose that $\phi\colon M \to N$ is such that the induced map $M/J^nM \to N/J^nN$ is pure for every $n > 0$. If $f\colon R \to M$ is a pure map, then the composition $R \xrightarrow{f} M \xrightarrow{\phi} N$ is pure.*

In particular, the same conclusion holds if $A = M \to N = B$ is a J-completely faithfully flat map of R-algebras (e.g., a g-completely faithfully flat map of R-algebras for some $g \in J$).

Proof. Let $x = f(1) \in M$. By Lemma 2.24 it suffices to assume that R is local, with $E \coloneqq E(R/\mathfrak{m})$, and show that $E \to E \otimes_R N$, $u \mapsto u \otimes \phi(x)$ is injective. By taking direct limit, it suffices to show that $K \to K \otimes_R N$ is injective for every finitely generated submodule $K \subseteq E$. But then $J^n K = 0$ for some $n \in \mathbb{N}$, so $K \otimes_R N \cong K \otimes_{R/J^n R} N/J^n N$ so the map $K \to K \otimes_R N$ is injective as a composition of injective maps. □

Lemma 2.27. *Let S be a ring and $\phi\colon (A, M) \to (B, N)$ be a map between S-algebras and pointed modules over these algebras. Then*

(a) If $f\colon R \to S$ is a homomorphism of rings which is pure as a map of R-modules (e.g., $\mathcal{I}^R(A, M) \neq A$), then $\mathcal{I}^R(A, M) \subseteq \mathcal{I}^S(A, M)$.

(b) If $f\colon (R, \mathfrak{m}) \to (S, \mathfrak{n})$ is a flat map of Noetherian local rings such that $\mathfrak{m}S = \mathfrak{n}$, then $\mathcal{I}^R(A, M) = \mathcal{I}^S(A, M)$.

(c) $\mathcal{I}^S(A, M) \subseteq \mathcal{I}^S(B, N) \cap A$.

(d) If $\{M_i\}_{i\in\mathbb{N}}$ is a directed system of pointed A-modules, then $\mathcal{I}^S(A, M_i)$ form an ascending sequence of sets and $\mathcal{I}^S(A, \varinjlim M_i) = \cup_i \mathcal{I}^S(A, M_i)$.

(e) If $M \xrightarrow{\phi} N$ is pure as S-modules, then $\mathcal{I}^S(A, M) = A \cap \mathcal{I}^S(B, N)$. The same conclusion holds if S is Noetherian and J is its Jacobson radical such that the induced map $M/J^n M \to N/J^n N$ is pure for all $n \in \mathbb{N}$.

(f) If $(S, \mathfrak{m})$ is Noetherian local, $\phi\colon A \to B$ is $\mathfrak{m}$-completely faithfully flat and $N = M \otimes_A B$, then $\mathcal{I}^S(A, M)B = \mathcal{I}^S(B, N)$.

(g) If S is Noetherian and $\mathbf{S}_2$, then $\mathcal{I}^S(A, M) = \mathcal{I}^S(A, M^{\mathrm{c1}})$.

Proof. First, if $\mathcal{I}^R(A, M) \neq A$ then there is $x \in A$ such that the composition $R \xrightarrow{f} S \xrightarrow{1\mapsto x\mathbf{1}_M} M$ is pure, so f must be pure. Now if f is pure and $x \notin \mathcal{I}^S(A, M)$, then $R \xrightarrow{f} S \xrightarrow{1\mapsto x\mathbf{1}_M} M$ must be pure as a composition of pure maps, so $x \notin \mathcal{I}^R(A, M)$.

Lemma 2.24 implies (b) after noting that $E(S/\mathfrak{n}) = E(R/\mathfrak{m}) \otimes_R S$ by the assumptions.

Next, take any $x \in A$. If $\phi(x) \notin \mathcal{I}^S(B, N)$ then, by definition, $S \xrightarrow{1\mapsto\phi(x)\mathbf{1}_N} N$ is a pure map that factors through $S \xrightarrow{1\mapsto x\mathbf{1}_M} M$. Hence, $x \notin \mathcal{I}^S(A, M)$, showing (c). Now, (d) is immediate from (c).

We now prove (e). Suppose $x \in A \setminus \mathcal{I}^S(A, M)$. By (c), it suffices to show that $\phi(x) \notin \mathcal{I}^S(B, N)$, i.e., $S \xrightarrow{\phi(x)\mathbf{1}_N} N$ is pure. This is immediate if $M \to N$ is pure and follows from Lemma 2.26 in the more general case.

For (f), we note that, since $B/\mathfrak{m}^n B$ is faithfully flat over $A/\mathfrak{m}^n A$ for every n, tensoring with B preserves injections of $\mathfrak{m}$-power-torsion A-modules. Applying this to the injection $A/\mathcal{I}^S(A, M) \cong A(u \otimes \mathbf{1}_M) \hookrightarrow E \otimes_S M$ in Lemma 2.24 shows that the annihilator of $u \otimes \mathbf{1}_N$ is $\mathcal{I}^S(A, M)B$.

The last claim follows from Lemma 2.21. □

2.4. **Compatible ideals.**

Definition 2.28 (*cf.* [Fay25])**.** Let R be a Noetherian ring with p in its Jacobson radical and let $I \subset R$ be an ideal. For a perfectoid R-algebra B and $\phi \in \mathrm{Hom}_R(B,R)$, we say that I is ϕ-compatible if $\phi((IB)_{\mathrm{perfd}}) \subset I$. Similarly, if $\mathscr{C} \subset \mathrm{Hom}_R(B,R)$ is an R-submodule then we say that I is $\mathscr{C}$-compatible if $\phi((IB)_{\mathrm{perfd}}) \subset I$ for all $\phi \in \mathscr{C}$. If I is ϕ-compatible for all $\phi \in \mathrm{Hom}_R(B,R)$ and all perfectoid R-algebras B, we say that it is uniformly compatible.

Remark 2.29. Unless R is complete local, the existence of a nonzero map $\phi \in \mathrm{Hom}_R(B,R)$ is not guaranteed, making the definition useful mostly in the complete local case (this is why the theory was only written for complete local rings in [Fay25]). Indeed, if $(R,\mathfrak{m},k)$ is complete local and $R \to B$ is pure then

$$\mathrm{Hom}_R(B,R) \cong \mathrm{Hom}_R(B,\mathrm{Hom}_R(E,E)) \cong \mathrm{Hom}_R(B \otimes_R E, E) \twoheadrightarrow \mathrm{Hom}_R(E,E) = R$$

where $E = E_R(k)$ and the surjection comes from the injection $E \to E \otimes_R B$ and the fact that E is an injective module. In our applications below, if R is not complete local, we assume a map exists.

Proposition 2.30. *Let R be a Noetherian ring with p in its Jacobson radical, B a perfectoid R-algebra and let $I \subset R$ be an ideal. Fix an R-submodule $\mathscr{C}$ of $\mathrm{Hom}_R(B,R)$.*

(a) I is $\mathscr{C}$-compatible if and only if $\mathscr{C}$ descends to a corresponding R/I-submodule $\overline{\mathscr{C}}$ of $\mathrm{Hom}_{R/I}(B/(IB)_{\mathrm{perfd}}, R/I)$. If some $\phi \in \mathscr{C}$ is surjective then so is the induced map on the quotient $\overline{\phi} \in \overline{\mathscr{C}}$.

(b) If I is ϕ-compatible for some surjective ϕ, then it is radical.

(c) If I is $\mathscr{C}$-compatible and $J \subset R$ is any ideal such that B admits a compatible system of p-th power roots for generators of $(I\colon J)$, then $(I\colon J)$ is also $\mathscr{C}$-compatible.

(d) If R is complete local and perfectoid pure then any uniformly compatible ideal is radical.

(e) If R is complete local then uniform compatibility can be checked on a single choice of $R^{A_\infty}_{\mathrm{perfd}}$ for $A \to R$ as in Notation 2.7.

(f) If R is normal, complete local and I is a radical $\mathscr{C}$-compatible ideal, then so is I^{**}.

Proof. Note that I is ϕ-compatible for $\phi \in \mathscr{C}$ if and only if the composition map

$$B \xrightarrow{\phi} R \longrightarrow R/I$$

factors through $B/(IB)_{\mathrm{perfd}}$ so (a) follows. In (b), (a) implies that ϕ descends to a surjective map $\overline{\phi} \in \mathrm{Hom}_R(B/(IB)_{\mathrm{perfd}}, R/I)$ so it follows that R/I injects into the perfectoid hence reduced ring $B/(IB)_{\mathrm{perfd}}$ and therefore R/I is reduced. For (c), by linearity and Krull's intersection theorem in the form [Mat89, Theorem 8.10], it suffices to show that $\phi(x^{1/p^e}B)j \in I$ for any $\phi \in \mathscr{C}$, $j \in J$ and generator x of $(I\colon J)$ having a p^e-th root in B. Then,

$$\phi(x^{1/p^e}B)j = \phi((xj^{p^e})^{1/p^e}B) \subset \phi(\sqrt{IB}) \subset I$$

where the last containment follows from the ϕ-compatibility of I and Corollary 2.4. Item (d) is [Fay25, Corollary 2.33] and item (e) is [Fay25, Corollary 2.29]. For item (f), since R is normal, $I^{**} = \bigcap_{\substack{\mathfrak{p}\in\mathrm{Min}(I)\\ \mathrm{ht}\,\mathfrak{p}=1}} \mathfrak{p}$. By (a), each $\phi \in \mathscr{C}$ induces an R/I-linear map $B/(IB)_{\mathrm{perfd}} \to R/I$, and from [Fay25, Corollary 2.36 and Proposition 2.31] every minimal prime of R/I is compatible with these induced maps, so the arguments of [Fay25, Propositions 2.34, 2.37] give the desired result. □

3. Perfectoid pure principal pairs

In this section we develop the theory of perfectoid pure pairs of the form (R, f^λ) where $f \in R$ and $\lambda \in \mathbb{Q}_{>0}$. Later, in Section 4 we will generalize this to pairs (R, Δ) where Δ is a $\mathbb{Q}$-divisor. We would recommend the reader to look at this case first before exploring the more technical divisor-pair setting.

We will give our definition of a perfectoid pure pair (R, f^t) shortly, but first we prove some small results to show it is meaningful.

Proposition 3.1 (cf. [BMP+24, Lemma 4.5])**.** *Let R be a Noetherian ring with p in its Jacobson radical, let $f \in R$, and let $a, b \in \mathbb{Z}_{>0}$. Suppose that B is a perfectoid R-algebra and $Y \in (fB)_{\mathrm{perfd}}$ satisfies $Y^b = f^a$. Let $B \to \widetilde{B}$ be a p-completely faithfully flat map with $\widetilde{B}$ absolutely integrally closed and perfectoid, as in Theorem 2.3. If $\widetilde{Y}$ denotes the image of Y, then $R \xrightarrow{1\mapsto Y} (fB)_{\mathrm{perfd}}$ is pure if and only if $R \xrightarrow{1\mapsto \widetilde{Y}} (f\widetilde{B})_{\mathrm{perfd}}$ is pure.*

Proof. The reverse implication is immediate because the second pointed map factors through the first. For the forward implication, write $\widehat{\otimes}^{\mathbb{L}}$ for the derived p-completion of the derived tensor product Since $B \to \widetilde{B}$ is p-completely faithfully flat, we have isomorphisms of $\widetilde{B}$-algebras

$$\big(B/(fB)_{\mathrm{perfd}}\big) \mathbin{\widehat{\otimes}}^{\mathbb{L}}_B \widetilde{B} \simeq \big((B/fB) \mathbin{\widehat{\otimes}}^{\mathbb{L}}_B \widetilde{B}\big)_{\mathrm{perfd}} \simeq (\widetilde{B}/f\widetilde{B})_{\mathrm{perfd}} = \widetilde{B}/(f\widetilde{B})_{\mathrm{perfd}}$$

using [BS22, Propositions 8.5 and 8.13] (as in the displayed calculation in the proof of [CLM+26, Lemma 2.19]). Taking their fibers and using the exactness of $\widehat{\otimes}^{\mathbb{L}}\widetilde{B}$ gives $(fB)_{\mathrm{perfd}} \mathbin{\widehat{\otimes}}^{\mathbb{L}}_B \widetilde{B} \simeq (f\widetilde{B})_{\mathrm{perfd}}$. Applying $- \otimes^{\mathbb{L}}_{\widetilde{B}} \widetilde{B}/p^n\widetilde{B}$ to this isomorphism removes the completion on the tensor product, since derived p-completion does not change the derived tensor product with a module killed by p^n. Thus

$$(f\widetilde{B})_{\mathrm{perfd}} \otimes^{\mathbb{L}}_{\widetilde{B}} \widetilde{B}/p^n\widetilde{B} \simeq \big((fB)_{\mathrm{perfd}} \otimes^{\mathbb{L}}_B B/p^nB\big) \otimes^{\mathbb{L}}_{B/p^nB} \widetilde{B}/p^n\widetilde{B}.$$

Since $\widetilde{B}/p^n\widetilde{B}$ is faithfully flat over B/p^nB, taking zeroth cohomology gives

$$(f\widetilde{B})_{\mathrm{perfd}}/p^n(f\widetilde{B})_{\mathrm{perfd}} \cong \big((fB)_{\mathrm{perfd}}/p^n(fB)_{\mathrm{perfd}}\big) \otimes_{B/p^nB} \widetilde{B}/p^n\widetilde{B}.$$

It follows that $(fB)_{\mathrm{perfd}}/p^n(fB)_{\mathrm{perfd}} \to (f\widetilde{B})_{\mathrm{perfd}}/p^n(f\widetilde{B})_{\mathrm{perfd}}$ is pure for every $n > 0$. The forward implication now follows from Lemma 2.26. □

Lemma 3.2. *Let R be a Noetherian ring with p in its Jacobson radical, let $f \in R$, and $a, b \in \mathbb{N}_{>0}$. Suppose that B is a perfectoid R-algebra and that $Y \in (fB)_{\mathrm{perfd}}$ satisfies $Y^b = f^a$ and admits a compatible system $\{Y^{1/p^e}\}_{e\in\mathbb{N}_{>0}}$ of p-power roots in B. Then the map $R \xrightarrow{1\mapsto Y} (fB)_{\mathrm{perfd}}$ is pure if and only if $R \xrightarrow{1\mapsto Y^{1-1/p^e}} B$ is pure for every $e \in \mathbb{N}$, or equivalently for every $e \gg 0$.*

Moreover, if B is absolutely integrally closed, then $R \xrightarrow{1\mapsto Y} (fB)_{\mathrm{perfd}}$ is pure if and only if for all $c, d \in \mathbb{N}_{>0}$ with $\frac{c}{d} < \frac{a}{b}$ and any $Z \in B$ such that $Z^d = f^c$, the map $R \xrightarrow{1\mapsto Z} B$ is pure.

Proof. Since $Y^b = f^a$, Lemma 2.5 (b) gives $(Y^{1/p^\infty})B = \sqrt{YB} = \sqrt{fB}$. Combining Lemma 2.24 and Corollary 2.6 shows that $R \to (fB)_{\mathrm{perfd}} = (YB)_{\mathrm{perfd}}$, $1 \mapsto Y$ is pure if and only if $R \to$

$(Y^{1/p^\infty})B, 1 \mapsto Y$ is pure. Lemma 2.5 (b) shows that this latter map is the colimit of $R \xrightarrow{1\mapsto Y^{1-1/p^e}} B$. These maps are all pure if and only if their colimit is pure.

For the moreover part, the "if" direction follows from the previous paragraph (and we only need to use $\frac{c}{d} = \frac{a(p^e-1)}{bp^e}$ and a particular choice of Z, namely Y^{1-1/p^e}). For the "only if" direction, note that

$$Z^{ad(p^e-1)} = f^{ac(p^e-1)} = (Y^{1-1/p^e})^{bcp^e}.$$

Since $ad > bc$, we have $ad(p^e-1) > bcp^e$ for $e \gg 0$. Thus by Lemma 2.5 (a), we see that $Y^{1-1/p^e} \in ZB$. It follows from the already established purity of $R \xrightarrow{1\mapsto Y^{1-1/p^e}} B$ that $R \xrightarrow{1\mapsto Z} B$ is also pure. □

Proposition 3.3. *Let R be a Noetherian ring with p in its Jacobson radical, and let $f \in R$. Let $a, b \in \mathbb{Z}_{>0}$. Let B be a perfectoid R-algebra and suppose that there are $Y, Z \in B$ with $Y^b = Z^b = f^a$. If $R \xrightarrow{1\mapsto Y} (fB)_{\mathrm{perfd}}$ is pure, then so is $R \xrightarrow{1\mapsto Z} (fB)_{\mathrm{perfd}}$.*

Proof. By Proposition 3.1, we may assume that B is absolutely integrally closed. Choose compatible systems of p-power roots of Y and Z. By Lemma 3.2, $R \xrightarrow{1\mapsto Y^{1-1/p^e}} B$ is pure for every $e \gg 0$. For $e \gg 0$, the equality $(Y^{1-1/p^{e+1}})^{bp^{e+1}(p^e-1)} = (Z^{1-1/p^e})^{bp^e(p^{e+1}-1)}$ and the inequality $bp^{e+1}(p^e-1) < bp^e(p^{e+1}-1)$ show by Lemma 2.5 (a) that $Y^{1-1/p^{e+1}} \in Z^{1-1/p^e}B$. Thus, the pure map $R \xrightarrow{1\mapsto Y^{1-1/p^{e+1}}} B$ factors through $R \xrightarrow{1\mapsto Z^{1-1/p^e}} B$, so the latter is pure. Another application of Lemma 3.2 proves that $R \xrightarrow{1\mapsto Z} (fB)_{\mathrm{perfd}}$ is pure. □

Definition 3.4. Let R be a Noetherian ring with p in its Jacobson radical, and let $f \in R$. Let $\lambda = \frac{a}{b}$ with $a, b \in \mathbb{N}_{>0}$. We say the pair (R, f^λ) is *perfectoid pure* if f is a nonzerodivisor and there exists a perfectoid R-algebra B together with $Y \in (fB)_{\mathrm{perfd}}$ such that $Y^b = f^a$ and the map $R \xrightarrow{1\mapsto Y} (fB)_{\mathrm{perfd}}$ is pure.

Remark 3.5. First note if (R, f^λ) is perfectoid pure then R itself is perfectoid pure (see Lemma 3.2).

(a) By Proposition 3.1 and Proposition 3.3, we may take B to be absolutely integrally closed. Furthermore, the definition is independent of the chosen Y and the representation $\lambda = a/b$. To see the latter, pass to a common denominator.

(b) Purity is not affected if f is replaced by uf for a unit $u \in R$: after taking B absolutely integrally closed, multiplication by a b-th root of u^a carries a witness Y for f^λ to one for $(uf)^\lambda$. Thus we may also speak of the perfectoid purity of $(R, (f)^\lambda)$ or $(R, \lambda \operatorname{div}_R(f))$. More general divisor pairs will be considered below in Section 4.

(c) By Lemma 3.2, after choosing compatible p-power roots of Y in an absolutely integrally closed B as in (a), the condition is equivalent to the purity of $R \xrightarrow{1\mapsto Y^{1-1/p^e}} B$ for every $e \gg 0$. Equivalently, R is perfectoid pure along these elements in the sense of [Fay25, Definition 3.1].

Lemma 3.6. *Let R be a Noetherian ring with p in its Jacobson radical and let $f \in R$ be a nonzerodivisor. The following are equivalent:*

(a) (R, f^λ) is perfectoid pure;

(b) (R, f^μ) *is perfectoid pure for all positive rational* $\mu < \lambda$*;*

(c) for all positive rational $\mu < \lambda$*, there is a perfectoid* R*-algebra* B_μ *and* $Y \in B_\mu$ *with* $Y = f^\mu$ *so that* $R \xrightarrow{1\mapsto Y} B_\mu$ *is pure;*

(d) there is a single perfectoid R*-algebra* B*, a sequence* $0 < \lambda_i = a_i/b_i < \lambda$ *of rational numbers converging to* λ*, and elements* $Y_i \in B$ *with* $Y_i^{b_i} = f^{a_i}$ *such that* $R \xrightarrow{1\mapsto Y_i} B$ *is pure for every* i*.*

Proof. First $(a) \Leftrightarrow (c)$ follows from Lemma 3.2 (we can take B to be absolutely integrally closed by Remark 3.5 (a), and conversely we can take B_μ to be absolutely integrally closed and set $B = \prod B_\mu$ which is perfectoid). Next $(c) \Rightarrow (b)$ is obvious as the map factors through $(fB_\mu)_{\mathrm{perfd}}$ and $(b) \Rightarrow (c)$ follows from $(a) \Rightarrow (c)$ applied to any λ' so that $\mu < \lambda' < \lambda$. $(c) \Rightarrow (d)$ is obvious by noting that we can assume each B_i is absolutely integrally closed and set $B = \prod B_i$. Finally, (d) obviously implies (R, f^{λ_i}) is perfectoid pure (as the map factors through $(fB)_{\mathrm{perfd}}$) and thus implies (b) (by applying $(a) \Rightarrow (b)$ to any $\mu < \lambda_i < \lambda$). □

3.1. **Localization, completion and choice of B for principal pairs.** In this subsection we verify the basic properties of perfectoid purity with regards to localization and completion.

Proposition 3.7. *Let* R *be a Noetherian ring with* p *in its Jacobson radical, and let* $\mathfrak{q}$ *be a prime ideal containing* p*. Let* $f \in R$ *be a nonzerodivisor and* $\lambda \in \mathbb{Q}_{>0}$*. If* (R, f^λ) *is perfectoid pure then* $(R_\mathfrak{q}, f^\lambda)$ *is perfectoid pure.*

Proof. If B is a perfectoid R-algebra, set $B_\mathfrak{q} = B \otimes_R R_\mathfrak{q}$ and $C = B_\mathfrak{q}^{\wedge p}$. The ring C is perfectoid by [BIM19, Example 3.8(7)] and $B_\mathfrak{q} \to C$ is an isomorphism modulo p^n for every n. Consequently, $B_\mathfrak{q} \to C$ is p-completely pure.

Suppose now that (R, f^λ) is perfectoid pure. By Lemma 3.6, there exist some perfectoid R-algebra B, rational numbers $\lambda_e = a_e/b_e < \lambda$ converging to λ, and elements $Y_e \in B$ such that $Y_e^{b_e} = f^{a_e}$ and $R \to B$, $1 \mapsto Y_e$, is pure for every e. By Lemma 2.26, $R_\mathfrak{q} \to C$, $1 \mapsto Y_e$, is pure for every e. Thus $(R_\mathfrak{q}, f^\lambda)$ is perfectoid pure by the same criterion. □

Proposition 3.8 (*cf.* [BMP+24, Lemma 4.10])**.** *Let* R *be a Noetherian ring with* p *in its Jacobson radical, let* $f \in R$ *be a nonzerodivisor, and let* $\lambda \in \mathbb{Q}_{>0}$*. Then* (R, f^λ) *is perfectoid pure if and only if* $(R_\mathfrak{m}, f^\lambda)$ *is perfectoid pure for every maximal ideal* $\mathfrak{m}$ *of* R*.*

Proof. The forward direction follows from Proposition 3.7. For the converse, suppose first that $(R_\mathfrak{m}, f^\lambda)$ is perfectoid pure for every maximal ideal $\mathfrak{m}$, and write $\lambda = a/b$ for some $a, b \in \mathbb{N}_{>0}$. By Proposition 3.1, for each maximal ideal $\mathfrak{m}$ there is an absolutely integrally closed perfectoid $R_\mathfrak{m}$-algebra $B(\mathfrak{m})$ as well as $Y_\mathfrak{m} \in B(\mathfrak{m})$ with $Y_\mathfrak{m}^b = f^a$ such that

$$R_\mathfrak{m} \longrightarrow (fB(\mathfrak{m}))_{\mathrm{perfd}}, \qquad 1 \longmapsto Y_\mathfrak{m},$$

is pure. Choose compatible p-power roots of every $Y_\mathfrak{m}$. By Lemma 3.2, the maps

$$R_\mathfrak{m} \longrightarrow B(\mathfrak{m}), \qquad 1 \longmapsto Y_\mathfrak{m}^{1-1/p^e}, \tag{3.8.1}$$

are pure for every $e \geq 1$. Set $B = \prod_\mathfrak{m} B(\mathfrak{m})$ and $Z_e = (Y_\mathfrak{m}^{1-1/p^e})_\mathfrak{m} \in B$. The ring B is perfectoid by [BIM19, Example 3.8(8)], and

$$Z_e^{bp^e} = f^{a(p^e-1)}.$$

Localizing at any $\mathfrak{m}$, projection onto $B(\mathfrak{m})$ gives the pure map of (3.8.1) so this localized map is pure. Since purity can be checked locally on R, the map $R \to B$, $1 \mapsto Z_e$, is also pure. By, Lemma 3.6 (R, f^λ) is perfectoid pure. □

Proposition 3.9. *Let $(R, \mathfrak{m})$ be a Noetherian local ring of residue characteristic $p > 0$. Let $f \in R$ be a nonzerodivisor and let $\lambda \in \mathbb{Q}_{>0}$. Then, (R, f^λ) is perfectoid pure if and only if $(\widehat{R}, f^\lambda)$ is perfectoid pure for $\widehat{R} = R^{\wedge \mathfrak{m}}$.*

Proof. This follows from Remark 3.5 (c) and [Fay25, Proof of proposition 3.4] (*cf.* [BMP+24, Lemma 4.8]). □

Now we show, at least if R is complete, that we can choose a B determining perfectoid purity of a pair quite explicitly, similarly to [BMP+24, Lemma 4.24].

Proposition 3.10. *Let $(R, \mathfrak{m}, k)$ be a complete Noetherian local ring of residue characteristic $p > 0$, let $f \in \mathfrak{m}$ be a nonzerodivisor, and let $\lambda \in \mathbb{Q}_{>0}$. Fix any $\varphi\colon A \to R$ as in Notation 2.7 with $\varphi(x_1) = f$ (such a map may be obtained by adjoining to a Cohen presentation or Noether–Cohen normalization a variable mapping to f). Then (R, f^λ) is perfectoid pure if and only if*

$$R \longrightarrow R^{A_\infty}_{\mathrm{perfd}}, \qquad 1 \longmapsto (x_1^{1/p^e})^{\lfloor (p^e-1)\lambda \rfloor}, \tag{3.10.1}$$

is pure for every $e \gg 0$.

Proof. For the backwards direction, note that $\lambda_e := \lfloor (p^e - 1)\lambda \rfloor / p^e < \lambda$ and $\lambda_e \to \lambda$ so the purity of (3.10.1) implies that (R, f^λ) is perfectoid pure by Lemma 3.6. Conversely, suppose that (R, f^λ) is perfectoid pure. Then by Lemma 3.2 there is an absolutely integrally closed perfectoid R-algebra B with a fixed compatible system of p-th power roots of f such that $R \to B$, $1 \mapsto (f^{1/p^e})^{\lfloor (p^e-1)\lambda \rfloor}$ is pure for all $e \gg 0$. Following the proof of [BMP+24, Lemma 4.24], choose a map $W(k^{1/p^\infty}) \to B$ compatible with $C_k \to B$, together with compatible p-power roots in B of p and of the images of the variables of A. Replacing B by its $\mathfrak{m}B$-adic completion, the universal property of perfectoidization then implies that $R \to B$, $1 \mapsto (f^{1/p^e})^{\lfloor (p^e-1)\lambda \rfloor}$ factors through (3.10.1), which must therefore also be pure. □

Finally, we compare perfectoid purity of principal pairs to the characteristic $p > 0$ definition that inspired it, see [HW02].

Proposition 3.11. *Let R be an excellent reduced ring of characteristic $p > 0$, let $f \in R$ be a nonzerodivisor, and let $\lambda \in \mathbb{Q}_{>0}$. Then (R, f^λ) is perfectoid pure if and only if the map*

$$R \longrightarrow R^{1/p^e}, \qquad 1 \longmapsto f^{\lfloor \lambda(p^e-1) \rfloor / p^e}$$

is pure for all $e \gg 0$.

Proof. Using Proposition 3.8 and Proposition 3.9 above, we see both conditions are local and invariant under completion. We may therefore assume $(R, \mathfrak{m}, k)$ is complete local. If f is a unit, this is the usual equivalence between perfectoid purity and F-purity [BMP+24, Remark 4.3], so further assume $f \in \mathfrak{m}$. Then for A as in Proposition 3.10, $R^{A_\infty}_{\mathrm{perfd}} = R_{\mathrm{perf}}$, so Proposition 3.10 tells us that (R, f^λ) is perfectoid pure if and only if $R \to R_{\mathrm{perf}}$, $1 \longmapsto f^{\lfloor \lambda(p^e-1) \rfloor / p^e}$ is pure for all $e \gg 0$. If this is true then certainly $R \longrightarrow R^{1/p^e}$, $1 \mapsto f^{\lfloor \lambda(p^e-1) \rfloor / p^e}$ is pure for all e. Conversely, suppose that these maps to R^{1/p^e} are pure for all $e \gg 0$. Each factors through the

e-iterated Frobenius map $F^e\colon R \to R^{1/p^e}$, so F^e is pure and hence Frobenius F is pure. The transition maps $R^{1/p^j} \to R^{1/p^{j+1}}$ are Frobenius twists of F and hence are pure. It follows that $R^{1/p^e} \to R_{\mathrm{perf}} = \varinjlim_j R^{1/p^j}$ is pure. Thus $R \to R_{\mathrm{perf}}$, $1 \mapsto f^{\lfloor \lambda(p^e-1)\rfloor/p^e}$, is pure for all $e \gg 0$, as desired. □

3.2. **Adjunction and inversion of adjunction for principal pairs.**

Lemma 3.12. *Let $(R,\mathfrak{m},k)$ be a Noetherian local ring of residue characteristic p, let $f,g \in R$ with f a nonzerodivisor, and let $\lambda \in \mathbb{Q}_{>0}$. The following are equivalent:*

(a) (R, fg^λ) is perfectoid pure.

(b) There is a perfectoid R-algebra B admitting a compatible system of p-th power roots of f and elements g_n with $g_n^{b_n} = g^{a_n}$ for some sequences $a_n, b_n \in \mathbb{N}_{>0}$ with $\frac{a_n}{b_n} < \lambda$, and $\frac{a_n}{b_n} \to \lambda$ such that

$$fR \to (fB)_{\mathrm{perfd}}, \quad fr \mapsto frg_n \tag{3.12.1}$$

is pure for all n.

Proof. Assume (a) and let B be an absolutely integrally closed perfectoid R-algebra witnessing the perfectoid purity of (R, fg^λ). Let $e > 0$ be arbitrary and fix a sequence g_n in B as in (b). By Lemma 3.6, for any n, $R \to B$, $1 \mapsto f^{1-p^{-e}}g_n$ is pure. Since e was arbitrary, taking the colimit over all e while fixing n and using Lemma 2.5(b) and Corollary 2.6, we get that $R \to (fB)_{\mathrm{perfd}}$, $1 \mapsto fg_n$ is pure. Part (b) then follows from the isomorphism $R \cong fR$, $1 \mapsto f$.

Now, assume (b) holds. By Theorem 2.3 and the argument in the proof of Proposition 3.1, we may assume B is absolutely integrally closed. Using the isomorphism $R \cong fR$, $1 \mapsto f$, we know that $R \to (fB)_{\mathrm{perfd}}$, $1 \mapsto fg_n$ is pure. This factors through $R \to (fgB)_{\mathrm{perfd}}$, $1 \mapsto f^{(a_n/b_n)/\lambda}g_n$, so this latter map must be pure. Part (a) then follows from another application of Lemma 3.6. □

Proposition 3.13 (*cf.* [BMP+24, Lemma 6.5])**.** *Suppose $(R,\mathfrak{m},k)$ is a Noetherian local ring of residue characteristic p, f is a nonzerodivisor on R, and $g \in R$. Denote the image of g in R/fR by $\overline{g}$. If (R, fg^λ) is perfectoid pure, then so is $(R/fR, \overline{g}^\lambda)$.*

Proof. Let g_n and B be as in Lemma 3.12(b). Let M be an R/fR-module. Let

$$\gamma_M\colon M \otimes_{R/fR} B/(fB)_{\mathrm{perfd}} \to M \otimes_R (fB)_{\mathrm{perfd}}, \quad m \otimes \overline{b} \mapsto m \otimes fb.$$

This is well defined: if $c \in (fB)_{\mathrm{perfd}}$ then $m \otimes fc = mf \otimes c = 0$. Since multiplication by f induces an isomorphism $M \otimes_R fR \cong M$, the composition

$$M \to M \otimes_{R/fR} B/(fB)_{\mathrm{perfd}} \xrightarrow{\gamma_M} M \otimes_R (fB)_{\mathrm{perfd}}, \quad m \mapsto m \otimes \overline{g_n} \mapsto m \otimes fg_n$$

is a base change of (3.12.1) so it is injective and therefore so is

$$M \to M \otimes_{R/fR} B/(fB)_{\mathrm{perfd}}, \quad m \mapsto m \otimes \overline{g_n}.$$

To conclude that $(R/fR, \overline{g}^\lambda)$ is perfectoid pure, we need to show that $\overline{g}$ is a nonzerodivisor on R/fR. Suppose not and let $r \in R \setminus fR$ be such that $rg \in fR$. Then, $(rg_n)^{b_n} = r^{b_n}g^{a_n} = 0$ in the reduced ring $B/(fB)_{\mathrm{perfd}}$. Therefore we must have $rg_n \in (fB)_{\mathrm{perfd}}$, contradicting the injectivity of

$$R/fR \to B/(fB)_{\mathrm{perfd}}. \quad r \mapsto rg_n.$$

□

Theorem 3.14 (*cf.* [BMP+24, Theorem 6.6])**.** *Suppose $(R, \mathfrak{m}, k)$ is a complete intersection of residue characteristic p and let $f \in \mathfrak{m}$ be a nonzerodivisor and $g \in R$. Then, (R, fg^λ) is perfectoid pure if and only if $(R/f, \overline{g}^\lambda)$ is perfectoid pure.*

Proof. The forward direction is Proposition 3.13 so we only need to show the backwards one. By Proposition 3.9, it suffices to show the theorem in the case $(R, \mathfrak{m})$ is complete. By Remark 3.5(b), if g is not in $\mathfrak{m}$ then we can assume $g = 1$, in which case this is [BMP+24, Theorem 6.6] so assume $g \in \mathfrak{m}$. We follow the proof of [BMP+24, Theorem 6.6].

Let $A \to R$ be as in Notation 2.7 and by potentially adding variables to A that get sent to f and g, we can assume $R^{A_\infty}_{\text{perfd}}$ has a fixed choice of compatible systems of p-th power roots for both f and g. Let $g_e := (g^{1/p^e})^{\lfloor (p^e-1)\lambda \rfloor}$. Let $d = \dim R$. By [BMP+24, Theorem 4.25], the following commutative diagram has exact rows.

$$\begin{array}{ccccccccc} 0 & \longrightarrow & H^{d-1}_{\mathfrak{m}}(R/fR) & \longrightarrow & H^d_{\mathfrak{m}}(fR) & \longrightarrow & H^d_{\mathfrak{m}}(R) & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \cdot\overline{g_e}} & & \downarrow{\scriptstyle \cdot g_e} & & \downarrow{\scriptstyle \cdot g_e} & & \\ 0 & \longrightarrow & H^{d-1}_{\mathfrak{m}}((R/fR)^{A_\infty}_{\text{perfd}}) & \longrightarrow & H^d_{\mathfrak{m}}((fR^{A_\infty}_{\text{perfd}})_{\text{perfd}}) & \longrightarrow & H^d_{\mathfrak{m}}(R^{A_\infty}_{\text{perfd}}) & \longrightarrow & 0. \end{array}$$

By Proposition 3.10,

$$R/fR \to (R/fR)^{A_\infty}_{\text{perfd}}, \quad 1 \mapsto \overline{g_e}$$

is pure for all $e \gg 0$ so the left vertical map in the above diagram is injective. Since R is Gorenstein and f is a nonzerodivisor, $H^d_{\mathfrak{m}}(fR) \cong E_R(R/\mathfrak{m})$. In fact, its socle is in the image of $H^{d-1}_{\mathfrak{m}}(R/fR)$ since multiplication by f kills it. Chasing this socle shows that the middle vertical map is injective so

$$fR \otimes_R H^d_{\mathfrak{m}}(R) \to (fR^{A_\infty}_{\text{perfd}})_{\text{perfd}} \otimes_R H^d_{\mathfrak{m}}(R), \quad f \otimes u \mapsto fg_e \otimes u$$

is injective. The result then follows from combining $H^d_{\mathfrak{m}}(R) \cong E_R(R/\mathfrak{m})$, Lemma 2.24 and Lemma 3.12. □

4. Perfectoid pure divisor pairs

We now extend the treatment of principal pairs to divisor pairs (R, Δ), where R is an excellent $\mathbf{S}_2$ (hence locally equidimensional [Sta26, Tag 0FIW]) ring and Δ is an effective Mumford $\mathbb{Q}$-divisor on $X = \operatorname{Spec} R$. We will repeatedly use finite flat R-algebras, which must be excellent and $\mathbf{S}_2$ and form a class closed under tensor products over R.

4.1. Definition and first properties.

Definition 4.1. Let R be an excellent $\mathbf{S}_2$ ring, Δ an effective Mumford $\mathbb{Q}$-divisor on R, and let B be an R-algebra.

(a) A *divisorial lift of* Δ, or simply a Δ-*lift*, is a pair (S, Δ_S), where S is a finite flat R-algebra and Δ_S is an effective Weil generalized divisor on S such that, for some integer $n > 0$ with $n\Delta$ integral, the flat pullback of $n\Delta$ to S is $n\Delta_S$. We often refer to such an n as a *witness* of the lift.

(b) A B-*realization* of the Δ-lift (S, Δ_S) is an R-algebra map $\iota\colon S \to B$, i.e., such that $R \to B$ factors as $R \to S \xrightarrow{\iota} B$.

(c) A Δ-*datum on* B is a triple $\mathfrak{d} = (S, \Delta_S, \iota)$ consisting of a Δ-lift together with a B-realization.

(d) Associated to $\mathfrak{d}$ is the pointed B-module

$$M_B(\mathfrak{d}) \coloneqq B \otimes_{S,\iota} S(\Delta_S), \qquad \mathbf{1}_{\mathfrak{d}} \coloneqq 1 \otimes 1,$$

and the distinguished map

$$\eta_{\mathfrak{d}} \colon R \longrightarrow M_B(\mathfrak{d}), \qquad 1 \longmapsto \mathbf{1}_{\mathfrak{d}}.$$

Here $\mathbf{1}_{\mathfrak{d}}$ is well-defined because the effectivity of Δ_S gives an inclusion $S \subseteq S(\Delta_S)$.

Remark 4.2. Because our rings are typically not normal, the divisors that appear in divisorial lifts need not be unique. Suppose that (S, Δ_S) and (S, Δ'_S) are two Δ-lifts on the same algebra S, witnessed by positive integers n and n', respectively. If N is any common multiple of n and n', then both $N\Delta_S$ and $N\Delta'_S$ are the flat pullback of $N\Delta$ to S. Thus, while Δ_S and Δ'_S need not be equal, their difference is torsion in the group of Weil generalized divisors on S (if S is normal, there is no such torsion, and so Δ_S and Δ'_S agree). Likewise, distinct R-algebra maps $\iota, \iota' \colon S \to B$ give distinct Δ-data

$$\mathfrak{d} = (S, \Delta_S, \iota) \qquad \text{and} \qquad \mathfrak{d}' = (S, \Delta_S, \iota')$$

on B. The pointed modules $M_B(\mathfrak{d})$ and $M_B(\mathfrak{d}')$, as well as the purity of their distinguished maps, may therefore depend on the chosen B-realization.

Example 4.3. Let R be an excellent reduced $\mathbf{S}_2$ ring with p in its Jacobson radical, let $f \in R$ be a nonzerodivisor that does not vanish along any codimension-one component of the singular locus, and let $a, b \in \mathbb{Z}_{>0}$. Set $\Delta = \frac{a}{b} \operatorname{div}_R(f)$ and $S = R[x]/(x^b - f^a)$. The algebra S is finite free over R, hence flat . Moreover, x is a nonzerodivisor since $x^b = f^a$ and f is a nonzerodivisor. Consequently, $b \operatorname{div}_S(x) = a \operatorname{div}_S(f)$ is the flat pullback of $b\Delta = a \operatorname{div}_R(f)$ to S. Thus $(S, \operatorname{div}_S(x))$ is a Δ-lift.

Any R-algebra B containing an element Y with $Y^b = f^a$ realizes this lift via $\iota_Y \colon S \to B$, $x \mapsto Y$ and $\mathfrak{d}_Y = (S, \operatorname{div}_S(x), \iota_Y)$ is a Δ-datum on B. Multiplication by x induces an isomorphism

$$M_B(\mathfrak{d}_Y) = B \otimes_{S,\iota_Y} S(\operatorname{div}_S(x)) \xrightarrow{\sim} B$$

under which $\mathbf{1}_{\mathfrak{d}_Y}$ corresponds to Y. Hence $\eta_{\mathfrak{d}_Y}$ corresponds to the map $R \xrightarrow{1 \mapsto Y} B$. Different choices of Y give different B-realizations of the same Δ-lift, and hence different Δ-data on B.

Example 4.4. In Construction 2.18, $(S_{f,m}, \Delta_{f,m})$ is a Δ-lift. For any R-algebra B admitting an element z with $z^m = f$ (for instance any absolutely integrally closed B), the map $\iota_z \colon S_{f,m} \to B$ given by $x \mapsto z$ is a B-realization and $(S_{f,m}, \Delta_{f,m}, \iota_z)$ is a Δ-datum on B. Different choices of z give different B-realizations of the same Δ-lift, and hence different Δ-data on B.

The purity of the maps $\eta_{\mathfrak{d}} : R \to M_B(\mathfrak{d})$ will appear in our definition of perfectoid purity. Before giving that definition, we record some compatibilities of this purity.

Lemma 4.5. *Let R be an excellent reduced $\mathbf{S}_2$ ring with p in its Jacobson radical J, let Δ be an effective Mumford $\mathbb{Q}$-divisor on R, let B be a perfectoid R-algebra, and let $\mathfrak{d} = (S, \Delta_S, \iota)$ be a Δ-datum on B.*

(a) If $f \colon B \to C$ is a map of perfectoid R-algebras, then

$$\mathfrak{d}_C \coloneqq (S, \Delta_S, S \xrightarrow{\iota} B \xrightarrow{f} C).$$

is a Δ-datum on C and there is a natural pointed isomorphism $M_C(\mathfrak{d}_C) \cong C \otimes_B M_B(\mathfrak{d})$. Moreover, if f is J-completely pure, then so is the natural map $M_B(\mathfrak{d}) \to M_C(\mathfrak{d}_C)$. In particular, $\eta_{\mathfrak{d}}$ is pure if and only if $\eta_{\mathfrak{d}_C}$ is pure.

(b) B admits a p-completely faithfully flat map to an absolutely integrally closed perfectoid algebra C for which $\eta_{\mathfrak{d}}$ is pure if and only if $\eta_{\mathfrak{d}_C}$ is pure.

(c) Suppose that ι factors as

$$S \longrightarrow T \xrightarrow{\iota_T} B,$$

where $S \to T$ is finite flat. If Δ_T is the flat pullback of Δ_S to T, then $\mathfrak{d}_T = (T, \Delta_T, \iota_T)$ is a Δ-datum on B, and there is a natural pointed isomorphism

$$M_B(\mathfrak{d}) \xrightarrow{\sim} M_B(\mathfrak{d}_T).$$

In particular, $\eta_{\mathfrak{d}}$ is pure if and only if $\eta_{\mathfrak{d}_T}$ is pure.

(d) Suppose that $R \to B$ factors through a flat excellent reduced $\mathbf{S}_2$ R-algebra R' and that there is an effective Mumford divisor Δ' on R' which is the flat pullback of Δ. Write $S' \coloneqq S \otimes_R R'$, and let $\Delta_{S'}$ be the flat pullback of Δ_S. Then, the natural map $S' \to B$ makes $\mathfrak{d}' \coloneqq (S', \Delta_{S'}, S' \to B)$ a Δ'-datum on B and there is a natural pointed isomorphism $M_B(\mathfrak{d}') \cong M_B(\mathfrak{d})$.

Proof. The first part of (a) is immediate from definition. The second part follows by noting that $M_B(\mathfrak{d}) \to M_C(\mathfrak{d}_C)$ is a base change of f and using Lemma 2.26. Part (b) is immediate from Theorem 2.3 and (a). Part (c) follows from Proposition 2.15 and associativity of tensor products. Part (d) follows by putting every divisor to a sufficiently high multiple of itself and Proposition 2.15. □

Recall from (2.16.1) that for effective Mumford divisors Δ and Δ', the notation $\Delta \prec \Delta'$ means $\Delta \leq (1-\varepsilon)\Delta'$ for some $0 < \varepsilon < 1$.

Lemma 4.6. *Let R be an excellent $\mathbf{S}_2$ ring with p in its Jacobson radical and let $\Delta' \prec \Delta$ be effective Mumford $\mathbb{Q}$-divisors on R. Let B be a perfectoid R-algebra admitting Δ and Δ'-data $\mathfrak{d} = (S, D, \iota)$ and $\mathfrak{d}' = (S', D', \iota')$, respectively. Let $J_{\mathfrak{d}'} \subseteq B$ be the radical of the image of $B \otimes_{S'} S'(-D') \to B$. Let $U \subseteq \operatorname{Spec} R$ be a big open such that both D and D' are Cartier on the respective inverse images of U. Then, there is a unique pointed $\widetilde{B}\big|_U$-linear map of quasicoherent sheaves*

$$\widetilde{M_B(\mathfrak{d}')}\Big|_U \longrightarrow \widetilde{M_B(\mathfrak{d})}\Big|_U$$

whose image is contained in $\widetilde{J_{\mathfrak{d}'} \cdot M_B(\mathfrak{d})}\,\Big|_U$.

Proof. The images in $\operatorname{Spec} R$ of the Cartier loci of D and D' are big opens since the two induced maps $\operatorname{Spec} S, \operatorname{Spec} S' \to \operatorname{Spec} R$ are finite flat and preserve heights. Thus open subsets U as in the statement exist. Set $T = S \otimes_R S'$, with the realization induced by ι and ι'. By Lemma 4.5(c), it suffices to show the lemma after pulling both data back to T. We continue to write D and D' for the pulled-back divisors. Let $\rho\colon \operatorname{Spec} T \to \operatorname{Spec} R$ be the natural morphism and let $V \coloneqq \rho^{-1}(U)$. By hypothesis, D and D' are Cartier on V.

Let $0 < \varepsilon < 1$ be such that $\Delta' \leq (1-\varepsilon)\Delta$ and let n be a witness to both divisorial lifts. Fix $a < b \in \mathbb{N}_{>0}$ such that $1 - \varepsilon < a/b < 1$. Let $H = n(aD - bD')$, which is the flat pullback of the integral Mumford divisor $n(a\Delta - b\Delta')$ and is effective because $b\Delta' \leq b(1-\varepsilon)\Delta \leq a\Delta$.

Since V is a big open, it admits a principal affine cover $\{V_\alpha = \operatorname{Spec} T_\alpha = \operatorname{Spec} T[1/h_\alpha]\}_\alpha$, where each h_α is a nonzerodivisor (since V contains the associated primes), such that

$$T_\alpha(-D') = x_\alpha T_\alpha \qquad \text{and} \qquad T_\alpha(-D) = y_\alpha T_\alpha.$$

Multiplying by powers of h_α, we may take $x_\alpha, y_\alpha \in T$. Write $B_\alpha \coloneqq B \otimes_T T_\alpha$, $M_B(\mathfrak{d})_\alpha \coloneqq M_B(\mathfrak{d}) \otimes_T T_\alpha$ and $M_B(\mathfrak{d}')_\alpha \coloneqq M_B(\mathfrak{d}') \otimes_T T_\alpha$. Since $T_\alpha(-H) = y_\alpha^{an} x_\alpha^{-bn} T_\alpha$ and H is effective, $y_\alpha^{an} B_\alpha \subseteq x_\alpha^{bn} B_\alpha$. Applying Lemma 2.5 gives a unique $c_\alpha \in \sqrt{x_\alpha B_\alpha}$ such that $y_\alpha = c_\alpha x_\alpha$. On the other hand, $T_\alpha(D') = x_\alpha^{-1} T_\alpha$ and $T_\alpha(D) = y_\alpha^{-1} T_\alpha$, so $M_B(\mathfrak{d}')_\alpha$ and $M_B(\mathfrak{d})_\alpha$ are freely generated over B_α by x_α^{-1} and y_α^{-1}, respectively. Thus $x_\alpha^{-1} \longmapsto c_\alpha y_\alpha^{-1}$ defines a pointed B_α-linear map between them.

To glue these maps, set $V_{\alpha\beta} = V_\alpha \cap V_\beta = \operatorname{Spec} T_{\alpha\beta}$ and $B_{\alpha\beta} = B \otimes_T T_{\alpha\beta}$. By Lemma 2.5, on $V_{\alpha\beta}$, the element c_α is the unique element of $\sqrt{x_\alpha B_{\alpha\beta}}$ satisfying $y_\alpha = c_\alpha x_\alpha$. If $x_\beta = u_{\alpha\beta} x_\alpha$ and $y_\beta = v_{\alpha\beta} y_\alpha$ for units $u_{\alpha\beta}, v_{\alpha\beta} \in T_{\alpha\beta}$, uniqueness gives $c_\alpha = v_{\alpha\beta}^{-1} u_{\alpha\beta} c_\beta$. Hence by restricting the map on V_α to $V_{\alpha\beta}$, we see $x_\alpha^{-1} \mapsto c_\alpha y_\alpha^{-1} = v_{\alpha\beta}^{-1} u_{\alpha\beta} c_\beta y_\alpha^{-1} = c_\beta u_{\alpha\beta} y_\beta^{-1}$ and likewise restricting the map on V_β, we see $x_\alpha^{-1} = u_{\alpha\beta} x_\beta^{-1}$ is also sent to $c_\beta u_{\alpha\beta} y_\beta^{-1}$. This implies that the local maps glue to a pointed morphism

$$\widetilde{M_B(\mathfrak{d}')}\big|_V \longrightarrow \widetilde{M_B(\mathfrak{d})}\big|_V$$

of T-modules. Pushing this map forward along the affine morphism $\rho|_V \colon V \to U$ gives the desired map of sheaves on U. Since Proposition 2.15, $T(-D') \cong T \otimes_{S'} S'(-D')$, $J_{\mathfrak{d}'} B_\alpha = \sqrt{x_\alpha B_\alpha}$. Moreover, since the localized modules are free, any local pointed map whose image is contained in $J_{\mathfrak{d}'} M_B(\mathfrak{d})_\alpha$ is determined by an element $c \in \sqrt{x_\alpha B_\alpha}$ such that $y_\alpha = c x_\alpha$. Uniqueness in Lemma 2.5 shows that it is the map constructed above. □

We now prove the divisor analog of the first part of Lemma 3.6. Also compare with Theorem 4.11 and Corollary 4.12.

Proposition 4.7. *Let R be an excellent $\mathbf{S}_2$ ring with p in its Jacobson radical and let $\Delta' \prec \Delta$ be effective Mumford $\mathbb{Q}$-divisors on R. Let B be a perfectoid R-algebra admitting Δ and Δ'-data, $\mathfrak{d} = (S, D, \iota)$ and $\mathfrak{d}' = (S', D', \iota')$, respectively. If $\eta_{\mathfrak{d}}$ is pure, then so is $\eta_{\mathfrak{d}'}$.*

Proof. Using Lemma 4.6 and its notation, we have a pointed map

$$\Gamma\big(U, \widetilde{M_B(\mathfrak{d}')}\big) \longrightarrow \Gamma\big(U, \widetilde{M_B(\mathfrak{d})}\big).$$

Since $\eta_{\mathfrak{d}}$ is pure, so is $\Gamma(U, \eta_{\mathfrak{d}})$ by Lemma 2.20. By the above, this factors through $\Gamma(U, \eta_{\mathfrak{d}'})$, which must therefore also be pure and the result follows from another application of Lemma 2.20. □

We are finally ready to define perfectoid purity for pairs (R, Δ).

Definition 4.8. Let R be an excellent reduced $\mathbf{S}_2$ ring with p in its Jacobson radical, and let Δ be an effective Mumford $\mathbb{Q}$-divisor on R. We say that (R, Δ) is *perfectoid pure* if, for every rational number $0 < \varepsilon < 1$, there exist a perfectoid R-algebra B_ε and a $(1-\varepsilon)\Delta$-datum $\mathfrak{d}_\varepsilon$ on B_ε such that $\eta_{\mathfrak{d}_\varepsilon} \colon R \to M_{B_\varepsilon}(\mathfrak{d}_\varepsilon)$ is pure.

For $\Delta = 0$, this definition recovers the notion of a perfectoid pure ring from [BMP+24, Definition 4.1(a)]. Indeed, a pure map $R \to B$ to a perfectoid R-algebra gives a trivial 0-datum on B that works for every ε, while the converse follows from Corollary 4.10.

Remark 4.9 ($\mathbb{R}$-divisors). Suppose R is an excellent reduced $\mathbf{S}_2$ ring with p in its Jacobson radical. Further suppose that Δ is a Mumford $\mathbb{R}$-divisor. We could then define (R, Δ) to be perfectoid pure if for every sequence of effective Mumford $\mathbb{Q}$-divisors $\{\Delta_n\}_{n\geq 1}$ with $\Delta_n \leq \Delta$ and $\Delta_n \to \Delta$ coefficientwise, we have that (R, Δ_n) is perfectoid pure. Many results in this paper likely can be extended to this setting.

Corollary 4.10. *Let R and Δ be as in Definition 4.8. Suppose that there are a perfectoid R-algebra B and a Δ-datum $\mathfrak{d}$ on B such that $\eta_{\mathfrak{d}}$ is pure. Then, the R-algebra map $R \to B$ is pure and (R, Δ) is perfectoid pure. In particular, R is perfectoid pure.*

Proof. Effectivity gives a natural map $B \to M_B(\mathfrak{d})$, $b \mapsto b \otimes 1$, through which $\eta_{\mathfrak{d}}$ factors. Hence $R \to B$ is pure so R is perfectoid pure. Using Lemma 4.5(b), we may assume that B is absolutely integrally closed. Then, by Example 4.4, for every rational number $0 < \varepsilon < 1$, there is a $(1-\varepsilon)\Delta$-datum $\mathfrak{d}_\varepsilon$ on B. Choose the trivial datum when $\Delta = 0$. Applying Proposition 4.7 with $\Delta' = (1-\varepsilon)\Delta$ shows that $\eta_{\mathfrak{d}_\varepsilon}$ is pure, as desired. □

Theorem 4.11. *Let R and Δ be as in Definition 4.8. Then (R, Δ) is perfectoid pure if and only if there is an absolutely integrally closed perfectoid R-algebra B such that for some sequence of rational numbers $0 < \{\varepsilon_n\} < 1$ converging to 0, there is a $(1-\varepsilon_n)\Delta$-datum $\mathfrak{d}_{\varepsilon_n}$ on B for all n for which $\eta_{\mathfrak{d}_{\varepsilon_n}}$ is pure. Moreover, B can be chosen so that every effective Mumford $\mathbb{Q}$-divisor Δ' satisfying $\Delta' \prec \Delta$ admits a datum on B, and $\eta_{\mathfrak{d}'}$ is pure for every Δ'-datum $\mathfrak{d}'$ on B.*

Proof. The reverse implication is immediate from Proposition 4.7 so suppose that (R, Δ) is perfectoid pure. If $\Delta = 0$, the result follows from [BMP+24, Lemma 4.5] and Proposition 4.7, so we may assume that $\Delta \neq 0$. Fix $f \in R$ as in Construction 2.18 and note that since $(m\alpha\Delta)_{\mathrm{red}} = \Delta_{\mathrm{red}}$ for any $\alpha \neq 0$, this same f works for any $(1-\varepsilon_n)\Delta$.

By Lemma 4.5(b), we may assume that every B_ε in the definition that (R, Δ) being perfectoid pure is absolutely integrally closed. Now for every rational $0 < \varepsilon < 1$, there is m_ε so that $m_\varepsilon(1-\varepsilon)\Delta$ is integral. Let $(S_{f,m_\varepsilon}, D_\varepsilon)$ be the $(1-\varepsilon)\Delta$-lift supplied by Example 4.4. Since B_λ is absolutely integrally closed for each $0 < \lambda < 1$, there exists a $(1-\varepsilon)\Delta$-datum $\mathfrak{d}_{\varepsilon,\lambda} = (S_{f,m_\varepsilon}, D_\varepsilon, \iota_{\varepsilon,\lambda})$ on B_λ. Then $\mathfrak{d}_\varepsilon := \prod_\lambda \mathfrak{d}_{\varepsilon,\lambda}$ is a $(1-\varepsilon)\Delta$-datum on $B := \prod_{0<\lambda<1} B_\lambda$.

B is perfectoid by [BIM19, Example 3.8(8)], and it is absolutely integrally closed because monic polynomials can be solved coordinatewise. For each $0 < \lambda < 1$, we have a composition pointed map (where the second map is simply projection onto the λ-factor)

$$\eta_{\mathfrak{d}_{\varepsilon,\lambda}} \colon R \to M_B(\mathfrak{d}_\varepsilon) \longrightarrow M_{B_\lambda}(\mathfrak{d}_{\varepsilon,\lambda}).$$

Note that for each $0 < \lambda < \varepsilon$, $\eta_{\mathfrak{d}_{\varepsilon,\lambda}}$ is pure: this is because $(1-\varepsilon)\Delta < (1-\lambda)\Delta$ and thus Proposition 4.7 shows that $\eta_{\mathfrak{d}_{\varepsilon,\lambda}}$ is pure. It follows that the pointed map $\eta_{\mathfrak{d}_\varepsilon} \colon R \to M_B(\mathfrak{d}_\varepsilon)$ is pure for every rational $0 < \varepsilon < 1$ and the first statement follows. The second part of the statement follows from Example 4.4. □

Corollary 4.12. *Let R be an excellent $\mathbf{S}_2$ ring with p in its Jacobson radical, and let $\Delta' \leq \Delta$ be effective Mumford $\mathbb{Q}$-divisors on R. If (R, Δ) is perfectoid pure, then (R, Δ') is perfectoid pure.*

Moreover, if $\{\Delta_n\}_{n\geq 1}$ is a sequence of effective Mumford $\mathbb{Q}$-divisors such that $\Delta_n \leq \Delta$, $\Delta_n \to \Delta$ coefficientwise, and (R, Δ_n) is perfectoid pure for all n, then (R, Δ) is perfectoid pure.

Proof. Direct from Theorem 4.11 after noting that for any $0 < \varepsilon < 1$, there is $n \gg 0$ with $\Delta_n \geq (1-\varepsilon/2)\Delta$ so that $(1-\varepsilon)\Delta \prec \Delta_n$. □

Corollary 4.13. *Let R and Δ be as in Definition 4.8. If (R, Δ) is perfectoid pure, then every coefficient of Δ is at most one.*

Proof. Suppose that Δ has coefficient $c > 1$ along a prime Mumford divisor P and let $0 < \varepsilon < 1$ be a small enough rational number so that $(1-\varepsilon)c > 1$. By Construction 2.18 and Theorem 4.11, there a pure datum $\mathfrak{d}_\varepsilon = (S_\varepsilon, D_\varepsilon, \iota_\varepsilon)$ for some $m \in \mathbb{N}_{>0}$, nonzerodivisor $f \in R$ and $S_\varepsilon \coloneqq R[x]/(x^m - f)$ mapping to a perfectoid ring B. In fact, we have $S_\varepsilon(D_\varepsilon) \otimes_R R_P = x^{-m(1-\varepsilon)c}(S_\varepsilon \otimes_R R_P)$. Write π for the uniformizer of R_P and $f = u\pi$ for some unit $u \in R_P$. Since $m(1-\varepsilon)c > m$, $1/\pi = ux^{-m} \in S_\varepsilon(D_\varepsilon) \otimes_R R_P$ so the distinguished element of $M_B(\mathfrak{d}_\varepsilon)_P$ is in $\pi M_B(\mathfrak{d}_\varepsilon)_P$. Base changing $\eta_{\mathfrak{d}_\varepsilon}$ by $R_P/\pi R_P$ gives a pure map that sends 1 to 0, a contradiction. □

4.2. **The limiting module from below.** Fix R and Δ as in Definition 4.8 and a perfectoid R-algebra B. In this subsection, we combine Definition 2.11 and Lemma 4.6 to show that perfectoid purity can be tested on only one module. Throughout, codimension-one envelopes are taken over R.

Lemma 4.14. *Let R be an excellent $\mathbf{S}_2$ ring with p in its Jacobson radical, and let $\Delta' \prec \Delta$ be effective Mumford $\mathbb{Q}$-divisors on R. Let $\mathfrak{d} = (S, D, \iota)$ and $\mathfrak{d}' = (S', D', \iota')$ be Δ and Δ'-data on a perfectoid ring B, respectively. There is a canonical pointed B-linear map*

$$\Phi_{\mathfrak{d}',\mathfrak{d}} \colon M_B(\mathfrak{d}')^{\mathrm{c1}} \longrightarrow M_B(\mathfrak{d})^{\mathrm{c1}}$$

that satisfies the following properties.

- *(a) Suppose that ι and ι' factor through finite flat S- and S'-algebras T and T', respectively, and let $\mathfrak{d}_T$ and $\mathfrak{d}'_T$ be the pullback data given by Lemma 4.5(c). The natural pointed isomorphisms of that lemma identify $\Phi_{\mathfrak{d}',\mathfrak{d}}$ with $\Phi_{\mathfrak{d}'_T,\mathfrak{d}_T}$.*
- *(b) If Δ'' is an effective Mumford $\mathbb{Q}$-divisor on R, $\mathfrak{d}'' = (S'', D'', \iota'')$ is a Δ''-datum on B, and $\Delta'' \prec \Delta'$, then*

$$\Phi_{\mathfrak{d}'',\mathfrak{d}} = \Phi_{\mathfrak{d}',\mathfrak{d}} \circ \Phi_{\mathfrak{d}'',\mathfrak{d}'}.$$

Proof. Note that if both U and U' are big opens as in Lemma 4.6, then so is $U \cap U'$. Moreover, the maps of Lemma 4.6 on U and U' must agree on $U \cap U'$ by uniqueness. Since big opens as in Lemma 4.6 are cofinal in the system defining the codimension-one envelope, we can pass to the filtered colimits to obtain a canonical map $\Phi_{\mathfrak{d}',\mathfrak{d}}$.

For (a), Proposition 2.15 and associativity show that the ideals $J_{\mathfrak{d}'}$ (which we recall is defined as $\sqrt{\mathrm{Image}(B \otimes_{S'} S'(-D') \to B)}$) and $J_{\mathfrak{d}'_T}$ (defined likewise) agree. This implies that on any common admissible big open U, the pointed isomorphism of Lemma 4.5(c) identifies the maps $\widetilde{M_B(\mathfrak{d}')}|_U \to \widetilde{M_B(\mathfrak{d})}|_U$ and $\widetilde{M_B(\mathfrak{d}'_T)}|_U \to \widetilde{M_B(\mathfrak{d}_T)}|_U$ of Lemma 4.6 so the result follows. Part (b) similarly follows from the uniqueness in Lemma 4.6. □

Definition 4.15. Let R be an excellent $\mathbf{S}_2$ ring with p in its Jacobson radical and B a perfectoid R-algebra. Suppose that, for every $e \geq 1$, there are effective Mumford $\mathbb{Q}$-divisors Δ_e admitting Δ_e-data $\mathfrak{d}_e$ on B such that

$$\Delta_e \prec \Delta_{e+1} \prec \Delta \quad \text{for every } e \qquad \text{and} \qquad \Delta_e \longrightarrow \Delta \quad \text{coefficientwise.}$$

We call $(\Delta_e, \mathfrak{d}_e)$ a limiting sequence for Δ. The maps of Lemma 4.14 form a direct system, and we define

$$M_B(<\Delta) \coloneqq \varinjlim_e M_B(\mathfrak{d}_e)^{\mathrm{c1}}.$$

The distinguished elements give $\mathbf{1}_{<\Delta} \in M_B(<\Delta)$ and a map $\rho_{B,\Delta} \colon R \to M_B(<\Delta)$, $1 \mapsto \mathbf{1}_{<\Delta}$.

The next lemma shows that this pointed module and map do not depend on the chosen sequence.

Lemma 4.16. *Let R be an excellent $\mathbf{S}_2$ ring with p in its Jacobson radical and B a perfectoid R-algebra. For any effective Mumford divisor Δ admitting limiting sequences $(\Delta_e, \mathfrak{d}_e)$ and $(\Gamma_j, \mathfrak{d}'_j)$ on B, write*

$$\lambda_e \colon M_B(\mathfrak{d}_e)^{\mathrm{c1}} \longrightarrow \varinjlim_e M_B(\mathfrak{d}_e)^{\mathrm{c1}}, \qquad \mu_j \colon M_B(\mathfrak{d}'_j)^{\mathrm{c1}} \longrightarrow \varinjlim_j M_B(\mathfrak{d}'_j)^{\mathrm{c1}}$$

for the maps defining the colimits. There is a canonical pointed isomorphism

$$\Theta \colon \varinjlim_e M_B(\mathfrak{d}_e)^{\mathrm{c1}} \xrightarrow{\sim} \varinjlim_j M_B(\mathfrak{d}'_j)^{\mathrm{c1}},$$

which is uniquely characterized by $\Theta \circ \lambda_e = \mu_j \circ \Phi_{\mathfrak{d}_e, \mathfrak{d}'_j}$ whenever $\Delta_e \prec \Gamma_j$ for $\Phi_{\mathfrak{d}_e, \mathfrak{d}'_j}$ as in Lemma 4.14. In particular

$$M_B(<\Delta) \cong \varinjlim_{\substack{\Gamma \prec \Delta \\ \mathfrak{d}' \text{ a } \Gamma\text{-datum on } B}} M_B(\mathfrak{d}')^{\mathrm{c1}}.$$

Moreover, for any $D \leq \Delta$ admitting a limiting sequence on B, there is a canonical pointed map

$$M_B(<D) \longrightarrow M_B(<\Delta).$$

Proof. Fix $e \geq 1$ and consider $\Delta_e \prec \Delta$. Since Δ has finite support and $\Gamma_j \to \Delta$ coefficientwise, it is easy to see that for $j \gg 0$ we have $\Delta_e \prec \Gamma_j$. We may therefore choose strictly increasing sequences of indices $e_1 < e_2 < \cdots$ and $j_1 < j_2 < \cdots$ recursively so that

$$\Delta_{e_1} \prec \Gamma_{j_1} \prec \Delta_{e_2} \prec \Gamma_{j_2} \prec \cdots .$$

Since this is an interleaved direct system, the colimit in Γ must agree with the colimit in Δ.

For each e, the transitivity of Lemma 4.14 implies that the composite $\mu_j \circ \Phi_{\mathfrak{d}_e, \mathfrak{d}'_j}$ is independent of the choice of j satisfying $\Delta_e \prec \Gamma_j$. These composites are also compatible as e varies, and hence induce the displayed map Θ. Note that Θ is independent of the chosen interleaving. Moreover, reversing the roles of the two sequences gives its inverse so it is indeed an isomorphism. All the maps are pointed and therefore so is Θ. For the last statement, let $(D_e, \mathfrak{d}'_{\mathfrak{e}})$ be a limiting sequence for D. For any e, $D_e \prec D \leq \Delta$ so D_e is part of a limiting sequence for Δ and we get a canonical pointed map $M_B(\mathfrak{d}'_{\mathfrak{e}}) \to M_B(<\Delta)$. This is compatible with transitions so going to the colimit gives the desired $M_B(<D) \longrightarrow M_B(<\Delta)$. □

Although any limiting sequence works to define $M_B(<\Delta)$, two of them are particularly useful.

Construction 4.17. Let R be an excellent $\mathbf{S}_2$ ring and let Δ be an effective Mumford $\mathbb{Q}$-divisor. Let $m \in \mathbb{N}_{>0}$ be such that $m\Delta$ has integer coefficients and fix $E = \Delta_{\mathrm{red}}$. Then there are sequences of data for $e \gg 0$

$$\mathfrak{d}_e = (S_e, \Delta_e, \iota_e) \quad \text{and} \quad \mathfrak{d}'_e = (S'_e, \Delta'_e, \iota'_e)$$

with the following properties.

(a) Δ_e and Δ'_e are effective Mumford divisors.

(b) Δ_e is the flat pullback of the effective divisor $\Delta - \frac{1}{mp^e}E \prec \Delta$ and Δ'_e is the flat pullback of the effective divisor $t_e\Delta$, where $t_e = \frac{m}{p^e}(\lfloor \frac{p^e}{m} \rfloor - 1)$. Moreover,

$$\lim_{e\to\infty} t_e\Delta = \lim_{e\to\infty} \left(\Delta - \frac{1}{mp^e}E\right) = \Delta,$$

where the limit is to be understood coefficientwise.

(c) For each $e > 0$, there are finite flat R-algebra maps $S_e \to S_{e+1}$ and $S'_e \to S'_{e+1}$ such that the following diagrams commute

$$\begin{array}{ccc} S_e & \xrightarrow{\iota_e} & B \\ \downarrow & \nearrow_{\iota_{e+1}} & \\ S_{e+1} & & \end{array} \qquad \begin{array}{ccc} S'_e & \xrightarrow{\iota'_e} & B' \\ \downarrow & \nearrow_{\iota'_{e+1}} & \\ S'_{e+1} & & \end{array}.$$

(d) If $h_e\colon \operatorname{Spec} S_{e+1} \to \operatorname{Spec} S_e$ and $h'_e\colon \operatorname{Spec} S'_{e+1} \to \operatorname{Spec} S'_e$ are induced from the previous maps, then $h_e^*\Delta_e \prec \Delta_{e+1}$ and $(h'_e)^*\Delta'_e \prec \Delta'_{e+1}$ for any $e \gg 0$. In particular there are natural maps $S_e(\Delta_e) \otimes_{S_e} S_{e+1} \to S_{e+1}(\Delta_{e+1})$ and $S'_e(\Delta'_e) \otimes_{S'_e} S'_{e+1} \to S'_{e+1}(\Delta'_{e+1})$ which are injective.

We now perform the constructions. Essentially we are simply going to form iterated versions of Construction 2.18. Pick $f \in R$ such that $\operatorname{div}(f) = E + G = \Delta_{\mathrm{red}} + G$ and where G has no common components with E. Set $S_e = R[X_e]/(X_e^{mp^e} - f)$ and $S'_e = R[Y_e]/(Y_e^{p^e} - f)$. Just as in Construction 2.18, then the flat pullback of Δ to S_e is a Mumford divisor, Δ_{S_e}. We set

$$\Delta_e \coloneqq \Delta_{S_e} - (\Delta_{S_e})_{\mathrm{red}}.$$

By construction, Δ_e is the flat pullback of $\Delta - \frac{1}{mp^e}E$ to S_e. Similarly, we set Δ'_e to be the flat pullback of $t_e\Delta$ to S'_e, which is possible since $t_e p^e \Delta = m(\lfloor \frac{p^e}{m} \rfloor - 1)\Delta$ has integer coefficients. There are certainly finite flat maps

$$S_e = R[X_e]/(X_e^{mp^e} - f) \xrightarrow{X_e \mapsto X_{e+1}^p} R[X_{e+1}]/(X_{e+1}^{mp^{e+1}} - f).$$

and

$$S'_e = R[Y_e]/(Y_e^{p^e} - f) \xrightarrow{Y_e \mapsto Y_{e+1}^p} R[Y_{e+1}]/(Y_{e+1}^{p^{e+1}} - f).$$

Fix B a perfectoid R-algebra such that f has an m-th root $Z \in B$ ($Z^m = f$) and such that Z has a compatible system of p-power roots, $Z^{1/p^e} \in B$. Fix B' a perfectoid R-algebra such that f has a compatible system of p-th power roots $W^{1/p^e} \in B'$. For instance, this holds if B and B' are absolutely integrally closed. Sending $X_e \mapsto Z^{1/p^e}$ and $Y_e \mapsto W^{1/p^e}$ produces maps $\iota_e\colon S_e \to B$ and $\iota'_e\colon S'_e \to B'$ so that

$$\mathfrak{d}_e = (S_e, \Delta_e, \iota_e) \quad \text{and} \quad \mathfrak{d}'_e = (S'_e, \Delta'_e, \iota'_e)$$

are datum for $\Delta - \frac{1}{mp^e}E$ and $t_e\Delta$, respectively. Write $a_e \coloneqq \lfloor \frac{p^e}{m} \rfloor - 1$ and note that $m \leq p^e - ma_e < 2m$ for all e so $p^{e+1} - ma_{e+1} < 2m \leq p(p^e - ma_e)$, which gives $t_e < t_{e+1}$. Then, the properties outlined above all follow formally.

Set $S_\infty \coloneqq \varinjlim_e S_e$ where the transition maps are the ones above and note that the induced map $S_e \to S_\infty \to B$ is ι_e. Set $M_{S_\infty}(<\Delta) \coloneqq \varinjlim_e (S_\infty \otimes_{S_e} S_e(\Delta_e))^{\mathrm{c1}}$. Since the formation of $(-)^{\mathrm{c1}}$ commutes with filtered colimits, we have canonical R- (and S_e)-module isomorphisms

$$M_{S_\infty}(<\Delta) \cong (\varinjlim_e S_e(\Delta_e))^{\mathrm{c1}}$$

and also a canonical pointed B-module isomorphism

$$M_B(<\Delta) \cong (M_{S_\infty}(<\Delta) \otimes_{S_\infty} B)^{\mathrm{c1}}.$$

Remark 4.18. It is worth noting that if $0 \leq \Delta \leq D$ is such that both mD and $m\Delta$ have integer coefficients, then we may use the same f and sequence of S_e's for both of them in Construction 4.17. Furthermore, setting $E' = D_{\mathrm{red}}$ and D_e to be the flat pullback of $D - \frac{1}{mp^e}E'$, we have $\Delta_e \leq D_e$ and that this is compatible with the inclusions of (d). One can also use the same S'_e and work with the flat pullbacks of $t_e\Delta$ and t_eD and keep the inclusions. This can also be done with any finite collection of divisors Δs such that $m\Delta$ has integer coefficients for a common m.

Lemma 4.19. *Let R be an excellent $\mathbf{S}_2$ ring with p in its Jacobson radical, let Δ and Δ' be effective Mumford $\mathbb{Q}$-divisors on R, and let B be a perfectoid R-algebra admitting limiting sequences that define both $M_B(<\Delta)$ and $M_B(<\Delta')$. If $\Delta|_V = \Delta'|_V$ on an open subset $V \subseteq \operatorname{Spec} R$, then there is a canonical pointed $\widetilde{B|_V}$-linear isomorphism*

$$\widetilde{M_B(<\Delta)}\Big|_V \xrightarrow{\sim} \widetilde{M_B(<\Delta')}\Big|_V,$$

which is compatible with restriction to smaller open subsets.

Proof. Choose limiting sequences $(\Delta_e, \mathfrak{d}_e)_{e\geq 1}$ and $(\Delta'_j, \mathfrak{d}'_j)_{j\geq 1}$ for $M_B(<\Delta)$ and $M_B(<\Delta')$, respectively. Now, $\Delta|_V = \Delta'|_V$ has finite support so coefficientwise convergence implies that $\Delta_e|_V$ and $\Delta'_e|_V$ are mutually cofinal with respect to $\prec$. Since restriction commutes with filtered colimits, the result follows from the fact that the construction of the map in Lemma 4.6 is local. □

Proposition 4.20. *Let R be an excellent $\mathbf{S}_2$ ring with p in its Jacobson radical and let Δ be an effective Mumford $\mathbb{Q}$-divisor on R. Let B be a perfectoid R-algebra admitting a limiting sequence for Δ. Then,*

$$\rho_{B,\Delta} \colon R \to M_B(<\Delta) \quad 1 \mapsto \mathbf{1}_{<\Delta},$$

is pure if and only if $\eta_{\mathfrak{d}'}$ is pure for every Δ'-datum $\mathfrak{d}'$ on B with $\Delta' \prec \Delta$. Consequently, (R, Δ) is perfectoid pure if and only if $\rho_{B,\Delta} \colon R \to M_B(<\Delta)$ is pure for some absolutely integrally closed perfectoid R-algebra B.

Proof. If every $\eta_{\mathfrak{d}'}$ is pure then Lemma 2.21 implies that all the maps $R \to M_B(\mathfrak{d}')^{\mathrm{c1}}$ are pure. Taking the filtered colimit shows that $\rho_{B,\Delta}$ is pure. Conversely, any $(\Delta', \mathfrak{d}')$, is part of a limiting sequence for Δ so Lemma 4.16 implies that $\rho_{B,\Delta}$ factors through $R \to M_B(\mathfrak{d}')^{\mathrm{c1}}$. The purity of $\rho_{B,\Delta}$ therefore forces that of $R \to M_B(\mathfrak{d}')^{\mathrm{c1}}$, and hence that of $\eta_{\mathfrak{d}'}$. The last assertion follows from Theorem 4.11. □

Proposition 4.21. *Let R be an excellent $\mathbf{S}_2$ ring with p in its Jacobson radical, let D be an effective integral Mumford divisor and let Γ be an effective Mumford $\mathbb{Q}$-divisor. Fix $f \in R$ an element vanishing to order 1 at the generic point of every prime Mumford divisor in the support of D or Γ and such that $\operatorname{div}(f)$ is a Mumford divisor. Choose $m > 0$ such that $m\Gamma$ has integer coefficients and pick B a perfectoid R-algebra such that f has an m-th root $Y \in B$ (i.e., $Y^m = f$) and such that Y has a compatible system of p-power roots in B (e.g., B is absolutely integrally closed). Set $I_D = R(-D)$, and let $T_{B,D} \coloneqq \left(\sqrt{I_D B} \otimes_R R(D)\right)^{\mathrm{c1}}$.*

(a) If Γ and D share no common components, then there is a canonical pointed B-linear isomorphism

$$(M_B(<(D+\Gamma)), \mathbf{1}_{<(D+\Gamma)}) \xrightarrow{\sim} ((T_{B,D} \otimes_B M_B(<\Gamma))^{\mathrm{c1}}, \mathbf{1}_D \otimes \mathbf{1}_{<\Gamma}),$$

where $\mathbf{1}_D$ is induced by the canonical identification $(I_D \otimes_R R(D))^{\mathrm{c1}} \cong R$.

(b) There is a canonical pointed B-linear map

$$\alpha_{D,\Gamma} \colon \left(M_B(<(D+\Gamma)), \mathbf{1}_{<(D+\Gamma)}\right) \longrightarrow \left(((B \otimes_R R(D)) \otimes_B M_B(<\Gamma))^{\mathrm{c1}}, \mathbf{1}_D \otimes \mathbf{1}_{<\Gamma}\right).$$

When Γ and D have no common component, this is given by composing the map in (a) with the natural pointed map

$$T_{B,D} \longrightarrow (B \otimes_R R(D))^{\mathrm{c1}}$$

coming from $\sqrt{I_D B} \subseteq B$.

Proof. Apply the construction of Construction 4.17 to $D+\Gamma$ noting that it also immediately applies to both D and Γ by Remark 4.18. Set $E = \Gamma_{\mathrm{red}}$ and $E' = D_{\mathrm{red}}$. On $\operatorname{Spec} S_e$, we have D_e (the flat pullback of $D - \frac{1}{mp^e}E'$), Γ_e (the flat pullback of $\Gamma - \frac{1}{mp^e}E$) and $\Delta_e \coloneqq \Gamma_e + D_e$ (the flat pullback of $D+\Gamma - \frac{1}{mp^e}(E+E')$). Write $\mathfrak{d}_e$ and $\mathfrak{d}'_e$ for the corresponding $\Gamma - \frac{1}{mp^e}E$ and $D+\Gamma - \frac{1}{mp^e}(E+E')$-data, respectively. We first show (a). Recall our construction also provides pointed isomorphisms

$$M_B(<(\Gamma+D)) \cong \left(M_{S_\infty}(<(\Gamma+D)) \otimes_{S_\infty} B\right)^{\mathrm{c1}}.$$

In fact, by construction and since D and E have no common components in their support, we have that

$$S_e(\Delta_e) \cong \left(\sqrt{I_{D+E}S_e} \otimes_{S_e} S_e(\Gamma+D)\right)^{\mathrm{c1}} \cong \left((\sqrt{I_D S_e} \otimes_R R(D))^{\mathrm{c1}} \otimes_{S_e} S_e(\Gamma_e)\right)^{\mathrm{c1}}. \tag{4.21.1}$$

Likewise $S_e(D_e) = \left(\sqrt{I_D S_e} \otimes_{S_e} S_e(D)\right)^{\mathrm{c1}}$. By taking filtered colimits and using the associativity of tensor product, we obtain a canonical pointed map

$$M_B(<(\Gamma+D)) \cong \varinjlim_e \left((\sqrt{I_D S_e} \otimes_R R(D))^{\mathrm{c1}} \otimes_{S_e} S_e(\Gamma_e) \otimes_{S_e} B\right)^{\mathrm{c1}} \to \left(T_{B,D} \otimes_B M_B(<\Gamma)\right)^{\mathrm{c1}}.$$

As tensor products and codimension-1-envelopes commute with filtered colimits, and using the associativity of tensor product yet again, it suffices to show that

$$\varinjlim_e \left(\left(\sqrt{I_D S_e} \otimes_R R(D)\right) \otimes_{S_e} B\right)^{\mathrm{c1}} \to T_{B,D}$$

is an isomorphism. Similarly, the result then follows if we can show that

$$\varinjlim_e \left(\sqrt{I_D S_e} \otimes_{S_e} B\right)^{\mathrm{c1}} \to \left(\sqrt{I_D B}\right)^{\mathrm{c1}}$$

is an isomorphism. By Proposition 2.12, it suffices to check this at codimension-1 points of $\operatorname{Spec} R$, at which it suffices to prove the result without the $(-)^{\mathrm{c1}}$. At those points outside of the support of D, there is nothing to show, and hence we may assume that R and all S_e are DVRs, $I_D = f^a R$ and $X_e^{mp^e} = f \in S_e$ where X_e is a uniformizer for S_e and f is a uniformizer for R. We write h for the image of X_1^p in B (thought of as $f^{1/m}$). We thus need to show that

$$\varinjlim_e (X_e S_e \otimes_{S_e} B) \to \sqrt{hB} = \sqrt{I_D B}$$

is an isomorphism. As S_e is a DVR, we have isomorphisms $X_e S_e \otimes_{S_e} B \cong B$ and with this identification our direct system becomes

$$B \xrightarrow{h^{1-1/p}} B \xrightarrow{h^{1/p-1/p^2}} B \xrightarrow{h^{1/p^2-1/p^3}} \dots.$$

That colimit is identified with $\sqrt{hB}$ by Lemma 2.5 (b), and (a) follows.

For (b), note that for any e, there is a natural map

$$M_B(\mathfrak{d}'_e)^{\mathrm{c1}} \to (R(D) \otimes_R M_B(\mathfrak{d}_e))^{\mathrm{c1}} \cong ((B \otimes_R R(D)) \otimes_B M_B(\mathfrak{d}_e))^{\mathrm{c1}}$$

coming from the natural morphism

$$(S_e(D_e + \Gamma_e))^{\mathrm{c1}} \to (S_e(D + \Gamma_e))^{\mathrm{c1}} \cong (S_e(\Gamma_e) \otimes_R R(D))^{\mathrm{c1}}. \tag{4.21.2}$$

This is compatible with transitions and codimension 1-envelopes commute with filtered colimits so we can go to the colimit and obtain $\alpha_{D,\Gamma}$. To see the last statement, combine (4.21.1) and (4.21.2) into

$$\left((\sqrt{I_D S_e} \otimes_R R(D))^{\mathrm{c1}} \otimes_{S_e} S_e(\Gamma_e)\right)^{\mathrm{c1}} \to (S_e(\Gamma_e) \otimes_R R(D))^{\mathrm{c1}} \tag{4.21.3}$$

and go to the colimit. □

4.3. **Non-pure ideals.**

Definition 4.22. Let $(R, \mathfrak{m})$ be an excellent local $\mathbf{S}_2$ ring of residue characteristic p and let Δ be an effective Mumford $\mathbb{Q}$-divisor on R. Let B be a perfectoid R-algebra admitting a limiting sequence for Δ. Let $E = E_R(R/\mathfrak{m})$ and fix a socle element $u \in E$. We define

$$\mathcal{I}(B, <\Delta) \coloneqq \mathcal{I}^R(B, M_B(<\Delta)) = \operatorname{Ann}_B \left(u \otimes \mathbf{1}_{<\Delta} \text{ in } E \otimes_R M_B(<\Delta)\right).$$

Recall that the non-pure ideal $\mathcal{I}^R(B, M)$ corresponding to an R-algebra B and pointed B-module M was introduced in Definition 2.23.

Corollary 4.23. *Let $(R, \mathfrak{m})$ be an excellent local $\mathbf{S}_2$ ring of residue characteristic p and let Δ be an effective Mumford $\mathbb{Q}$-divisor on R. Let B be a perfectoid R-algebra admitting a limiting sequence for Δ. Then, any limiting sequence $\{\mathfrak{d}_e\}$ for Δ computes $\mathcal{I}(B, <\Delta)$ by*

$$\mathcal{I}(B, <\Delta) = \bigcup_e \mathcal{I}(B, M_B(\mathfrak{d}_e)) = \bigcup_e \operatorname{Ann}_B \left(u \otimes \mathbf{1}_{\mathfrak{d}_e} \text{ in } E \otimes_R M_B(\mathfrak{d}_e)\right). \tag{4.23.1}$$

where the unions are ascending. Consequently, the pair (R, Δ) is perfectoid pure if and only if $\mathcal{I}(B, <\Delta) \neq B$ for some absolutely integrally closed perfectoid R-algebra B.

Proof. Follows immediately from Lemma 2.27(d), (g) and Proposition 4.20. □

Proposition 4.24. *With notation as in Definition 4.22, let $B \to C$ be a map of perfectoid R-algebras. Then, C admits a limiting sequence for Δ and $\mathcal{I}(B, <\Delta) \subseteq \mathcal{I}(C, <\Delta) \cap B$. If, additionally, $B \to C$ is $\mathfrak{m}$-completely faithfully flat, then*

$$\mathcal{I}(C, <\Delta) = \mathcal{I}(B, <\Delta)C, \qquad \mathcal{I}(C, <\Delta) \cap B = \mathcal{I}(B, <\Delta)$$

and $\rho_{B,\Delta}$ is pure if and only if $\rho_{C,\Delta}$ is pure.

Proof. By Lemma 4.5, we can compose the realizations of a limiting sequence for Δ in B with $B \to C$ to obtain realizations on C, so C also admits a limiting sequence for Δ. Then, we use (4.23.1) to compute the non-pure ideals. In this way the containment and the two equalities follow from Lemma 4.5 and Lemma 2.27. The purity assertion follows from Corollary 4.23, since the contraction equality shows that $\mathcal{I}(B, <\Delta)$ is proper if and only if $\mathcal{I}(C, <\Delta)$ is proper. □

Remark 4.25. Let R be an excellent $\mathbf{S}_2$ ring with p in its Jacobson radical and B a perfectoid R-algebra. Suppose that $f \in R$ is a nonzerodivisor that does not vanish along any codimension-one component of the singular locus (thus $\Delta = \lambda \operatorname{div}_R(f)$ is a Mumford $\mathbb{Q}$-divisor). Write $\lambda = a/b$ with $a, b \in \mathbb{Z}_{>0}$, and suppose there is $Y \in B$ with $Y^b = f^a$ that admits a fixed compatible system of p-power roots in B. By Lemma 4.16, we may compute $M_B(<\Delta)$ using the $(1 - 1/p^e)\Delta$-data $\mathfrak{d}_e = (S_e, \Delta_e, \iota_e)$ as described in Example 4.3, where $S_e = R[x]/(x^{bp^e} - f^{a(p^e-1)})$, $\Delta_e = \operatorname{div}(x)$, and ι_e sends x to Y^{1-1/p^e}. Thus for each e, we have a pointed identification $M_B(\mathfrak{d}_e) \cong B$ with distinguished element Y^{1-1/p^e} in B, and the natural map $M_B(\mathfrak{d}_e) \to M_B(\mathfrak{d}_{e+1})$ is identified with multiplication by $Y^{1/p^e - 1/p^{e+1}}$ on B. It follows from Lemma 2.5 (b) that we have a pointed isomorphism

$$M_B(<\Delta) \cong \left((Y^{1/p^\infty})B\right)^{\mathrm{c1}}$$

which sends $\mathbf{1}_{<\Delta}$ to Y. Since $\sqrt{YB} = \sqrt{fB}$, by Lemma 2.5 (b) and Corollary 2.4 we have that

$$\left((Y^{1/p^\infty})B\right)^{-} = (fB)_{\mathrm{perfd}}.$$

It follows from Lemma 2.21 and Lemma 3.2 that $\rho_{B,\Delta}$ is pure if and only if $R \to (fB)_{\mathrm{perfd}}$ sending 1 to Y is pure. In other words, (R, f^λ) is perfectoid pure in the sense of Definition 3.4 if and only if (R, Δ) is perfectoid pure in the sense of Definition 4.8[4].

In fact, by Lemma 2.27 (e) and unwinding the definition of pointed B-modules we have

$$\mathcal{I}(B, <\Delta) = \left\{ b \in B \;\middle|\; R \xrightarrow{1 \mapsto bY} (fB)_{\mathrm{perfd}} \text{ is not pure} \right\}.$$

Remark 4.26. Let R be an excellent $\mathbf{S}_2$ ring with p in its Jacobson radical, let Δ be an effective Mumford $\mathbb{Q}$-divisor on R, let $f \in R$ be a nonzerodivisor that does not vanish along any codimension-one component of the singular locus, and let $t \in \mathbb{Q}_{>0}$. In light of Remark 4.25, it is natural to say that the triple (R, Δ, f^t) is *perfectoid pure* if $(R, \Delta + t \operatorname{div}_R(f))$ is perfectoid pure. Equivalently, there is a perfectoid R-algebra B that admits both a fixed compatible system of p-th power roots of f and a limiting sequence for Δ such that for every $e \gg 0$ (equivalently $e \geq 1$),

$$R \longrightarrow M_B(<\Delta), \qquad 1 \longmapsto (f^{1/p^e})^{\lfloor t(p^e-1) \rfloor} \mathbf{1}_{<\Delta}, \tag{4.26.1}$$

is pure: this can be seen with a similar argument as in Remark 4.25. The sequence $\lfloor t(p^e - 1) \rfloor / p^e$ may be replaced by any sequence $t_i = a_i/b_i$ of positive rational numbers, with $a_i, b_i \in \mathbb{Z}_{>0}$,

[4] If (R, f^λ) is perfectoid pure for some $\lambda > 0$, then Proposition 4.33 (a), together with Corollary 4.13, shows that f cannot vanish along any codimension-one component of the singular locus.

such that $t_i < t$ and converges to t, as long as f^{t_i} has a representative in B (for instance, if B is absolutely integrally closed). If $(R, \mathfrak{m})$ is local, this is also equivalent to $(f^{1/p^e})^{\lfloor t(p^e-1)\rfloor} \notin \mathcal{I}(B, <\Delta)$ for every $e \gg 0$.

Corollary 4.27. *Keep the hypotheses and notation of Proposition 4.21. Let $\phi \in \mathrm{Hom}_R(B, R)$. Consider the following conditions.*

(a) I_D is ϕ-compatible.

(b) $\phi(\sqrt{I_D B}) \subseteq I_D$.

(c) There is $\widetilde{\phi} \in \mathrm{Hom}_R(M_B(<D), R)$ whose composite with the pointed map $\iota_{B,D}\colon B \to M_B(<D)$ is ϕ.

Then (a) $\Longrightarrow$ (b) $\Longleftrightarrow$ (c). All three conditions are equivalent whenever $(I_D B)_{\mathrm{perfd}} = (\sqrt{I_D B})^-$. By Corollary 2.4, this holds if I_D has a finite generating set whose elements admit compatible systems of p-power roots in B. Whenever any of these conditions holds and ϕ is surjective, (R, D) is perfectoid pure.

Proof. Suppose first that (b) holds, i.e., $\phi(\sqrt{I_D B}) \subseteq I_D$. The rule $k \otimes r \mapsto r\phi(k)$ defines a map $\sqrt{I_D B} \otimes_R R(D) \to R$. Taking codimension-one envelopes and applying Proposition 4.21 gives $\widetilde{\phi}$. If h is a local equation for D, then $\iota_{B,D}(a) = ah \otimes h^{-1}$, so $\widetilde{\phi} \circ \iota_{B,D} = \phi$ in codimension one and hence everywhere, which shows (b) $\Longrightarrow$ (c).

Conversely, suppose that $\widetilde{\phi}$ exists as in (c). At any codimension-one point P where h is a local equation for D, every $k \in \sqrt{I_D B}$ satisfies $\phi(k) = h\widetilde{\phi}(k \otimes h^{-1}) \in hR_P$. As I_D is $\mathbf{S}_2$, this gives $\phi(\sqrt{I_D B}) \subseteq I_D$. This proves (c) $\Longleftrightarrow$ (b). Moreover, if ϕ is surjective with $\phi(b) = 1$ for some $b \in B$, $m \mapsto \widetilde{\phi}(bm)$ sends $\mathbf{1}_{<D}$ to $\phi(b) = 1$, and hence splits the distinguished map $R \to M_B(<D)$. Thus (R, D) is perfectoid pure by Proposition 4.20.

Finally, the implication (a) $\Longrightarrow$ (b) follows as $\sqrt{I_D B} \subseteq (I_D B)_{\mathrm{perfd}}$ (see Corollary 2.4); when $(I_D B)_{\mathrm{perfd}} = (\sqrt{I_D B})^-$, the converse follows as in [Fay25, proof of Lemma 2.21], since I_D is p-adically closed. □

4.4. **Localization, completion, and choice of B for divisor pairs.** We now show that perfectoid purity of pairs can be detected locally (Proposition 4.28 and Proposition 4.29) or after completion (Proposition 4.30). In Proposition 4.31 we also show that our notions coincide with notion of F-purity introduced in [HW02]. In the complete local case, we show how to construct a specific B that can determine perfectoid purity for pairs (Proposition 4.32). Finally, we show that perfectoid pure pairs remain so under normalization in Proposition 4.33.

Proposition 4.28. *Let R be an excellent $\mathbf{S}_2$ ring with p in its Jacobson radical, and let $\mathfrak{q}$ be a prime ideal containing p. For an effective Mumford $\mathbb{Q}$-divisor Δ on R, if (R, Δ) is perfectoid pure then so is $(R_\mathfrak{q}, \Delta_\mathfrak{q})$.*

Proof. Suppose that (R, Δ) is perfectoid pure, and fix a rational number $0 < \varepsilon < 1$. Choose a perfectoid R-algebra B and a $(1-\varepsilon)\Delta$-datum $\mathfrak{d} = (S, D, \iota)$ on B for which $\eta_\mathfrak{d}$ is pure. As in the proof of Proposition 3.7, we set $B_\mathfrak{q} = B \otimes_R R_\mathfrak{q}$, $C = B_\mathfrak{q}^{\wedge p}$, and note that C is perfectoid and $B_\mathfrak{q} \to C$ is p-completely pure. Lemma 4.5 gives a $(1-\varepsilon)\Delta_\mathfrak{q}$-datum $\mathfrak{d}_\mathfrak{q}$ on C such that $\eta_{\mathfrak{d}_\mathfrak{q}}$ is pure. Since ε was arbitrary, $(R_\mathfrak{q}, \Delta_\mathfrak{q})$ is perfectoid pure. □

Proposition 4.29 (*cf.* [BMP+24, Lemma 4.10])**.** *Let R be an excellent $\mathbf{S}_2$ ring with p in its Jacobson radical. Let Δ be an effective Mumford $\mathbb{Q}$-divisor on R. Then (R, Δ) is perfectoid pure if and only if $(R_{\mathfrak{m}}, \Delta_{\mathfrak{m}})$ is perfectoid pure for every maximal ideal $\mathfrak{m}$ of R.*

Proof. The forward direction follows from Proposition 4.28.

For the converse, suppose that $(R_{\mathfrak{m}}, \Delta_{\mathfrak{m}})$ is perfectoid pure for every maximal ideal $\mathfrak{m}$, and fix a rational number $0 < \varepsilon < 1$. Use Construction 2.18 to construct a $(1-\varepsilon)\Delta$-lift (S, D), taking the trivial lift when $\Delta = 0$. By Theorem 4.11, for every $\mathfrak{m}$, there is an absolutely integrally closed perfectoid $R_{\mathfrak{m}}$-algebra $B(\mathfrak{m})$ admitting the datum $\mathfrak{d}(\mathfrak{m}) = (S_{\mathfrak{m}}, D_{\mathfrak{m}}, \iota_{\mathfrak{m}})$ with $R \to M_B(\mathfrak{d}_{\mathfrak{m}})$ pure. Combining these gives an R-algebra map $S \to B := \prod_{\mathfrak{m}} B(\mathfrak{m})$ where B is perfectoid. Let $\mathfrak{d}$ be the resulting $(1-\varepsilon)\Delta$-datum on B. Projecting onto the $\mathfrak{m}$-th factor gives a pointed map $M_B(\mathfrak{d})_{\mathfrak{m}} \to M_{B(\mathfrak{m})}(\mathfrak{d}(\mathfrak{m}))$ whose composite with $(\eta_{\mathfrak{d}})_{\mathfrak{m}}$ is the pure map $\eta_{\mathfrak{d}(\mathfrak{m})}$ so $(\eta_{\mathfrak{d}})_{\mathfrak{m}}$ is also pure. Since purity can be checked locally and ε was arbitrary, (R, Δ) is perfectoid pure. □

Proposition 4.30 (*cf.* [BMP+24, Lemma 4.8])**.** *Let $(R, \mathfrak{m})$ be an excellent $\mathbf{S}_2$ ring of residue characteristic $p > 0$. Let Δ be an effective Mumford $\mathbb{Q}$-divisor on R. Then (R, Δ) is perfectoid pure if and only if $(\widehat{R}, \widehat{\Delta})$ is perfectoid pure, where $\widehat{\Delta}$ is the flat pullback of Δ along the regular map $R \to \widehat{R}$. More precisely, the following comparisons hold.*

(a) If C is an absolutely integrally closed perfectoid $\widehat{R}$-algebra, $\mathcal{I}^R(C, <\Delta) = \mathcal{I}^{\widehat{R}}(C, <\widehat{\Delta})$.

(b) Let B be an absolutely integrally closed perfectoid R-algebra, set $\widehat{B} = B^{\wedge \mathfrak{m}}$, which is perfectoid, and let $\widehat{B} \to C$ be a p-completely faithfully flat map to an absolutely integrally closed perfectoid ring. Then C is naturally an $\widehat{R}$-algebra and

$$\mathcal{I}^{\widehat{R}}(C, <\widehat{\Delta}) = \mathcal{I}^R(B, <\Delta)C, \qquad \mathcal{I}^{\widehat{R}}(C, <\widehat{\Delta}) \cap B = \mathcal{I}^R(B, <\Delta).$$

Proof. Suppose first that (R, Δ) is perfectoid pure, and fix a rational number $0 < \varepsilon < 1$. Choose a perfectoid R-algebra B and a $(1-\varepsilon)\Delta$-datum $\mathfrak{d} = (S, D, \iota)$ on B such that $\eta_{\mathfrak{d}}$ is pure. Let C be the $\mathfrak{m}B$-adic completion of B. Since $p \in \mathfrak{m}B$, Proposition 2.2 shows that C is perfectoid and agrees with the derived $\mathfrak{m}B$-adic completion. The map $R \to C$ extends to $\widehat{R} \to C$, and hence, by (a) and (d) of Lemma 4.5, there is a pointed isomorphism $M_C\big(\widehat{\mathfrak{d}}\big) \cong C \otimes_B M_B(\mathfrak{d})$. Since ε was arbitrary (b) follows from Corollary 4.23 together with Lemma 2.27 (b) (e) and (f). In particular, $(\widehat{R}, \widehat{\Delta})$ is perfectoid pure.

Conversely, suppose that $(\widehat{R}, \widehat{\Delta})$ is perfectoid pure, and fix a rational number $0 < \varepsilon < 1$. Choose a $(1-\varepsilon)\Delta$-lift (S, D) supplied by Construction 2.18, taking the trivial lift when $\Delta = 0$. Its flat pullback $(\widehat{S}, \widehat{D})$, where $\widehat{S} = S \otimes_R \widehat{R}$, is a $(1-\varepsilon)\widehat{\Delta}$-lift. Choose an absolutely integrally closed perfectoid $\widehat{R}$-algebra C as in Theorem 4.11. By Example 4.4, C realizes $(\widehat{S}, \widehat{D})$, say with datum $\widehat{\mathfrak{d}}$. Note that $\eta_{\widehat{\mathfrak{d}}}$ is pure by Theorem 4.11 since $(1-\varepsilon)\widehat{\Delta} \prec \widehat{\Delta}$. Via $S \to \widehat{S} \to C$, this realization also gives a $(1-\varepsilon)\Delta$-datum $\mathfrak{d}$ on C. By Lemma 4.5 $M_C(\mathfrak{d}) = M_C(\widehat{\mathfrak{d}})$, so Lemma 2.27 (b) then implies (a). In particular, (R, Δ) is perfectoid pure. □

We now compare perfectoid purity with the characteristic p definition of [HW02]. The purity that appears in (4.31.1) was originally called *F-purity* of the pair (R, Δ) in [HW02]. Sometimes it is also called *weak F-purity* to distinguish it from stronger notions.

Proposition 4.31. *Let R be an excellent reduced $\mathbf{S}_2$ ring of characteristic $p > 0$, and Δ an effective Mumford $\mathbb{Q}$-divisor on R. Then (R, Δ) is perfectoid pure if and only if the map*

$$R \longrightarrow R\big(\lfloor (p^e - 1)\Delta \rfloor\big)^{1/p^e}, \qquad 1 \longmapsto 1, \tag{4.31.1}$$

is pure for all $e \gg 0$. When R is F-finite, purity here is equivalent to splitting.

Proof. Set $\Delta_e = p^{-e}\lfloor (p^e - 1)\Delta \rfloor$. Construction 4.17 shows that $R_{\mathrm{perf}} = \varinjlim_j R^{1/p^j}$ admits a limiting sequence for Δ so if (R, Δ) is perfectoid pure, $R \to M_{R_{\mathrm{perf}}}(<\Delta)$ is pure. Since for any e, $\Delta_e \prec \Delta$, we must have that $R \to R_{\mathrm{perf}} \otimes_{R^{1/p^e}} R(p^e\Delta_e)^{1/p^e}$ is pure. Since this factors through $R \to R(p^e\Delta_e)^{1/p^e}$, this is also pure, as desired.

Conversely, suppose that $R \to R(p^e\Delta_e)^{1/p^e}$ is pure for all $e \gg 0$. Since this implies that R is F-pure, $R^{1/p^e} \to R_{\mathrm{perf}}$ is pure. Tensoring this with $R(p^e\Delta_e)^{1/p^e}$ and precomposing with the pure map $R \to R(p^e\Delta_e)^{1/p^e}$ gives that $R \to R_{\mathrm{perf}} \otimes_{R^{1/p^e}} R(p^e\Delta_e)^{1/p^e}$ is pure. Since a subsequence of the Δ_es gives a limiting sequence for Δ, (R, Δ) is perfectoid pure by Proposition 4.20. □

We also obtain the analogous result to Proposition 3.10 for divisor pairs.

Proposition 4.32. *Let $(R, \mathfrak{m}, k)$ be a complete Noetherian $\mathbf{S}_2$ local ring of residue characteristic $p > 0$, and let Δ be an effective Mumford $\mathbb{Q}$-divisor on R. Choose $f \in \mathfrak{m}$ for Δ as in Construction 2.18. For any $\varphi\colon A \to R$ as in Notation 2.7 with $\varphi(x_1) = f$ (such a map may be obtained by adjoining a variable mapping to f to any A as in Notation 2.7), there are rational numbers $0 < t_e < t_{e+1} < 1$ for $e \gg 0$, with $t_e \to 1$, and $t_e\Delta$-data $\mathfrak{d}_e = (S_e, D_e, \iota_e)$ on $R^{A_\infty}_{\mathrm{perfd}}$. For these data, (R, Δ) is perfectoid pure if and only if $\eta_{\mathfrak{d}_e}\colon R \longrightarrow M_{R^{A_\infty}_{\mathrm{perfd}}}(\mathfrak{d}_e)$ is pure for every $e \gg 0$. Equivalently, as these data also form a limiting sequence for Δ, (R, Δ) is perfectoid pure if and only if $\rho_{R^{A_\infty}_{\mathrm{perfd}}, \Delta}\colon R \to M_{R^{A_\infty}_{\mathrm{perfd}}}(<\Delta)$ is pure.*

Proof. The existence of the t_e data is Construction 4.17 and the backwards direction is direct from the definition. To show the forward direction, suppose that (R, Δ) is perfectoid pure and use Theorem 4.11 to fix an absolutely integrally closed perfectoid R-algebra B on which we can test the perfectoid purity of (R, Δ). Just as in Proposition 3.10, $R \to B$ factors through $R^{A_\infty}_{\mathrm{perfd}}$ and base changing allows us to realize every (S_e, D_e) on B, say with data $\mathfrak{d}_{e,B}$. Since each resulting $\eta_{\mathfrak{d}_{e,B}}$ is pure and factors through $\eta_{\mathfrak{d}_e}$ by Lemma 4.5(a), these latter maps must also be pure, as desired. □

Finally, we generalize [Fay25, Corollary 5.5] to the context of pairs.

Proposition 4.33. *Let R be an excellent $\mathbf{S}_2$ ring with p in its Jacobson radical, R^N its normalization, $\mathfrak{c}$ the conductor of R in R^N. Let C be the divisorial part of $\mathfrak{c}$ so that $R^N(-C) = \mathfrak{c}^{**} =: I_C$.*

(a) If $f \in R$ is a nonzerodivisor, $\lambda \in \mathbb{Q}_{>0}$, and (R, f^λ) is perfectoid pure, then $(R^N, C + \lambda \operatorname{div}_{R^N}(f))$ is perfectoid pure and $\operatorname{div}_{R^N}(f)$ and C have no component in common.

(b) If $\Delta = \sum_i a_i P_i$ is an effective Mumford $\mathbb{Q}$-divisor, let P_i^N be the unique prime divisor of R^N lying over P_i and set $\Delta^N = \sum_i a_i P_i^N$. If (R, Δ) is perfectoid pure, then $(R^N, \Delta^N + C)$ is perfectoid pure.

Proof. By Remark 4.25, part (b) for $\Delta = 0$ follows from (a) with $f = 1$; hence assume $\Delta \neq 0$.

The formation of R^N and C commutes with localization. If $(R,\mathfrak{m})$ is local and $\widehat{R} = R^{\wedge\mathfrak{m}}$, then $R^N \otimes_R \widehat{R} = (\widehat{R})^N$ by [Gro65, Scholie 7.8.3(vii)] and in (a) principal divisors commute with this base change. In (b), the ring R_{P_i} is a DVR for every i, and hence $R^N \otimes_R R_{P_i} = R_{P_i}$. Thus P_i^N is well-defined, and the conductor is the unit ideal there. In particular, Δ^N and C have no common prime component. If $\widehat{\Delta}$ is the flat pullback of Δ, the coefficient-wise definition gives $(\widehat{\Delta})^N = \widehat{\Delta^N}$; here a component may split, but every prime above it receives the same coefficient. Flat base commutes with the construction of the conductor and, by Proposition 2.15, so does the construction of I_C. Hence Proposition 4.29, [Fay25, Proposition 3.4], and Proposition 4.30, applied also to the local factors of R^N, reduce both (a) and (b) to the case where $(R,\mathfrak{m})$ is complete local.

If (R,Δ) or (R,f^λ) is perfectoid pure, R is perfectoid pure so $\mathfrak{c}$ is radical in both R and R^N by [Fay25, Corollary 2.33, Proposition 5.2, and Corollary 5.5] and C is reduced hence has integer coefficients. Note that the localization of $\mathfrak{c}$ at every prime component Q of C is the maximal ideal of the DVR $(R^N)_Q$. On the other hand, the localization of $\mathfrak{c}$ at every P_i^N in (b) is the unit ideal. In the semilocal ring obtained by localizing R^N at these primes and at the generic points of the remaining irreducible components, $\mathfrak{c}$ is invertible, hence principal. Clearing a denominator from a generator gives a nonzerodivisor $g \in \mathfrak{c} \subseteq R$ having order one at every such Q and, in (b), order zero at every P_i^N. Thus $\operatorname{div}_{R^N}(g) = C + G$ for an effective divisor G having no component in common with C. For (b), since Δ and C have no component in common, we can similarly choose $h \in R$ with $\operatorname{div}_{R^N}(h) = (\Delta^N + C)_{\mathrm{red}} + G'$ for an effective divisor G' having no component in common with $\Delta + C$: instead of looking at $\mathfrak{c}$ in the semilocal ring above, look at $\mathfrak{c}\prod_i P_i^N$.

Let $A \to R$ be as in Notation 2.7. By potentially enlarging A, we can assume $R^{A_\infty}_{\mathrm{perfd}}$ contains compatible systems of p-th power roots for g, generators of I_C, and in the case of (b), h. By Proposition 4.32 and Proposition 3.10, in both (a) and (b), we can assume our pure (hence split since R is complete) maps are with the perfectoid ring $R^{A_\infty}_{\mathrm{perfd}}$. By [Fay25, Corollary 5.5], any $\phi \in \operatorname{Hom}_R(R^{A_\infty}_{\mathrm{perfd}}, R)$ extends to $\phi^N \in \operatorname{Hom}_{R^N}(R^{N,A_\infty}_{\mathrm{perfd}}, R^N)$. Moreover, [Fay25, Lemma 5.1] shows that $\mathfrak{c}$ is compatible with such extension. By [Fay25, Proposition 2.35] applied on each complete local factor of R^N, all the height 1 associated primes of $\mathfrak{c}$ are compatible with such extensions and therefore so is their intersection, I_C ([Fay25, Proposition 2.34]).

To prove (a), fix $e \gg 0$. By Proposition 3.10, there is $\phi_e \in \operatorname{Hom}_R(R^{A_\infty}_{\mathrm{perfd}}, R)$ with

$$\phi_e\big((f^{1/p^e})^{\lfloor \lambda(p^e-1)\rfloor}\big) = 1.$$

This map extends to $\phi_e^N \in \operatorname{Hom}_{R^N}(R^{N,A_\infty}_{\mathrm{perfd}}, R^N)$ with $\phi_e^N\big((f^{1/p^e})^{\lfloor \lambda(p^e-1)\rfloor}\big) = 1$. Since I_C is ϕ_e^N-compatible, Corollary 4.27 gives $\widetilde{\phi}_e \in \operatorname{Hom}_{R^N}\big(M_{R^{N,A_\infty}_{\mathrm{perfd}}}(<C), R^N\big)$ such that

$$\widetilde{\phi}_e\big((f^{1/p^e})^{\lfloor \lambda(p^e-1)\rfloor}\mathbf{1}_{<C}\big) = 1.$$

Applying Remark 4.26 proves (a). If f and C had a common height-one component Q, then localizing ϕ_e^N at Q would give $1 \in Q(R^N)_Q$, a contradiction.

For (b), if $C = 0$ then since R is $\mathbf{S}_2$, we see that $R = R^N$ and there is nothing to show. Thus we may assume that $C \neq 0$. Since R is complete, perfectoid purity of (R,Δ) gives a splitting

$$\widetilde{\phi}_\Delta \colon M_{R^{A_\infty}_{\mathrm{perfd}}}(<\Delta) \longrightarrow R$$

of $\rho_{R^{A_\infty}_{\mathrm{perfd}},\Delta}$ and so set $\phi \coloneqq \widetilde{\phi}_\Delta \circ \iota_{R^{A_\infty}_{\mathrm{perfd}},\Delta} \colon R^{A_\infty}_{\mathrm{perfd}} \longrightarrow R$. Let ϕ^N be its extension to $R^{N,A_\infty}_{\mathrm{perfd}}$. Since I_C is ϕ^N-compatible, Corollary 4.27 gives a splitting

$$\widetilde{\phi}_C \colon M_{R^{N,A_\infty}_{\mathrm{perfd}}}(<C) \longrightarrow R^N, \qquad \widetilde{\phi}_C \circ \iota_{R^{N,A_\infty}_{\mathrm{perfd}},C} = \phi^N.$$

Set

$$U_\Delta = \operatorname{Spec} R^N \setminus V(\mathfrak{c}), \qquad U_C = \operatorname{Spec} R^N \setminus \operatorname{Supp}(\Delta^N), \qquad U = U_\Delta \cup U_C.$$

Over U_Δ, C vanishes and the pullback square in [Fay25, Lemma 5.1] shows that $R^{A_\infty}_{\mathrm{perfd}}|_{U_\Delta} \to R^{N,A_\infty}_{\mathrm{perfd}}|_{U_\Delta}$ is an isomorphism. By Lemma 4.19,

$$\widetilde{M_{R^{N,A_\infty}_{\mathrm{perfd}}}(<\Delta^N + C)}\Big|_{U_\Delta} \cong \widetilde{M_{R^{A_\infty}_{\mathrm{perfd}}}(<\Delta)}\Big|_{U_\Delta}$$

so $\widetilde{\phi}_\Delta$ gives a splitting over U_Δ. Similarly,

$$\widetilde{M_{R^{N,A_\infty}_{\mathrm{perfd}}}(<\Delta^N + C)}\Big|_{U_C} \cong \widetilde{M_{R^{N,A_\infty}_{\mathrm{perfd}}}(<C)}\Big|_{U_C}$$

so $\widetilde{\phi}_C$ gives a splitting over U_C. Moreover, since both divisors vanish on $U_\Delta \cap U_C$,

$$\widetilde{M_{R^{N,A_\infty}_{\mathrm{perfd}}}(<\Delta^N + C)}\Big|_{U_\Delta \cap U_C} \cong \widetilde{(R^{N,A_\infty}_{\mathrm{perfd}})^{\mathrm{c1}}}\Big|_{U_\Delta \cap U_C},$$

and $\widetilde{\phi}_C\big|_{U_\Delta \cap U_C} = \widetilde{\phi}_\Delta\big|_{U_\Delta \cap U_C} = \widetilde{(\phi^N)^{\mathrm{c1}}}\big|_{U_\Delta \cap U_C}$. In particular, $\widetilde{\phi}_C$ and $\widetilde{\phi}_\Delta$ glue to a splitting over the big open $U_\Delta \cup U_C$ so the result follows from Lemma 2.20 and Proposition 4.20. The compositions of the two splittings with $R^{N,A_\infty}_{\mathrm{perfd}} \to \left(R^{N,A_\infty}_{\mathrm{perfd}}\right)^{\mathrm{c1}}$ both equal ϕ^N. Since this map is an isomorphism in codimension at most one and $\mathcal{O}_{U_\Delta \cap U_C}$ is $\mathbf{S}_1$, the two splittings agree. They therefore glue to a splitting of the distinguished map over U. Since C and Δ^N have no common prime component, $\operatorname{Spec} R^N \setminus U$ has codimension at least two and thus $\Gamma(U, \mathcal{O}_U) = R^N$, Lemma 2.20 shows that

$$\rho_{R^{N,A_\infty}_{\mathrm{perfd}},\Delta^N + C} \colon R^N \longrightarrow M_{R^{N,A_\infty}_{\mathrm{perfd}}}(<\Delta^N + C)$$

is pure. The result follows from Proposition 4.20. □

5. Comparison with log canonical singularities

In this subsection, we show that a perfectoid pure pair (R, Δ) for which $K_R + \Delta$ is $\mathbb{Q}$-Cartier and R is deminormal is semi-log canonical. We briefly sketch the idea of the overall argument here to make it a bit easier to follow. After reducing to the complete local mixed characteristic setting, let $0 \neq r \in R$ be such that $(R[r^{-1}], \Delta R[r^{-1}])$ is log canonical. By Lemma 5.1, it is enough to show $r \in \mathscr{J}\left(R, \Delta + (1 - p^{-e}) \operatorname{div}_R(r)\right)$ for every $e \gg 0$. Furthermore, as the BCM test ideal τ_B of [MS21, Definitions 6.2 and 6.9] is contained in the multiplier ideal [MS21, Theorem 6.21], it suffices to show $r \in \tau_B(R, \Delta + (1 - p^{-e}) \operatorname{div}_R(r))$ for a sufficiently large perfectoid big Cohen–Macaulay R^+-algebra B. For carefully chosen $r = fg$ as in Notation 5.2, this is verified in two steps, proving the membership (Lemma 5.4) and containment (Proposition 5.5) in

$$r \in \operatorname{im}\left(\operatorname{Hom}_R\left((C \otimes_R R(-K_R))^{\mathrm{c1}}, R\right) \xrightarrow{\mathrm{ev}_{\widetilde{q}_e}} R\right) \subseteq \tau_B\left(R, \Delta + (1 - p^{-e}) \operatorname{div}_R(r)\right),$$

where C is an appropriately chosen absolutely integrally closed perfectoid R-algebra for which $\rho_{C,\Delta}$ is pure and $\widetilde{q}_e$ is to be understood as some $K_R + \Delta$ twist of $f^{1-p^{-e}} g^{1-p^{-2e}}$.

Lemma 5.1. *Let $X = \operatorname{Spec} R$ be an excellent normal scheme with a dualizing complex, and let Δ be an effective $\mathbb{Q}$-divisor on X such that $K_X + \Delta$ is $\mathbb{Q}$-Cartier. Suppose that there is a nonzerodivisor $r \in R$ such that for every proper birational morphism $\pi\colon Y \to X$ with Y normal,*

$$r \in \pi_* \mathscr{O}_Y \left(\lceil K_Y - \pi^*(K_X + \Delta) - (1-\varepsilon) \operatorname{div}_Y(r) \rceil \right) \tag{5.1.1}$$

for arbitrarily small $\varepsilon > 0$. Then (R, Δ) is log canonical.

Proof. For any such π, (5.1.1) is equivalent to

$$\begin{aligned} 1 &\in \pi_* \mathscr{O}_Y \left(\operatorname{div}_Y(r) + \lceil K_Y - \pi^*(K_X + \Delta) - (1-\varepsilon) \operatorname{div}_Y(r) \rceil \right) \\ &= \pi_* \mathscr{O}_Y \left(\lceil K_Y - \pi^*(K_X + \Delta) + \varepsilon \operatorname{div}_Y(r) \rceil \right). \end{aligned}$$

This could happen only if, at every prime divisor E on Y, the coefficient of $\lceil K_Y - \pi^*(K_X + \Delta) + \varepsilon \operatorname{div}_Y(r) \rceil$ at E is ≥ 0. Equivalently, the coefficient $b_{E,\varepsilon}$ of $K_Y - \pi^*(K_X + \Delta) + \varepsilon \operatorname{div}_Y(r)$ at E is > -1. Letting a_E be the coefficient of $K_Y - \pi^*(K_X + \Delta)$ at E, we want to show $a_E \geq -1$. If $a_E < -1$ then for any sufficiently small ε, $b_{E,\varepsilon} = a_E + \varepsilon \operatorname{ord}_E(r) < -1$, a contradiction. □

Notation 5.2. Let $(R, \mathfrak{m})$ be a complete normal local domain of mixed characteristic $(0, p)$, and let $\Delta \geq 0$ be a $\mathbb{Q}$-divisor. We fix an effective canonical divisor K_R and assume that $K_R + \Delta$ is $\mathbb{Q}$-Cartier.

- $\boxed{n, h, f}$**:** Choose $n > 0$ and $h \in R$ such that $\operatorname{div}_R(h) = n(K_R + \Delta)$, taking $n = 1$ and $h = 1$ if $K_R + \Delta = 0$. Choose a nonzero $f \in \mathfrak{m}$ such that $\operatorname{div}_R(f) = (K_R + \Delta)_{\mathrm{red}} + G$, where $G \geq 0$ has no component in common with K_R or Δ.
- $\boxed{A, g}$**:** Choose a Noether–Cohen normalization $A \subseteq R$ and $0 \neq g \in \mathfrak{m} \cap A$ such that $A[g^{-1}] \to R[g^{-1}]$ is finite étale, and p, n, f, h are units in $R[g^{-1}]$. Form $R^{A_\infty}_{\mathrm{perfd}}$ as in Notation 2.7.
- $\boxed{C, y}$**:** Choose a p-completely faithfully flat map $R^{A_\infty}_{\mathrm{perfd}} \to C$, where C is absolutely integrally closed and perfectoid, and choose $y \in C$ with $y^n = h$. For every $e \geq 1$, put $q_e = y f^{1-p^{-e}} g^{1-p^{-2e}} \in C$. Since C is absolutely integrally closed, it admits a limiting sequence for Δ by Construction 4.17 so we can define $M_C(<\Delta)$ with distinguished map $\rho_{C,\Delta}\colon R \to M_C(<\Delta)$.

Lemma 5.3. *Let C and R be as in Notation 5.2. Then every nonzero element of R is a nonzerodivisor on C. In particular, $(C \otimes_R R(-K_R))^{\mathrm{c1}}$ is isomorphic to its image in C^{c1}, $(R(-K_R)C)^{\mathrm{c1}}$.*

Proof. Let $0 \neq r \in R$ be arbitrary. By Lemma 2.8, there exists an integer $N \geq 0$ such that

$$(p^n R^{A_\infty}_{\mathrm{perfd}} :_{R^{A_\infty}_{\mathrm{perfd}}} r) \subseteq p^{n-N} R^{A_\infty}_{\mathrm{perfd}}$$

for every $n > N$. Since $R^{A_\infty}_{\mathrm{perfd}} \to C$ is faithfully flat modulo p^n, we get

$$(p^n C :_C r) \subseteq p^{n-N} C.$$

In particular, if $rz = 0$ in C, then $z \in \bigcap_n p^{n-N} C = 0$ so C is r-torsion free. To show the second part of the lemma, it suffices to show that $(C \otimes_R R(-K_R))^{\mathrm{c1}} \to C^{\mathrm{c1}}$ is injective. The proof of Proposition 2.12(d) shows that it is enough to check that this map is injective at every height ≤ 1 prime of R. Let $P \in \operatorname{Spec} R$ be any such prime and let $r \in R$ be such that $R(-K_R)_P \cong rR_P$. Then, $(C \otimes_R R(-K_R))_P \cong C \otimes_R rR_P$, which injects into C_P since C is R-torsion free. □

Lemma 5.4. *In the setting of Notation 5.2, for every $e \geq 1$, there are C-linear maps*

$$M_C(<\Delta) \xrightarrow{\theta_e} (R(-K_R)C)^{\mathrm{c1}} \xrightarrow{\beta_e} M_C(<\Delta)$$

such that $\theta_e(\mathbf{1}_{<\Delta}) = yf^{1-1/p^e}g^{1-1/p^{2e}}$ and $\beta_e\theta_e = fg\,\mathrm{id}$. In particular, if (R,Δ) is perfectoid pure and if we set $\widetilde{q_e}$ to be the element corresponding to $yf^{1-1/p^e}g^{1-1/p^{2e}}$ in $(C \otimes_R R(-K_R))^{\mathrm{c1}} \cong (R(-K_R)C)^{\mathrm{c1}}$, then

$$fg \in \operatorname{im}\left(\operatorname{Hom}_R\big((C \otimes_R R(-K_R))^{\mathrm{c1}}, R\big) \xrightarrow{\mathrm{ev}_{\widetilde{q_e}}} R\right).$$

Proof. Work with $S_i = R[x_i]/(x_i^{mp^i} - f)$ as in Construction 4.17, using the f chosen in Notation 5.2 and m so that $m\Delta$ has integer coefficients.

Define $\theta_{e,i}\colon M_C(\mathfrak{d}_i)^{\mathrm{c1}} \to (R(-K_R)C)^{\mathrm{c1}}$ by $\alpha \otimes t \mapsto \alpha t y f^{1-1/p^e} g^{1-1/p^{2e}}$. We need to check that the image lands in $(R(-K_R)C)^{\mathrm{c1}}$, which can be done locally at any prime of height ≤ 1 of R. Let P be such a prime. If P is not in the support of $K_R + \Delta$ then both $M_C(\mathfrak{d}_i)_P$ and $(R(-K_R)C)_P$ are isomorphic to C_P so there is nothing to check. Suppose now that P is in the support of $K_R + \Delta$ and write k, δ, s for the respective coefficients of $K_R, \Delta, \operatorname{div}_R(g)$ at P. Note that the coefficient of f at P is 1. Then, Construction 4.17 applied with our choice of f gives $(S_i(-K_R)C)_P = C_P x_i^{mp^i k}$ and $M_C(\mathfrak{d}_i)_P = C \otimes_{S_i} S_{i,P} x_i^{-(\delta mp^i - \varepsilon_P)}$ where $\varepsilon_P = 1$ if P is in the support of Δ and $\varepsilon_P = 0$ if not, in which case $\delta = 0$ so that $M_C(\mathfrak{d}_i)_P = C_P$. Abusing notation and denoting by x_i its image in C, we claim that there is $c_i \in C_P$ with

$$yf^{1-1/p^e}g^{1-1/p^{2e}} = c_i x_i^{\delta mp^i - \varepsilon_P + mp^i k}. \tag{5.4.1}$$

Then, $\theta_{e,i}(\alpha \otimes t) = c_i \alpha t x_i^{\delta mp^i - \varepsilon_P + mp^i k}$ which must be in $R(-K_R)C_P$ so $\theta_{e,i}$ has the correct target. Using that $\operatorname{div} h = n(K_R + \Delta)$, we have

$$(yf^{1-1/p^e}g^{1-1/p^{2e}})^{np^{2e}} = (u_h x_i^{mp^i(k+\delta)} u_f x_i^{mp^i(1-1/p^e)} u_g x_i^{mp^i s(1-1/p^{2e})})^{np^{2e}} = u x_i^b$$

for some integer $b > (\delta mp^i - \varepsilon_P + mp^i k)(np^{2e})$ and units $u,\ u_h,\ u_f,\ u_g$. Applying Lemma 2.5 (c) gives a unique $c_i \in \sqrt{x_i^{\delta mp^i - \varepsilon_P + mp^i k} C_P}$ that satisfies (5.4.1), as desired. Note that $\theta_{e,i}$ behaves well with the transition maps $M_C(\mathfrak{d}_i) \to M_C(\mathfrak{d}_j)$ so we can go to the colimit to define θ_e.

We proceed similarly and construct $\beta_{e,i}\colon (R(-K_R)C)^{\mathrm{c1}} \to M_C(\mathfrak{d}_i)^{\mathrm{c1}}$ by $\alpha \mapsto \alpha y^{-1} f^{1/p^e} g^{1/p^{2e}} \otimes 1$ assuming $i \gg e$. We need to show that the image indeed lands in $M_C(\mathfrak{d}_i)^{\mathrm{c1}}$, which can be done locally at any prime of height ≤ 1 of R. Let P be such a prime. If P is not in the support of $K_R + \Delta$ then there is nothing to check. Else, we claim that there is $d_i \in C_P$ with $x_i^{mp^i(k+\delta) - \varepsilon_P} f^{1/p^e} g^{1/p^{2e}} = d_i y$. This will allow us to rewrite our morphism as $\alpha \mapsto \alpha d_i x_i^{-(mp^i k)} \otimes x_i^{-(mp^i \delta - \varepsilon_P)}$ whose image is clearly in $M_C(\mathfrak{d}_i)_P$. As before, raising everything to the np^{2e} gives

$$(x_i^{mp^i(k+\delta)-\varepsilon_P} f^{1/p^e} g^{1/p^{2e}})^{np^{2e}} = (x_i^{mp^i(k+\delta)-\varepsilon_P} u_f x_i^{mp^i/p^e} u_g x_i^{mp^i s/p^{2e}})^{np^{2e}} = u x_i^b$$

for some $b > mp^i(k+\delta)np^{2e}$ since $i \gg e$. Similarly, $y^{np^{2e}} = u_h x_i^{mp^i(k+\delta)np^{2e}}$ so the existence of d_i follows from Lemma 2.5 (c). Again, this behaves well with the transition maps $M_C(\mathfrak{d}_i) \to M_C(\mathfrak{d}_j)$ so we can pass to the colimit to define β_e.

For the final statement, suppose that (R,Δ) is perfectoid pure. Consider $A_1 = A[\![X]\!] \xrightarrow{X \mapsto f} R$, and set $C' = \left(C \otimes_{R^{A_\infty}_{\mathrm{perfd}}} R^{(A_1)_\infty}_{\mathrm{perfd}}\right)^{\wedge p}$ which is perfectoid by [ČS24, Proposition 2.1.11(b)] and

$R^{(A_1)_\infty}_{\mathrm{perfd}} \to C'$ remains p-completely faithfully flat since $R^{A_\infty}_{\mathrm{perfd}} \to C$ is. By Proposition 4.32, $\rho_{R^{(A_1)_\infty}_{\mathrm{perfd}},\Delta}$ is pure, hence so is $\rho_{C',\Delta}$ by Lemma 4.5, and so $\rho_{C,\Delta}$ is pure by Proposition 4.24. As R is complete, we have an R-linear map $\psi\colon M_C(<\Delta) \to R$ sending $\mathbf{1}_{<\Delta} \mapsto 1$. Hence $fg = \psi\beta_e(\theta_e(\mathbf{1}_{<\Delta}))$. But now we are done thanks to the isomorphism $(C \otimes_R R(-K_R))^{\mathrm{c1}} \cong (R(-K_R)C)^{\mathrm{c1}}$. □

Proposition 5.5. *In the setting of Notation 5.2, suppose in addition that (R,Δ) is perfectoid pure. Then, for every $e \geq 1$ and every integral perfectoid balanced big Cohen–Macaulay R^+-algebra B,*

$$fg \in \tau_B\left(R, \Delta + (1-p^{-e})\operatorname{div}_R(fg)\right).$$

Proof. Fix e and. Using Lemma 5.4 and its notation, we want to show that

$$\operatorname{im}\left(\operatorname{Hom}_R\big((C\otimes_R R(-K_R))^{\mathrm{c1}}, R\big) \xrightarrow{\mathrm{ev}_{\tilde{q}_e}} R\right) \subseteq \tau_B\big(R, \Delta + (1-p^{-e})\operatorname{div}_R(fg)\big).$$

By [MS21, Definitions 6.2, 6.9] and Lemma 2.19, it suffices to show

$$\ker\left(H^d_{\mathfrak{m}}(R) \xrightarrow{\cdot q_e^+} H^d_{\mathfrak{m}}(B)\right) \subseteq \ker\left(H^d_{\mathfrak{m}}(R) \xrightarrow{\cdot q_e} H^d_{\mathfrak{m}}(C)\right), \tag{5.5.1}$$

for some specific choice of $q_e^+ \in B$ with $(q_e^+)^{np^{2e}} = h^{p^{2e}}((fg)^{1-1/p^e})^{np^{2e}}$. Choose $y_+ \in R^+$ with $y_+^n = h$ and $u_e, v_{2e} \in R^+$ with $u_e^{p^e} = f$ and $v_{2e}^{p^{2e}} = g$, and put $q_e^+ = y_+ u_e^{p^e-1} v_{2e}^{p^{2e}-p^e} \in R^+$. Set

$$T_e = R[U,V,Y]/(U^{p^e} - f,\ V^{p^{2e}} - g,\ Y^n - h)$$

and put $a_e = YU^{p^e-1}V^{p^{2e}-p^e}$ and $b_e = YU^{p^e-1}V^{p^{2e}-1}$. Mapping U,V,Y to $f^{1/p^e}, g^{1/p^{2e}}, y$ in C sends b_e to q_e, while mapping them to $u_e, v_{2e}, y_+ \in R^+ \subseteq B$ sends a_e to q_e^+. By construction, the finite free R-algebra T_e is complete local with residue field[5] $R/\mathfrak{m}$ and $A[g^{-1}] \to T_e[g^{-1}]$ is finite étale (this is where we use that the fraction field has characteristic 0). Thus Proposition 2.9 applies to $A \to T_e$.

Let $\eta \in H^d_{\mathfrak{m}}(R)$, and suppose that $q_e^+\eta = 0$ in $H^d_{\mathfrak{m}}(B)$. To show (5.5.1), we need to show $q_e\eta = 0$ in $H^d_{\mathfrak{m}}(C)$. We first apply Proposition 2.9 to $N = T_e \otimes_R H^d_{\mathfrak{m}}(R)$ and $a_e \otimes \eta$ to show that $(gA_\infty)_{\mathrm{perfd}}(1 \otimes (a_e \otimes \eta)) = 0$ in $(T_e)^{A_\infty}_{\mathrm{perfd}} \otimes_{T_e} N$. For every minimal prime P_i of T_e, finite freeness gives $P_i \cap R = (0)$, so $R_i = T_e/P_i$ is a complete local domain and a finite extension of R. Choose an R-embedding $R_i \to R^+$. Then $R_i^+ = R^+$, and B is balanced big Cohen–Macaulay over R_i by [Die07, Corollary 4.6]. The images of U,V,Y differ from u_e, v_{2e}, y_+ by roots of unity, so under the identification $B \otimes_{T_e} N \cong H^d_{\mathfrak{m}}(B)$, the image of $a_e \otimes \eta$ is a unit multiple of $q_e^+\eta$, and is therefore zero. This verifies the componentwise hypothesis of Proposition 2.9 so $(gA_\infty)_{\mathrm{perfd}}(1 \otimes (a_e \otimes \eta)) = 0$ in $(T_e)^{A_\infty}_{\mathrm{perfd}} \otimes_{T_e} N$.

The maps $A_\infty \to C$ and $T_e \to C$ together with the universal property of perfectoidization induce $(T_e)^{A_\infty}_{\mathrm{perfd}} \to C$ so $(gA_\infty)_{\mathrm{perfd}}C$ annihilates $a_e\eta$ in $H^d_{\mathfrak{m}}(C)$. By [CLM+26, Lemma 2.19], the p-adic closure of this ideal is $(gC)_{\mathrm{perfd}}$. Since $a_e\eta$ is killed by a power of p, this closure also annihilates it, and Corollary 2.6, applied to the cyclic R-module generated by $a_e\eta$, shows that $(g^{1/p^\infty})C$ annihilates $a_e\eta$. In particular, $b_e\eta = V^{p^e-1}a_e\eta = 0$, and hence $q_e\eta = 0$ in $H^d_{\mathfrak{m}}(C)$, as desired. □

[5] this follows from Notation 5.2 because $g, f \in \mathfrak{m}$ and if $h \notin \mathfrak{m}$ then $h = 1$ and $n = 1$

The next theorem generalizes [BMP+24, Corollary 5.11] to the pair setting and removes the index not divisible by p assumption in op.cit.

Theorem 5.6. *Let R be an excellent normal ring admitting a dualizing complex, with p in its Jacobson radical. Let Δ be an effective $\mathbb{Q}$-divisor on R such that $K_R + \Delta$ is $\mathbb{Q}$-Cartier. If (R, Δ) is perfectoid pure, then it is log canonical.*

Proof. By Proposition 4.29 and Proposition 4.30, we may pass to the completion at each maximal ideal. Log canonicity is local and descends from completion: base changing any proper birational modification along the faithfully flat regular map $R \to \widehat{R}$ preserves its discrepancy divisor. We may therefore assume that R is complete local.

If R has characteristic p, we use Hochster's Γ-construction as developed in [Mur21, Construction 3.1, Lemma 3.2, and Theorem 3.4]. Write $\Delta = \sum_i a_i P_i$ and choose Γ sufficiently small simultaneously for R and the finitely many R/P_i. Then R^Γ is F-finite and normal and each $P_i R^\Gamma$ is prime. Thus $\Delta^\Gamma = \sum_i a_i P_i R^\Gamma$ is the flat pullback of Δ. By Proposition 4.31 and [CRST21, Lemma 3.5], $(R^\Gamma, \Delta^\Gamma)$ is F-pure, and hence log canonical by [HW02, Proposition 2.2(4) and Theorem 3.3]. If (R, Δ) were not log canonical, choose a normal proper birational map $\mu\colon Y \to \operatorname{Spec} R$ and a prime divisor E on Y with discrepancy less than -1, and shrink Γ further so that the construction also applies to Y and E. Then Y^Γ is normal and E^Γ is a reduced prime divisor by the simultaneous construction used in [Mur21, Corollary 3.7(ii) and the proof of Theorem 5.6]. Normalized dualizing complexes commute with $R \to R^\Gamma$ and this flat proper base change [Sta26, Tags 0AU3, 0AWD, and 0AA8], while reducedness of E^Γ shows that the induced local map of DVRs at the generic points sends a uniformizer to a uniformizer. Consequently, writing $\mu^\Gamma\colon Y^\Gamma \to \operatorname{Spec} R^\Gamma$ for the base change, one has

$$\operatorname{coeff}_{E^\Gamma}\big(K_{Y^\Gamma} - (\mu^\Gamma)^*(K_{R^\Gamma} + \Delta^\Gamma)\big) = \operatorname{coeff}_E(K_Y - \mu^*(K_R + \Delta)) < -1,$$

contradicting log canonicity of $(R^\Gamma, \Delta^\Gamma)$.

If R has mixed characteristic $(0, p)$, we work in Notation 5.2 and put $X = \operatorname{Spec} R$ and $r = fg$. Fix $e \geq 1$ and a proper birational map $\pi\colon Y \to X$ with Y normal. For a sufficiently large integral perfectoid balanced big Cohen–Macaulay R^+-algebra B, Proposition 5.5 and [MS21, Theorem 6.21] give

$$r \in \tau_B\big(R, \Delta + (1 - p^{-e}) \operatorname{div}_R(r)\big) \subseteq \pi_* \mathscr{O}_Y\left(\left\lceil K_Y - \pi^*\big(K_X + \Delta + (1 - p^{-e}) \operatorname{div}_R(r)\big)\right\rceil\right).$$

Letting $e \to \infty$, the result follows from Lemma 5.1. □

Remark 5.7. Note that one can run the proof [Fay25, Theorem 4.12] using Theorem 5.6. This shows that any log canonical center of a $\mathbb{Q}$-Gorenstein normal complete local perfectoid pure ring of residue characteristic $p > 0$ is a center of perfectoid purity in the sense of [Fay25].

Corollary 5.8. *Let R be an excellent demi-normal ring with p in its Jacobson radical and admitting a dualizing complex, and let Δ be an effective Mumford $\mathbb{Q}$-divisor on R such that $K_R + \Delta$ is $\mathbb{Q}$-Cartier. If (R, Δ) is perfectoid pure, then it is semi-log canonical.*

Proof. Let R^N, C, and Δ^N be as in Proposition 4.33 (b). By that proposition, $(R^N, \Delta^N + C)$ is perfectoid pure. For compatible choices of canonical divisors, $K_{R^N} + \Delta^N + C$ is the pullback of $K_R + \Delta$ under the normalization, and hence is $\mathbb{Q}$-Cartier. Thus Theorem 5.6 shows

that $(R^N, \Delta^N + C)$ is log canonical. By the normalization characterization of semi-log canonicity [Kol13, Definition–Lemma 5.10; see also the assumptions on page 188], (R, Δ) is semi-log canonical. □

6. Adjunction for divisor pairs

In this section, we prove a version of adjunction for divisor pairs. We begin by noting that a weak version of adjunction already follows from Corollary 4.27.

Proposition 6.1. *Let R be an excellent $\mathbf{S}_2$ ring of residue characteristic $p > 0$, and let D be a reduced effective Mumford divisor on $\operatorname{Spec} R$. If (R, D) is perfectoid pure, then R/I_D is perfectoid pure.*

Proof. Without loss of generality, we may assume that R is complete and local by Proposition 4.29 and Proposition 4.30. By Proposition 4.20, there is an absolutely integrally closed B with $R \to B \to M_B(<D)$ split, say by $\widetilde{\phi}$. By Corollary 4.27 I_D is ϕ-compatible so the result follows by Proposition 2.30. □

For the rest of the section, we aim to incorporate a boundary and the corresponding different. We work in the following setup.

Setup 6.2. Let R be an excellent equidimensional demi-normal ring with p in its Jacobson radical. In particular, R is reduced, $\mathbf{S}_2$, and $\mathbf{G}_1$. Suppose that R admits a dualizing complex $\omega_R^\bullet$ and let ω_R be the canonical module associated to it. Let D be a reduced effective Mumford divisor on R and let $i\colon D = V(I_D) \hookrightarrow \operatorname{Spec} R$. Assume that $S := R/I_D$ is deminormal. Write $\omega_S := \operatorname{Ext}^1_R(S, \omega_R^\bullet[-d])$, which is a canonical sheaf for S by [Sta26, Tag 0AX1]. Let Δ be an effective Mumford $\mathbb{Q}$-divisor on X having no prime component in common with D. Since R is excellent, it has at most finitely many singular primes of height 1 so we can fix a canonical divisor $K_R = -D + G$ for some effective integral Mumford divisor G having no common component with D. Use this choice of K_R to fix an embedding $\omega_R \subseteq K(R)$. Assume that $K_R + D + \Delta$ is $\mathbb{Q}$-Cartier at every height 1-prime of S and that $(R, D + \Delta)$ is perfectoid pure.

Lemma 6.3 (*cf.* [Kol13, Definition 4.1])**.** *Work in Setup 6.2 and let $\omega_R(D) := \operatorname{Hom}_R(I_D, \omega_R) = R(K_R + D)$. There is an exact sequence of R-modules*

$$0 \longrightarrow \omega_R \longrightarrow \omega_R(D) \longrightarrow \omega_S.$$

Furthermore, the map $\omega_R(D) \to \omega_S$ is surjective at any prime of height ≤ 2 in the support of D. In other words, viewing $\omega_R(D)/\omega_R$ as an S-module, we have an isomorphism

$$(\omega_R(D)/\omega_R)^{\mathrm{c1}} \xrightarrow{\sim} \omega_S.$$

Proof. Since ω_R is $\mathbf{S}_2$ and I_D has height 1, we have $\operatorname{depth}_{I_D} \omega_R = 1$ and thus $\operatorname{Hom}_R(S, \omega_R) = 0$. Applying $\operatorname{Hom}_R(-, \omega_R)$ to

$$0 \longrightarrow I_D \longrightarrow R \longrightarrow S \longrightarrow 0$$

gives

$$0 \longrightarrow \omega_R \longrightarrow \omega_R(D) \xrightarrow{\delta} \operatorname{Ext}^1_R(S, \omega_R) \longrightarrow 0,$$

Let $\mathrm{CM}(-)$ denote the open subset on which the given ring is Cohen–Macaulay and note that $U \coloneqq \mathrm{CM}(S) \cap i^{-1}\,\mathrm{CM}(R)$ is a big open subset of $\operatorname{Spec} S$ since R is $\mathbf{S}_2$. On U, [Kol23, Definition 11.13 and (11.13.1)] gives an isomorphism $\operatorname{Ext}^1_R(S,\omega_R)|_U \cong \omega_S|_U$. Since ω_S is $\mathbf{S}_2$, it has the codimension 2 extension property so we obtain an isomorphism $(\operatorname{Ext}^1_R(S,\omega_R))^{\mathrm{c1}} \cong \omega_S$. □

Lemma 6.4. *The hypotheses of* [Kol13, Definition 4.2] *are satisfied on S. In particular, we can define the different, $\Delta_D = \operatorname{Diff}_D(\Delta)$, which is an effective Mumford $\mathbb{Q}$-divisor on S. Moreover, for any $m \in \mathbb{N}_{>0}$ for which $m(K_R + D + \Delta)$ is Cartier at every height 1 prime of S, there is a canonical isomorphism*

$$\left(\omega_R^{[m]}(mD + m\Delta) \otimes_R S\right)^{\mathrm{c1},S} \xrightarrow{\sim} \omega_S^{[m]}(m\Delta_D),$$

where $\omega_R^{[m]}(mD + m\Delta) = R(m(K_R + D + \Delta))$.

Proof. The fact that the hypotheses of [Kol13, Definition 4.2] hold is direct from Setup 6.2: since S is $\mathbf{G}_1$, one can take $S' = S$ in [Kol13, Definition 4.2]. The second part of the statement is in the proof of [Kol13, Definition 4.2]: following the notation in that proof, S_0 is a big open of S. □

Theorem 6.5 (Adjunction for perfectoid pure pairs)**.** *Keeping the notation of Lemma 6.4, (S, Δ_D) is perfectoid pure.*

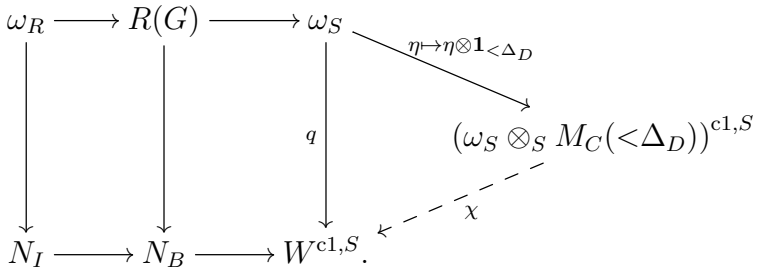
Proof. By Proposition 4.29, we can assume that R and S are local, say with maximal ideals $\mathfrak{m}$ and $\mathfrak{n}$, respectively. Let $I = I_D$ and $d = \dim R$. The case $d \leq 1$ is immediate, so assume $d \geq 2$. Since $\omega_R(D) = R(G)$, Lemma 6.3 gives an isomorphism $(R(G)/\omega_R)^{\mathrm{c1},S} \cong \omega_S$. Let B be an absolutely integrally closed perfectoid R-algebra such that $\rho_{B,D+\Delta}$ is pure. Define

$$\begin{aligned} N_I &= \left(((IB)_{\mathrm{perfd}} \otimes_R R(G)) \otimes_B M_B(<\Delta)\right)^{\mathrm{c1}}, \\ N_B &= \left((B \otimes_R R(G)) \otimes_B M_B(<\Delta)\right)^{\mathrm{c1}}, \\ L &= (\operatorname{im}(N_I \longrightarrow N_B))^{\mathrm{c1}} \subseteq N_B, \\ W &= N_B/L, \end{aligned}$$

where the codimension-1 envelopes are taken over R. On any common big open used to define N_I and N_B, $(IB)_{\mathrm{perfd}} N_B \subseteq \operatorname{im}(N_I \to N_B)$ so $(IB)_{\mathrm{perfd}} W = 0$. Thus W and $W^{\mathrm{c1},S}$ are naturally $C \coloneqq B/(IB)_{\mathrm{perfd}}$-modules. Define a map $R(G) \to N_B$ by sending $\gamma \in R(G)$ to $1 \otimes \gamma \otimes \mathbf{1}_{<\Delta}$ and taking the codimension-1 envelope. The restriction of this map to $\omega_R = (IR(G))^{\mathrm{c1}}$ has image in L. In particular, it defines a morphism $R(G)/\omega_R \to W$ which, after using $(R(G)/\omega_R)^{\mathrm{c1},S} \simeq \omega_S$, yields $q\colon \omega_S \longrightarrow W^{\mathrm{c1},S}$.

To sum up, we have a commutative diagram of solid arrows

(6.5.1)



Proposition 6.6. *The dashed arrow in the diagram exists. More precisely, there is a C-linear map*

$$\chi\colon\ (\omega_S \otimes_S M_C(<\Delta_D))^{\mathrm{c1},S} \longrightarrow W^{\mathrm{c1},S}$$

which makes the triangle on the right of the preceding diagram commute.

Proof. We construct this map locally around every height 1 point of S before showing that it glues. For an arbitrary $P \in \mathrm{Supp}(\Delta_D)$, S_P is a DVR since Δ_D is Mumford. Moreover, since S is $\mathbf{G}_1$, $\omega_{S,P}$ is free of rank 1. In a sufficiently small neighborhood of P, prime avoidance allows us to find a generator κ of ω_S along with a lift r of κ in $R(G)\cap K(R)^\times$ such that $G_r \coloneqq G+\mathrm{div}_R(r) \geq 0$ has no component in common with D and is Mumford. Note that $R(G_r) = r^{-1}R(G)$ and that $G_r + \Delta = K_R + D + \Delta + \mathrm{div}_R(r)$ is $\mathbb{Q}$-Cartier. Choose a sufficiently divisible $n > 0$ and $h \in R$ such that $n(G_r+\Delta) = \mathrm{div}_R(h)$ sufficiently close to P. The image $\overline{h} \in S$ is a nonzerodivisor and Lemma 6.4 gives $n\Delta_D = \mathrm{div}_S(\overline{h})$. Fix an n-th root z of h in B. By Remark 4.25 and Lemma 4.19, on that neighborhood of P, there are pointed isomorphisms $M_B(< G_r + \Delta) \cong (z^{1/p^\infty}B)^{\mathrm{c1}}$, $\mathbf{1}_{<G_r+\Delta} \mapsto z$ and $M_C(<\Delta_D) \cong (\overline{z}^{1/p^\infty}C)^{\mathrm{c1},S}$, $\mathbf{1}_{<\Delta_D} \mapsto \overline{z}$. By Lemma 2.5 (b), there is a pointed isomorphism $C \otimes_B (z^{1/p^\infty}B) \cong (\overline{z}^{1/p^\infty}C)$. Recall from Proposition 4.21(b) that there is a canonical pointed B-linear map

$$\alpha_r \coloneqq \alpha_{G_r,\Delta}\colon M_B(<(G_r+\Delta)) \longrightarrow ((B \otimes_R R(G_r)) \otimes_B M_B(<\Delta))^{\mathrm{c1}}\,. \tag{6.6.1}$$

Composing it with multiplication by r and quotienting by L gives a B-linear map $\beta_r\colon M_B(<(G_r+\Delta)) \to N_B \to W$. Since $(IB)_{\mathrm{perfd}}W = 0$, the composition

$$(z^{1/p^\infty}B) \to M_B(<(G_r+\Delta)) \to W$$

descends to $(\overline{z}^{1/p^\infty})C \to W$. Taking codimension-1 envelopes with respect to S gives a map $\psi_\kappa\colon M_C(<\Delta_D) \to W^{\mathrm{c1},S}$.

Chasing the maps around, we see that $\psi_\kappa(\mathbf{1}_{<\Delta_D}) = q(\kappa)$. We can finally define a C-linear map on the neighborhood of P

$$\chi_P\colon \omega_S \otimes_S M_C(<\Delta_D) \longrightarrow W^{\mathrm{c1},S}, \qquad \kappa \otimes m \longmapsto \psi_\kappa(m).$$

Note that for any $\eta = s\kappa$ in ω_S, $\chi_P(\eta \otimes \mathbf{1}_{<\Delta_D}) = s\psi_\kappa(\mathbf{1}_{<\Delta_D}) = sq(\kappa) = q(\eta)$. We want to show that it glues. Let $P_1, \ldots, P_m$ be the finitely many height-1 points in $\mathrm{Supp}(\Delta_D)$ and for each i, let $f_i \in (\bigcap_{j\neq i} P_j) \setminus P_i$ where the empty intersection is S. For any choice of i, we can pick $U_i \subset D(f_i)$ so χ_P as above can be defined on U_i. Let $U_0 = D \setminus \bigcup_i V(P_i)$ and note that $\bigcup_{m\geq i\geq 0} U_i$ is a big open in S. Moreover, since U_0 does not contain any prime in the support of Δ_D, one can define χ_0 on U_0 by $\eta \otimes c \mapsto cq(\eta)$.

By the choice of the f_i, no height-one point of $\mathrm{Supp}(\Delta_D)$ lies in an overlap. At every height-1 point of an overlap, the restrictions of χ_{P_i} and χ_{P_j} (or χ_0) must therefore agree so they glue over U, note that by construction, the transition maps are equalities. Taking codimension 1-envelopes finishes the proposition. □

We return to the proof of the theorem. Let $N_0 \coloneqq \left((\sqrt{IB} \otimes_R R(G)) \otimes_B M_B(<\Delta)\right)^{\mathrm{c1}}$. The pointed map

$$\rho_{B,D+\Delta}\colon R \longrightarrow M_B(<(D+\Delta)), \qquad 1 \longmapsto \mathbf{1}_{<(D+\Delta)},$$

was chosen to be pure. Tensoring this map with ω_R, using $(\omega_R \otimes_R R(D))^{c1} = R(G)$, Proposition 4.21, and passing to top local cohomology gives an injection

$$E_R \hookrightarrow H^d_{\mathfrak{m}}(N_0). \tag{6.6.2}$$

Since the generators of I have compatible p-power roots in B, the proof of Corollary 2.6, together with Corollary 2.4, gives, for every p^∞-torsion R-module T,

$$T \otimes_R \sqrt{IB} \xrightarrow{\sim} T \otimes_R (IB)_{\mathrm{perfd}}.$$

In particular, this is the case for $T = H^d_{\mathfrak{m}}(R) \otimes_R R(G)$. Apply the above isomorphism and tensoring with $M_B(<\Delta)$ over B gives an isomorphism

$$H^d_{\mathfrak{m}}\Big((\sqrt{IB} \otimes_R R(G)) \otimes_B M_B(<\Delta)\Big) \cong H^d_{\mathfrak{m}}(((IB)_{\mathrm{perfd}} \otimes_R R(G)) \otimes_B M_B(<\Delta)).$$

Taking the codimension-one envelopes does not change top local cohomology so

$$H^d_{\mathfrak{m}}(N_0) \xrightarrow{\sim} H^d_{\mathfrak{m}}(N_I).$$

Note that for any minimal prime $\mathfrak{p} \in \operatorname{Spec} R$, $IR_{\mathfrak{p}} = R_{\mathfrak{p}}$ so $\ker(N_I \to L)$ has support of dimension at most $d-1$. On the other hand, $\operatorname{coker}(N_I \to L)$ has support of dimension at most $d-2$ hence $H^d_{\mathfrak{m}}(N_I) \xrightarrow{\sim} H^d_{\mathfrak{m}}(L)$. Thus the injection in (6.6.2) can be rewritten as

$$a\colon E_R \hookrightarrow H^d_{\mathfrak{m}}(N_0) \xrightarrow{\sim} H^d_{\mathfrak{m}}(N_I) \xrightarrow{\sim} H^d_{\mathfrak{m}}(L).$$

With the same maps as in (6.5.1), the following diagram has exact rows and commutes

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \omega_R & \longrightarrow & R(G) & \longrightarrow & R(G)/\omega_R & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & L & \longrightarrow & N_B & \longrightarrow & W & \longrightarrow & 0. \end{array} \tag{6.6.3}$$

The codimension-1 envelope (over S) of the last vertical arrow is the map q constructed above. Since $K_R = -D + G$,

$$\operatorname{Hom}_R(R(G), \omega_R) = R(-D) = I, \qquad \operatorname{Hom}_R(\omega_R, \omega_R) = R.$$

It follows from duality that the natural map $H^d_{\mathfrak{m}}(\omega) \to H^d_{\mathfrak{m}}(R(G))$ can be identified with the Matlis dual of $I \hookrightarrow R$. On the other hand since $(R(G)/\omega_R)^{c1,S} \cong \omega_S$, invariance of top local cohomology under codimension-1 envelope and local duality give $H^{d-1}_{\mathfrak{m}}(R(G)/\omega_R) \cong E_S$. Consider the exact sequence

$$H^{d-1}_{\mathfrak{m}}(R(G)) \xrightarrow{\delta} H^{d-1}_{\mathfrak{m}}(R(G)/\omega_R) \to H^d_{\mathfrak{m}}(\omega_R) \to H^d_{\mathfrak{m}}(R(G)) \to 0.$$

We claim that $\delta = 0$, to see this, note that the Matlis dual of the sequence can be identified with $0 \to \hat{I} \to \hat{R} \to \hat{S} \xrightarrow{\delta^\vee} H^{d-1}_{\mathfrak{m}}(R(G))^\vee$. Since $R(G)$ is $\mathbf{S}_2$, $\dim(H^{d-1}_{\mathfrak{m}}(R(G))^\vee) \le d-3$ while the cokernel of the induced map $\hat{R}/\hat{I} \to \hat{S}$ (which is necessarily multiplication by some $r \in \hat{S}$), if nonzero, must have dimension $\ge d-2$ and thus cannot embed into $H^{d-1}_{\mathfrak{m}}(R(G))^\vee$. It follows that $\delta^\vee = 0$ and thus so is δ.

Applying local cohomology to (6.6.3) therefore gives

$$\begin{array}{ccc} E_S & \xhookrightarrow{\ j\ } & E_R \\ \downarrow{\scriptstyle \widetilde{q}} & & \downarrow{\scriptstyle a} \\ H^{d-1}_{\mathfrak{m}}(W^{\mathrm{c1},S}) & \xrightarrow[\partial]{} & H^d_{\mathfrak{m}}(L). \end{array}$$

The top and right arrows are injective; hence $\widetilde{q}$ is injective. The map χ of Proposition 6.6 factors $\widetilde{q}$ which gives

$$E_S \longrightarrow H^{d-1}_{\mathfrak{m}}\left((\omega_S \otimes_S M_C(<\Delta_D))^{\mathrm{c1},S}\right) \longrightarrow H^{d-1}_{\mathfrak{m}}(W^{\mathrm{c1},S}). \tag{6.6.4}$$

Again, by invariance of top local cohomology under codimension-1 envelopes,

$$H^{d-1}_{\mathfrak{m}}\left((\omega_S \otimes_S M_C(<\Delta_D))^{\mathrm{c1},S}\right) \cong H^{d-1}_{\mathfrak{m}}(\omega_S) \otimes_S M_C(<\Delta_D) \cong E_S \otimes_S M_C(<\Delta_D).$$

So the first arrow in (6.6.4) can be rewritten as

$$E_S \longrightarrow E_S \otimes_S M_C(<\Delta_D), \qquad u \longmapsto u \otimes \mathbf{1}_{<\Delta_D}$$

and is injective since $\widetilde{q}$ is injective. Now Lemma 2.24 shows that the pointed map

$$S \longrightarrow M_C(<\Delta_D), \qquad 1 \longmapsto \mathbf{1}_{<\Delta_D},$$

is pure so the result follows from Proposition 4.20. □

Finally we show that by imposing natural geometric hypotheses one can remove the assumptions on S from Theorem 6.5.

Theorem 6.7. *Let R be an equidimensional demi-normal ring essentially of finite type over an excellent DVR, with p in its Jacobson radical. Let D be a reduced effective Mumford divisor on $\operatorname{Spec}(R)$, and let Δ be an effective Mumford $\mathbb{Q}$-divisor having no prime component in common with D. Assume that $(R, D+\Delta)$ is perfectoid pure and that $K_R + D + \Delta$ is $\mathbb{Q}$-Cartier. Put $S = R/I_D$. Then S is at worst nodal in codimension one. Furthermore, if either S is $\mathbf{S}_2$ or D is $\mathbb{Q}$-Cartier, then S is demi-normal and (S, Δ_D) is perfectoid pure.*

Proof. Letting R^N be the normalization of R, and using Proposition 4.33 (b) and its notation, $(R^N, C + D^N + \Delta^N)$ is perfectoid pure. Put $E = C + D^N$ with its reduced divisor structure. Applying Proposition 6.1 at the maximal ideals shows that D and E are perfectoid pure. Thus R, D and E are weakly normal by [BMP+24, Corollary 4.30]. Note that Corollary 5.8 shows that $(R, D+\Delta)$ is semi-log canonical and so $(R^N, C + D^N + \Delta^N)$ is log canonical.

Fix a codimension-one point x of D. Let B be its strict henselization of R at x, and put $A = B/I_D B = \mathscr{O}^{\mathrm{sh}}_{D,x}$ and $k = B/\mathfrak{m}_B$. Weak normality is preserved by localization and this ind-étale base change [Ryd10, Proposition B.6]. Normalization commutes with strict henselization [Sta26, Tag 0CBM]; log canonicity is preserved as well, since it is étale local (see [Kol23, Section 2.16.1]). Write $B^N = \prod_{i=1}^r B_i$. The rings B_i are normal strictly henselian local surfaces, and their residue fields k_i are finite purely inseparable extensions of the separably closed field k.

We show that there are at most two branches of D at x. Let C_i and D_i^N be C and D^N on $\operatorname{Spec} B_i$, respectively, and let c_i and d_i be their respective numbers of branches. Since $C_i + D_i^N$ is the reduced boundary on a log canonical surface, it is nodal by [Kol13, Theorem 2.31], and since it is weakly normal, every node is separable [Pos25, Lemma 2.8], and hence split. Consequently $c_i + d_i \leq 2$.

Note that B is a Cohen-Macaulay local ring of dimension two, thus the punctured spectrum of B is connected. Since each B_i corresponds to a minimal prime $\mathfrak{p}_i$ of B, the Hochster-Huneke graph of B has at least $r-1$ edges. For an edge connecting $\mathfrak{p}_i$ and $\mathfrak{p}_j$, corresponding to a height 1 prime $\mathfrak{p}$ of B, we know that $B_{\mathfrak{p}}$ has exactly two minimal primes since B is deminormal (in particular, different edges correspond to different height 1 primes). It follows that $\mathfrak{p}$ gives a component of C_i in B_i and a component of C_j in B_j. Therefore the $r-1$ edges contributes to at least $2(r-1)$ components of various C_i's and thus $\sum_{i=1}^{r} c_i \geq 2(r-1)$. Hence we have

$$\sum_{i=1}^{r} d_i \leq 2r - \sum_{i=1}^{r} c_i \leq 2$$

as required.

Firstly, if there is one branch, the normalization $D^N \to D$ is a universal homeomorphism and so D is normal (hence regular) by weak normality. Suppose now there are two boundary branches, with normalization $A^N = T_1 \times T_2$ where T_i are DVRs. Since the node on B_i is separable and k is separably closed, the residue fields of the branches are a common finite purely inseparable extension L/k. The natural map

$$\operatorname{Spec}(T_1 \times_L T_2) \longrightarrow \operatorname{Spec} A$$

is a finite birational universal homeomorphism. Weak normality therefore gives $A = T_1 \times_L T_2$. Comparing residue fields yields $L = k$, so $A = T_1 \times_k T_2$.

It is a quotient of a regular ring by [Pos24, Lemma 3.3.1(c)], and so [Pos24, Lemma 3.2.1] shows that A is nodal. Finally [Pos24, Corollary 3.3.2] shows that the original D (before strict henselization) was also nodal.

It now remains to show that S is $\mathbf{S}_2$ in the case where D is $\mathbb{Q}$-Cartier. We may localize to assume that R is local. By Corollary 4.12, the pair (R, D) is perfectoid pure. Therefore, by Proposition 6.1, the underlying scheme D is perfectoid pure, and hence weakly normal by [BMP+24]. Since R is $\mathbf{S}_2$, it is topologically $\mathbf{S}_2$ [Kol16, Definition 6.8]. Choose $m > 0$ such that mD is Cartier. The connectedness theorem for Cartier divisors [Kol23, Lemma 2.93], applied étale-locally to mD, shows that mD is topologically $\mathbf{S}_2$. Since D and mD have the same underlying topological space, D is topologically $\mathbf{S}_2$ as well. Finally, a weakly normal scheme is $\mathbf{S}_2$ if and only if it is topologically $\mathbf{S}_2$ [Kol16, Definition 6.8], so D is $\mathbf{S}_2$. Since the maximal ideal $\mathfrak{n}$ is arbitrary, S is $\mathbf{S}_2$, and Theorem 6.5 applies. □

7. Perfectoid pure thresholds

7.1. **A threshold for perfectoid rings.** We present an analogue of F-threshold using perfectoid rings, in an approach similar to BCM thresholds defined in [RS25]. Compare also with [Yos25] as well as [HMTW08].

Definition 7.1. Let R be a ring and $\mathfrak{a}$ an ideal. For a positive rational number a/r we define

$$\overline{\mathfrak{a}^{a/r}} := \{x \in R \mid x^r \in \overline{\mathfrak{a}^a}\}.$$

It is easy to see from the equation of integral dependence that $\overline{\mathfrak{a}^{a/r}}$ does not depend on the presentation of the rational number a/r as the proof of [HS06, Proposition 10.5.2(1)] does not use that R is Noetherian.

Definition 7.2 ([RS25])**.** Let R be a ring and $I, \mathfrak{a}$ be ideals (we allow I to be the unit ideal). If $\mathfrak{a} \subseteq \sqrt{I}$, then we define[6] the *rational powers threshold* of $\mathfrak{a}$ with respect to I as

$$\mathrm{Pt}(\mathfrak{a}, I) = \sup\{r \in \mathbb{Q}_{\geq 0} \mid \overline{\mathfrak{a}^r} \not\subseteq I\}.$$

We state some elementary properties.

Lemma 7.3. *Let R be a ring and $\mathfrak{a}, \mathfrak{b}, I$ be ideals with $\mathfrak{a}, \mathfrak{b} \subseteq \sqrt{I}$. Then*

(a) if $\mathfrak{a} \subset \mathfrak{b}$ then $\mathrm{Pt}(\mathfrak{a}, I) \leq \mathrm{Pt}(\mathfrak{b}, I)$;

(b) if $\overline{\mathfrak{a}} = \overline{\mathfrak{b}}$ then $\mathrm{Pt}(\mathfrak{a}, I) = \mathrm{Pt}(\mathfrak{b}, I)$;

(c) $\mathrm{Pt}(\mathfrak{a}^n, I) = n^{-1}\mathrm{Pt}(\mathfrak{a}, I)$ for $n \in \mathbb{Z}_{\geq 1}$;

(d) if $I \subset J$ is an ideal, then $\mathrm{Pt}(\mathfrak{a}, I) \geq \mathrm{Pt}(\mathfrak{a}, J)$.

Proof. These properties are immediate from the definition. □

We will now switch to the case where R is perfectoid so that we benefit from the following Briançon–Skoda result that will allow us to replace rational powers by regular powers.

Proposition 7.4 ([MMRS25, Corollary 3.20])**.** *Let R be a perfectoid ring and let $I \subseteq R$ be an ideal generated by n elements. For every $k \geq 1$, one has $I_{\mathrm{perfd}}\overline{I^{n+k-1}} \subseteq I^k$ and $\overline{I^{n+k}} \subseteq I^k$.*

The above proposition provides the following characterization of the rational powers threshold in perfectoid rings.

Theorem 7.5. *Let R be a perfectoid ring and $x_1, \ldots, x_d$ be elements that admit compatible systems of p-power roots in R. Set $\mathfrak{a} = (x_1, \ldots, x_d)$. If I is an ideal such that $x_1, \ldots, x_d \in \sqrt{I}$ then*

$$\begin{aligned}
\mathrm{Pt}(\mathfrak{a}, I) &= \sup\left\{\frac{n}{p^e} \mid n, e \in \mathbb{Z}_{\geq 0}, \overline{(x_1^{1/p^e}, \ldots, x_d^{1/p^e})^n} \not\subseteq I\right\} \\
&= \sup\left\{\frac{|a|}{p^e} \mid a \in \mathbb{Z}^d_{\geq 0}, x_1^{a_1/p^e} \cdots x_d^{a_d/p^e} \notin I\right\} \\
&= \lim_{e \to \infty} p^{-e} \sup\left\{|a| \mid a \in \mathbb{Z}^d_{\geq 0}, x_1^{a_1/p^e} \cdots x_d^{a_d/p^e} \notin I\right\},
\end{aligned}$$

where $|a| = \sum_{i=1}^d a_i$.

Proof. We start by observing that

$$\sup\{r \in \mathbb{Q}_{\geq 0} \mid \overline{\mathfrak{a}^r} \not\subseteq I\} = \sup\left\{\frac{n}{p^e} \mid n, e \in \mathbb{Z}_{\geq 0}, \overline{\mathfrak{a}^{\frac{n}{p^e}}} \not\subseteq I\right\}.$$

Namely, the $\geq$ inequality is obvious and its converse holds because we may approximate from below any positive real r by a sequence of rational numbers of the form n/p^e, explicitly, by taking $n = \lfloor p^e r \rfloor$.

[6] We recall that $\sup \emptyset = -\infty$.

The first equality follows from $\overline{\mathfrak{a}^{n/p^e}} = \overline{(x_1^{1/p^e}, \dots, x_d^{1/p^e})^n}$. To prove the second equality, note that $\supseteq$ is obvious. The other inequality follows from Proposition 7.4 after noting that for every sufficiently large $e' \geq 0$

$$\overline{(x_1^{1/p^e}, \dots, x_d^{1/p^e})^n} = \overline{(x_1^{1/p^{e+e'}}, \dots, x_d^{1/p^{e+e'}})^{p^{e'}n}} \subseteq (x_1^{1/p^{e+e'}}, \dots, x_d^{1/p^{e+e'}})^{p^{e'}n-d}.$$

The last equality is obvious since the sequence

$$e \mapsto p^{-e}\left\{|a| \mid a \in \mathbb{Z}^d_{\geq 0}, x_1^{a_1/p^e} \cdots x_d^{a_d/p^e} \notin I\right\}$$

is non-decreasing. □

Corollary 7.6. *Let R be a perfectoid ring and $x_1, \dots, x_s$, $y_1, \dots, y_t$ be elements that admit compatible systems of p-power roots in R. Set $\mathfrak{a} = (x_1, \dots, x_s)$ and $\mathfrak{b} = (y_1, \dots, y_t)$. If $I \subseteq R$ is an ideal such that $\mathfrak{a} + \mathfrak{b} \subseteq \sqrt{I}$, then*

$$\mathrm{Pt}(\mathfrak{a} + \mathfrak{b}, I) \leq \mathrm{Pt}(\mathfrak{a}, I) + \mathrm{Pt}(\mathfrak{b}, I).$$

Proof. Consider a monomial of the form $x_1^{a_1/p^e} \cdots x_s^{a_s/p^e} y_1^{b_1/p^e} \cdots y_t^{b_t/p^e}$. If $\frac{|a|}{p^e} + \frac{|b|}{p^e} > \mathrm{Pt}(\mathfrak{a}, I) + \mathrm{Pt}(\mathfrak{b}, I)$, then either $\frac{|a|}{p^e} > \mathrm{Pt}(\mathfrak{a}, I)$ or $\frac{|b|}{p^e} > \mathrm{Pt}(\mathfrak{b}, I)$, hence, by Theorem 7.5,

$$x_1^{a_1/p^e} \cdots x_s^{a_s/p^e} y_1^{b_1/p^e} \cdots y_t^{b_t/p^e} \in I.$$

Applying Theorem 7.5 again we find that $\mathrm{Pt}(\mathfrak{a} + \mathfrak{b}, I) \leq \mathrm{Pt}(\mathfrak{a}, I) + \mathrm{Pt}(\mathfrak{b}, I)$ as wanted. □

Corollary 7.7. *Let $R \to S$ be a map of perfectoid rings and $\mathfrak{a} \subseteq \sqrt{I} \subseteq R$ be ideals. Suppose that $\mathfrak{a}$ is generated by finitely many elements admitting compatible systems of p-power roots. Then*

(a) $\mathrm{Pt}(\mathfrak{a}, I) \geq \mathrm{Pt}(\mathfrak{a}S, IS)$;

(b) if $J \subseteq S$ is an ideal such that $I = J \cap R$ then $\mathrm{Pt}(\mathfrak{a}, I) = \mathrm{Pt}(\mathfrak{a}S, J)$.

Proof. Both properties are immediate from Theorem 7.5. □

Proposition 7.8 (Reduction to principal ideals)**.** *Let R be a perfectoid ring and $\mathfrak{a} = (f_1, \dots, f_n)$ an ideal generated by elements f_i admitting compatible systems of p-power roots. If I is an ideal such that $\mathfrak{a} \subseteq \sqrt{I}$, then*

$$\mathrm{Pt}(\mathfrak{a}, I) = \sup\left\{|\alpha|\mathrm{Pt}((f_1^{\alpha_1} \cdots f_n^{\alpha_n}), I) \mid 0 \neq \alpha \in \mathbb{Z}^n_{\geq 0}\right\}.$$

Proof. Let $\mathcal{M}_k$ be the set of degree-k monomials on the f_i. For any $f \in \mathcal{M}_k$

$$k\mathrm{Pt}((f), I) \leq k\mathrm{Pt}(\mathfrak{a}^k, I) = \mathrm{Pt}(\mathfrak{a}, I),$$

since $f \in \mathfrak{a}^k$. The opposite inequality follows from Theorem 7.5 and the definition of Pt:

$$\mathrm{Pt}(\mathfrak{a}, I) = \sup\left\{\frac{k}{p^e} \mid f \in \mathcal{M}_k, f^{1/p^e} \notin I\right\} \leq \sup\left\{k\mathrm{Pt}((f), I) \mid f \in \mathcal{M}_k\right\}.$$

□

7.2. **Perfectoid pure thresholds.** In this subsection we introduce perfectoid thresholds and prove some of their basic properties.

Definition 7.9. Let $(R,\mathfrak{m})$ be a Noetherian local ring of residual characteristic $p>0$ and $\mathfrak{a}$ be a proper ideal of R. If B is a perfectoid R-algebra such that $\mathfrak{a}B=(x_1,\ldots,x_g)$ where the elements $x_1,\ldots,x_g$ have compatible systems of p-power roots, then we will call $\operatorname{ppt}(\mathfrak{a};B)\coloneqq \operatorname{Pt}(\mathfrak{a}B,\mathcal{I}(B))$ the *perfectoid pure threshold of* $\mathfrak{a}$ with respect to B.

We set the *perfectoid pure threshold* of $\mathfrak{a}$, $\operatorname{ppt}(\mathfrak{a})$, to be $\sup\{\operatorname{ppt}(\mathfrak{a};B)\}$, where B ranges through all perfectoid R-algebras such that $\mathfrak{a}B$ can be generated by elements that have compatible systems of p-power roots. By Proposition 2.2 the latter condition is automatic if $p\in\mathfrak{a}$ (e.g., when R is local and $\mathfrak{a}$ is the maximal ideal), or when B is absolutely integrally closed.

Definition 7.10. Suppose $(R,\mathfrak{m})$ is an excellent reduced $\mathbf{S}_2$ local ring of residual characteristic $p>0$, $\mathfrak{a}$ a proper ideal of R, and Δ is an effective Mumford $\mathbb{Q}$-divisor on R. If B is a perfectoid R-algebra such that $M_B(<\Delta)$ is defined (Definition 4.15) and $\mathfrak{a}B=(x_1,\ldots,x_g)$ where the elements $x_1,\ldots,x_g$ have compatible systems of p-power roots in B, then from Definition 4.22 we define $\operatorname{ppt}(\mathfrak{a};\Delta;B)\coloneqq \operatorname{Pt}(\mathfrak{a}B,\mathcal{I}(B,<\Delta))$. We further define $\operatorname{ppt}(\mathfrak{a};\Delta)=\sup\{\operatorname{ppt}(\mathfrak{a};\Delta;B)\}$, where B ranges through perfectoid R-algebras such that $\operatorname{ppt}(\mathfrak{a};\Delta;B)$ is defined.

Remark 7.11. We collect some basic observations.

(a) The definition of $\operatorname{ppt}(\mathfrak{a};\Delta;B)$ does not depend on the choice of generators of $\mathfrak{a}B$.[7]

(b) The pair (R,Δ) is perfectoid pure if and only if $\operatorname{ppt}(\mathfrak{a};\Delta)\neq-\infty$ for some (equivalently, all) ideals $\mathfrak{a}\subseteq R$.

(c) It follows from the above that for two perfectoid R-algebras B_1,B_2, it may happen that $\operatorname{ppt}(\mathfrak{a};B_1)\neq\operatorname{ppt}(\mathfrak{a};B_2)$: e.g., take a perfectoid pure R that is not a splinter, and take $B_1=R^{A_\infty}_{\text{perfd}}$ and $B_2=\widehat{R^+}$.

(d) On the other hand, we do not know whether it is possible that $\operatorname{ppt}(\mathfrak{a};B_1)\neq\operatorname{ppt}(\mathfrak{a};B_2)$ when $R\to B_1$ and $R\to B_2$ are both pure. In fact, we will show that this is not possible in positive characteristic, see Proposition 7.22.

Remark 7.12. In the setting of Definition 4.15, it easily follows from Corollary 4.23 that

$$\operatorname{ppt}(\mathfrak{a};\Delta;B)=\liminf_{e\to\infty}\operatorname{Pt}(\mathfrak{a}B,\mathcal{I}(B,M_B(\mathfrak{d}_e)))=\lim_{e\to\infty}\operatorname{Pt}(\mathfrak{a}B,\mathcal{I}(B,M_B(\mathfrak{d}_e))).$$

Namely, since $\mathcal{I}(B,M_B(\mathfrak{d}_e))\subseteq\mathcal{I}(B,<\Delta)$, the sequence $\operatorname{Pt}(\mathfrak{a}B,\mathcal{I}(B,M_B(\mathfrak{d}_e)))$ is non-increasing and is bounded below by $\operatorname{ppt}(\mathfrak{a};\Delta;B)$. The existence of the limit and $\leq$ inequality follow, and the opposite inequality is an easy consequence of Theorem 7.5.

As corollaries to Corollary 7.7 we get comparison results for perfectoid pure thresholds.

Corollary 7.13. *Let $(R,\mathfrak{m})$ be a Noetherian local ring of residual characteristic $p>0$ and suppose that $f\colon A\to B$ is a map of perfectoid R-algebras such that $R\to B$ is pure. Suppose that*

[7]The quantity $\operatorname{Pt}(\mathfrak{a}B,\mathcal{I}(B))$ makes sense even without the hypothesis that $\mathfrak{a}B$ can be generated by elements that have compatible systems of p-power roots. We make this assumption so that we can freely apply Theorem 7.5. Note that, in practice we will often assume (or reduce to the case) that B is absolutely integrally closed so this assumption is automatic.

$x_1, \dots, x_d \in R$ have compatible systems of p-power roots in A and set $\mathfrak{a} = (x_1, \dots, x_d)$. Then $\operatorname{ppt}(\mathfrak{a}; A) \geq \operatorname{ppt}(\mathfrak{a}; B)$ and the equality holds if f is $\mathfrak{m}$-completely pure.

Moreover, if R is an excellent reduced $\mathbf{S}_2$ ring and Δ is an effective Mumford $\mathbb{Q}$-divisor on R, then $\operatorname{ppt}(\mathfrak{a}; \Delta; A) \geq \operatorname{ppt}(\mathfrak{a}; \Delta; B)$ whenever both are defined and the equality holds if f is $\mathfrak{m}$-completely pure.

Proof. The first part follows from Corollary 7.7 and Lemma 2.27 (e) and the second part from Proposition 4.24. □

Corollary 7.14. *Let $f\colon (R, \mathfrak{m}) \to (S, \mathfrak{n})$ be a pure map of Noetherian local rings of residual characteristic $p > 0$ and let A be a perfectoid S-algebra. Suppose that $x_1, \dots, x_d \in R$ have compatible systems of p-power roots in A and set $\mathfrak{a} = (x_1, \dots, x_d)$. Suppose $g\colon A \to B$ is any map of perfectoid S-algebras (e.g., $A = B$). Then*

$$\operatorname{ppt}(\mathfrak{a}; A) \geq \operatorname{ppt}(\mathfrak{a}S; B).$$

Moreover, the equality holds if f is faithfully flat, g is p-completely pure, and $\mathfrak{m}S = \mathfrak{n}$.

Proof. For the first claim, (a) and (c) of Lemma 2.27 show that $\mathcal{I}^R(A)B \subseteq \mathcal{I}^R(B) \subseteq \mathcal{I}^S(B)$, proving the required inequality by Corollary 7.7. For the second claim, we use instead (b) and (e) of Lemma 2.27 to deduce that $\mathcal{I}^S(B) \cap A = \mathcal{I}^R(A)$. □

We record the following version of Corollary 7.14 for pairs.

Corollary 7.15. *Let $f\colon (R, \mathfrak{m}) \to (R', \mathfrak{m}')$ be a flat map between excellent reduced $\mathbf{S}_2$ local rings of residual characteristic $p > 0$. Let Δ be an effective Mumford $\mathbb{Q}$-divisor on R and suppose that its flat pullback Δ' is also a Mumford $\mathbb{Q}$-divisor (Remark 2.17). Suppose that P is a perfectoid R'-algebra such that $M_P(<\Delta)$ is defined (e.g., P is absolutely integrally closed) and $x_1, \dots, x_d \in R$ have compatible p-power roots in P. Set $\mathfrak{a} = (x_1, \dots, x_d)$. Then*

$$\operatorname{ppt}(\mathfrak{a}; \Delta; P) \geq \operatorname{ppt}(\mathfrak{a}R'; \Delta'; P).$$

Moreover, the equality holds if $\mathfrak{m}R' = \mathfrak{m}'$.

Proof. By Lemma 4.5, $M_P(<\Delta) \cong M_P(<\Delta')$. Lemma 2.27 (a) then shows that $\mathcal{I}(P, <\Delta) \subseteq \mathcal{I}(P, <\Delta')$ and the first claim follows. Likewise, Lemma 2.27 (b) implies the second claim. □

Corollary 7.16. *Let $(R, \mathfrak{m})$ be a Noetherian local ring of residual characteristic $p > 0$ and $\mathfrak{a}$ be an ideal. Then $\operatorname{ppt}(\mathfrak{a}) = \operatorname{ppt}(\widehat{\mathfrak{a}})$.*

Moreover, if R is an excellent reduced $\mathbf{S}_2$ ring and Δ is an effective Mumford $\mathbb{Q}$-divisor on R, then $\operatorname{ppt}(\mathfrak{a}; \Delta) = \operatorname{ppt}(\widehat{\mathfrak{a}}; \widehat{\Delta})$, where $\widehat{\Delta}$ is the flat pullback of Δ.

Proof. Any $\widehat{R}$-pure perfectoid $\widehat{R}$-algebra B is also an R-algebra, so $\operatorname{ppt}(\mathfrak{a}) \geq \operatorname{ppt}(\widehat{\mathfrak{a}})$ by Corollary 7.14. Furthermore, if B is a perfectoid R-algebra then $B^{\wedge \mathfrak{m}}$ is a perfectoid $\widehat{R}$-algebra thus $\operatorname{ppt}(\mathfrak{a}; B) = \operatorname{ppt}(\widehat{\mathfrak{a}}; B^{\wedge \mathfrak{m}})$ by Corollary 7.14. Therefore, $\operatorname{ppt}(\mathfrak{a}) \leq \operatorname{ppt}(\widehat{\mathfrak{a}})$.

We now proceed to pairs. For every absolutely integrally closed perfectoid $\widehat{R}$-algebra C, Proposition 4.30 (a) gives that $\operatorname{ppt}(\widehat{\mathfrak{a}}; \widehat{\Delta}; C) = \operatorname{ppt}(\mathfrak{a}; \Delta; C)$, and hence $\operatorname{ppt}(\widehat{\mathfrak{a}}; \widehat{\Delta}) \leq \operatorname{ppt}(\mathfrak{a}; \Delta)$. Conversely, for an absolutely integrally closed perfectoid R-algebra B, set $\widehat{B} = B^{\wedge \mathfrak{m}}$ and choose

$\widehat{B} \to C$ as in Proposition 4.30 (b). Then Proposition 4.30 (b) and Corollary 7.7 show that $\operatorname{ppt}(\mathfrak{a};\Delta;B) = \operatorname{ppt}(\widehat{\mathfrak{a}};\widehat{\Delta};C)$; taking suprema over B gives the converse inequality. □

Proposition 7.17. *Let $(R,\mathfrak{m},k)$ be a perfectoid pure local ring of residual characteristic $p>0$, let $\mathfrak{a}$ be an ideal of R, and let Δ be an effective Mumford $\mathbb{Q}$-divisor on R. Assume that either $\Delta = 0$ or R is an excellent reduced $\mathbf{S}_2$ ring. Then*

(a) $\operatorname{ppt}(\mathfrak{a}) = \operatorname{ppt}(\mathfrak{a};\widehat{R}^{A_\infty}_{\mathrm{perfd}})$ whenever $\mathfrak{a}\widehat{R}^{A_\infty}_{\mathrm{perfd}}$ can be generated by elements admitting compatible systems of p-power roots (here A and A_∞ are as in Notation 2.7);

(b) likewise, if $f = f_\Delta \in R$ is chosen as in Construction 2.18, then we have $\operatorname{ppt}(\mathfrak{a};\Delta) = \operatorname{ppt}(\mathfrak{a};\Delta;\widehat{R}^{A_\infty}_{\mathrm{perfd}})$ whenever f admits a compatible system of p-power roots in $\widehat{R}^{A_\infty}_{\mathrm{perfd}}$ and $\mathfrak{a}\widehat{R}^{A_\infty}_{\mathrm{perfd}}$ can be generated by elements admitting compatible systems of p-power roots;

(c) there is an absolutely integrally closed perfectoid R-algebra B such that $\operatorname{ppt}(\mathfrak{a};\Delta) = \operatorname{ppt}(\mathfrak{a};\Delta;B)$ for all $\mathfrak{a}\subseteq R$.

Proof. First, we note that, by Corollary 7.16, we may assume that R is complete and that there always exist relative perfectoidizations $R^{A_\infty}_{\mathrm{perfd}}$ on which $\operatorname{ppt}(\mathfrak{a};R^{A_\infty}_{\mathrm{perfd}})$ is defined. Namely, we can take any coefficient ring $\mathrm{C}(k)\subseteq R$ and build a surjection $\phi\colon A := \mathrm{C}(k)[\![x_1,\dots,x_n]\!] \twoheadrightarrow R$ so that $\mathfrak{a} = (x_1,\dots,x_g)R$ for some integer $g\in\{1,\dots,n\}$. Then x_i admit compatible systems of p-power roots in $R^{A_\infty}_{\mathrm{perfd}}$. The same argument applies to any finite number of ideals and to pairs via Proposition 4.32, by setting $x_{g+1}=f$.

We now show the first two claims by proving that $\operatorname{ppt}(\mathfrak{a};\Delta;R^{A_\infty}_{\mathrm{perfd}}) \geq \operatorname{ppt}(\mathfrak{a};\Delta;P)$ for any R-pure perfectoid ring P and A constructed as in the first paragraph. By Theorem 2.3 and Corollary 7.13, we may assume that P is absolutely integrally closed, so, as explained in [BMP+24, Lemma 4.23], there is a map $R^{A_\infty}_{\mathrm{perfd}}\to P$. Thus $\operatorname{ppt}(\mathfrak{a};\Delta;R^{A_\infty}_{\mathrm{perfd}}) \geq \operatorname{ppt}(\mathfrak{a};\Delta;P)$ by Corollary 7.13. By symmetry, this argument also shows that any two appropriate relative perfectoidizations have equal thresholds.

The final claim follows from Corollary 7.13 after applying Theorem 2.3 to build a p-completely faithfully flat map $R^{A_\infty}_{\mathrm{perfd}}\to P$ to an absolutely integrally closed perfectoid R-algebra. □

Corollary 7.18. *Let $(R,\mathfrak{m})$ be an excellent reduced $\mathbf{S}_2$ ring of residual characteristic $p>0$, Δ an effective Mumford $\mathbb{Q}$-divisor on R, and $\mathfrak{a}$ an ideal not contained in any minimal prime. Then we have*

(a) If $\mathfrak{a}\subseteq\mathfrak{p}$ for some height one prime $\mathfrak{p}$ such that $R_\mathfrak{p}$ is singular, then we have

$$\begin{cases} \operatorname{ppt}(\mathfrak{a};\Delta) = 0 \text{ if } (R,\Delta) \text{ is perfectoid pure;} \\ \operatorname{ppt}(\mathfrak{a};\Delta) = -\infty \text{ if } (R,\Delta) \text{ is not perfectoid pure.} \end{cases}$$

(b) Otherwise, there are minimal generators $(x_1,\dots,x_m)=\mathfrak{a}$ such that

$$\operatorname{ppt}(\mathfrak{a};\Delta) = \sup\left\{ \frac{|a|}{p^e} \mid e\in\mathbb{Z}_{\geq 0}, a\in\mathbb{Z}^m_{\geq 0}, (R,\Delta+\frac{1}{p^e}\operatorname{div}(x_1^{a_1}\cdots x_m^{a_m})) \text{ is perfectoid pure}\right\}.$$

Proof. For (a), it is clear that $\operatorname{ppt}(\mathfrak{a};\Delta) = -\infty$ if (R,Δ) is not perfectoid pure. So assume (R,Δ) is perfectoid pure. If $\operatorname{ppt}(\mathfrak{a};\Delta)\neq 0$, then $\operatorname{ppt}(\mathfrak{a};B)\neq 0$ for some perfectoid R-algebra

B. By Proposition 7.8 there is some $f \in \mathfrak{a}$ so that $\operatorname{ppt}((f); B) \neq 0$. This implies that for all $0 < \lambda < \operatorname{ppt}((f); B)$, the pair (R, f^λ) is perfectoid pure.[8] But then Proposition 4.33 (a) implies that $\operatorname{div}(f)$ cannot contain any height one prime $\mathfrak{p}$ so that $R_\mathfrak{p}$ is singular; this is a contradiction to $f \in \mathfrak{a}$.

We next treat (b). Since R is reduced, there are finitely many height-one prime ideals in the singular locus of R. Let $\mathfrak{p}_1, \ldots, \mathfrak{p}_N$ be these ideals along with the minimal primes of R. Since $\mathfrak{a}$ is not contained in any $\mathfrak{p}_i$, by prime avoidance we can choose a generating set $\{x_1, \ldots, x_m\}$ of $\mathfrak{a}$ so that x_i is not in $\mathfrak{p}_j$ for all j and i. Thus a monomial on $\{x_1, \ldots, x_m\}$ is not contained in any $\mathfrak{p}_i$. By Proposition 7.17 we may compute $\operatorname{ppt}(\mathfrak{a}; \Delta)$ in a fixed perfectoid absolutely integrally closed algebra B. The conclusion then follows from Theorem 7.5 and Remark 4.26. (Note that $\operatorname{ppt}(\mathfrak{a}; \Delta)$ could still be $-\infty$ in case (b) since when (R, Δ) is not perfectoid pure the right hand side is the supremum over an empty set, so it equals $-\infty$ by convention). □

Proposition 7.19 (Elementary properties)**.** *Let $(R, \mathfrak{m})$ be a perfectoid pure local ring of residual characteristic $p > 0$ and $\mathfrak{a}, \mathfrak{b}$ be ideals. In what follows, either assume $\Delta = 0$ or assume that R is an excellent reduced $\mathbf{S}_2$ ring and Δ is an effective Mumford $\mathbb{Q}$-divisor on R. Then*

(a) $\operatorname{ppt}(\mathfrak{a} + \mathfrak{b}; \Delta) \leq \operatorname{ppt}(\mathfrak{a}; \Delta) + \operatorname{ppt}(\mathfrak{b}; \Delta)$*;*

(b) if $\mathfrak{a} \subset \mathfrak{b}$ *then* $\operatorname{ppt}(\mathfrak{a}; \Delta) \leq \operatorname{ppt}(\mathfrak{b}; \Delta)$*;*

(c) if $\overline{\mathfrak{a}} = \overline{\mathfrak{b}}$ *then* $\operatorname{ppt}(\mathfrak{a}; \Delta) = \operatorname{ppt}(\mathfrak{b}; \Delta)$*;*

(d) $\operatorname{ppt}(\mathfrak{a}^n; \Delta) = n^{-1} \operatorname{ppt}(\mathfrak{a}; \Delta)$ *for* $n \in \mathbb{Z}_{\geq 1}$*;*

(e) if Δ' *is an effective Mumford* $\mathbb{Q}$*-divisor with* $\Delta' \leq \Delta$*, then* $\operatorname{ppt}(\mathfrak{a}; \Delta) \leq \operatorname{ppt}(\mathfrak{a}; \Delta')$*;*

(f) if $(p, \mathfrak{a}) \subseteq \mathfrak{q} \in \operatorname{Spec}(R)$ *then* $\operatorname{ppt}(\mathfrak{a}; \Delta) \leq \operatorname{ppt}(\mathfrak{a}R_\mathfrak{q}; \Delta R_\mathfrak{q})$.

Proof. By Proposition 7.17 there is an absolutely integrally closed perfectoid ring B that computes $\operatorname{ppt}(\mathfrak{a}; \Delta)$, $\operatorname{ppt}(\mathfrak{b}; \Delta)$, and $\operatorname{ppt}(\mathfrak{a} + \mathfrak{b}; \Delta)$. The first four properties now follow from Lemma 7.3 and Corollary 7.6. The next property follows immediately from the definitions and Lemma 4.16.

The last property follows immediately from Corollary 7.18 and Proposition 4.28 when $\Delta \neq 0$ and from a similar argument when $\Delta = 0$. □

Corollary 7.20. *Let $(R, \mathfrak{m})$ be a perfectoid pure local ring and $\mathfrak{a}$ be a proper ideal. Furthermore, either assume $\Delta = 0$ or assume that R is an excellent reduced $\mathbf{S}_2$ ring and Δ is an effective Mumford $\mathbb{Q}$-divisor on R such that (R, Δ) is perfectoid pure. Then for every $\varepsilon > 0$ there exists $n \in \mathbb{Z}_{>0}$ such that for every ideal $\mathfrak{b} \equiv \mathfrak{a} \pmod{\mathfrak{m}^n}$ we have*

$$|\operatorname{ppt}(\mathfrak{a}; \Delta) - \operatorname{ppt}(\mathfrak{b}; \Delta)| < \varepsilon.$$

In particular, for any $f \in \mathfrak{m}$ and any $\varepsilon > 0$ there exists $n \in \mathbb{Z}_{>0}$ such that for all $g \in \mathfrak{m}^n$ we have $|\operatorname{ppt}(f) - \operatorname{ppt}(f + g)| \leq \varepsilon$.

Proof. Suppose that $\mathfrak{b} \equiv \mathfrak{a} \pmod{\mathfrak{m}^n}$. Then we have $\mathfrak{a} \subseteq \mathfrak{b} + \mathfrak{m}^n$. Hence by Proposition 7.19

$$\operatorname{ppt}(\mathfrak{a}; \Delta) \leq \operatorname{ppt}(\mathfrak{b} + \mathfrak{m}^n; \Delta) \leq \operatorname{ppt}(\mathfrak{b}; \Delta) + \operatorname{ppt}(\mathfrak{m}^n; \Delta) = \operatorname{ppt}(\mathfrak{b}; \Delta) + n^{-1} \operatorname{ppt}(\mathfrak{m}; \Delta),$$

and so by symmetry $|\operatorname{ppt}(\mathfrak{a}; \Delta) - \operatorname{ppt}(\mathfrak{b}; \Delta)| \leq n^{-1} \operatorname{ppt}(\mathfrak{m}; \Delta)$. □

[8] Note that f must be a nonzerodivisor as otherwise f becomes zero in R_Q for some minimal prime Q, and $R_Q \to B_Q$ sending $1 \to f^{1/p^e} = 0$ cannot split for any e.

Corollary 7.21. *Let $(R, \mathfrak{m}, k)$ be a regular local ring of residual characteristic $p > 0$. Then* $\operatorname{ppt}(\mathfrak{m}) = \dim(R)$.

Proof. We may assume that R is complete. If R is unramified of mixed characteristic then $R \cong \mathrm{C}(k)[\![x_1, \ldots, x_d]\!]$, so we may compute

$$\operatorname{ppt}(\mathfrak{m}) = \sup\left\{\frac{|a|}{p^e} \mid a \in \mathbb{Z}_{\geq 0}^{d+1}, x_1^{a_1/p^e} \cdots x_d^{a_d/p^e} p^{a_{d+1}/p^e} \notin (x_1, \ldots, x_d, p)R_\infty\right\} = d+1 = \dim(R),$$

since $p, x_1, \ldots, x_d$ form a regular sequence in R_∞.

Otherwise, R can be written as $R \cong S/(p - f(x_1, \ldots, x_d))$ where $S = \mathrm{C}(k)[\![x_1, \ldots, x_d]\!]$. Since $\mathfrak{m} = (x_1, \ldots, x_d)$ in this case, we may compute

$$\operatorname{ppt}(\mathfrak{m}) = \sup\left\{\frac{|a|}{p^e} \mid a \in \mathbb{Z}_{\geq 0}^{d}, x_1^{a_1/p^e} \cdots x_d^{a_d/p^e} \notin (x_1, \ldots, x_d)R_{\mathrm{perfd}}^{S_\infty}\right\} = d = \dim(R),$$

since again $x_1, \ldots, x_d$ form a regular sequence on $R_{\mathrm{perfd}}^{S_\infty}$. □

We end this subsection by showing that in characteristic $p > 0$, $\operatorname{ppt}(\mathfrak{a}; B)$ does not depend on the choice of B *as long as* $R \to B$ *is pure*.

Proposition 7.22. *If $(R, \mathfrak{m})$ is a Noetherian F-finite local ring of characteristic $p > 0$, $\mathfrak{a} = (x_1, \ldots, x_d)$, and $R \to B$ is a pure map to a perfect ring B, then*

$$\operatorname{fpt}(\mathfrak{a}) = \operatorname{ppt}(\mathfrak{a}; B) = \operatorname{ppt}(\mathfrak{a}; R_{\mathrm{perf}}).$$

Furthermore, if R is a normal domain and Δ is an effective $\mathbb{Q}$-divisor on R, then

$$\operatorname{fpt}(\mathfrak{a}; \Delta) = \operatorname{ppt}(\mathfrak{a}; \Delta; B).$$

Proof. We may assume that R is F-pure. Note that the purity of $R \to B$ implies purity of $R_{\mathrm{perf}} \to B$ since B is perfect. Thus $\operatorname{ppt}(\mathfrak{a}; \Delta; R_{\mathrm{perf}}) = \operatorname{ppt}(\mathfrak{a}; \Delta; B)$ by Corollary 7.13. Hence, it suffices to show the statements for $B = R_{\mathrm{perf}}$.

First, assume that $\Delta = 0$. Recall that by [DSNB18] we have

$$\operatorname{fpt}(\mathfrak{a}) = \lim_{e \to \infty} \frac{\sup\{m \mid \mathfrak{a}^m \not\subseteq I_e\}}{p^e} = \lim_{e \to \infty} \frac{\sup\{|a| \mid a \in \mathbb{Z}_{\geq 0}^d, x_1^{a_1} \cdots x_d^{a_d} \notin I_e\}}{p^e},$$

where $I_e = \{x \in R \mid \phi(x) \in \mathfrak{m}, \forall \phi \in \operatorname{Hom}_R(F_*^e R, R)\}$, so $F_*^e I_e = \mathcal{I}(F_*^e R)$. By Lemma 2.27

$$\mathcal{I}(R_{\mathrm{perf}}) \cap F_*^e R = \mathcal{I}(F_*^e R) = F_*^e I_e,$$

thus the conclusion follows from Theorem 7.5.

We now assume $\Delta \neq 0$. Set $\Delta_e = p^{-e}\lfloor (p^e - 1)\Delta \rfloor$. Adapting [DSNB18], we may compute

$$\operatorname{fpt}(\mathfrak{a}; \Delta) = \lim_{e \to \infty} p^{-e} \sup\left\{|a| \mid a \in \mathbb{Z}_{\geq 0}^d, x_1^{a_1/p^e} \cdots x_d^{a_d/p^e} \notin \mathcal{I}^R(R^{1/p^e}, R(p^e\Delta_e)^{1/p^e})\right\}.$$

Set $J_e = \mathcal{I}(R_{\mathrm{perf}}, R_{\mathrm{perf}} \otimes_{R^{1/p^e}} R(p^e\Delta_e)^{1/p^e})$. Since $J_e \cap R^{1/p^e} = \mathcal{I}^R(R^{1/p^e}, R(p^e\Delta_e)^{1/p^e})$, it follows that

$$\operatorname{fpt}(\mathfrak{a}; \Delta) = \lim_{e \to \infty} p^{-e} \sup\left\{|a| \mid a \in \mathbb{Z}_{\geq 0}^d, x_1^{a_1/p^e} \cdots x_d^{a_d/p^e} \notin J_e\right\}.$$

The argument of Proposition 4.31 shows that a subsequence $\Delta_e, e \in S \subseteq \mathbb{N}$ forms a limiting sequence for Δ, so $\mathcal{I}(R_{\mathrm{perf}}, <\Delta) = \cup_{e\in S} J_e$. Thus, Theorem 7.5 immediately shows that

$$\begin{aligned}\mathrm{ppt}(\mathfrak{a};\Delta) &= \lim_{n\to\infty} p^{-n} \sup\left\{|a| \mid a \in \mathbb{Z}^d_{\geq 0}, x_1^{a_1/p^n}\cdots x_d^{a_d/p^n} \notin J_e \text{ for all } e \in S\right\}\\ &\leq \lim_{n\in S} p^{-n} \sup\left\{|a| \mid a \in \mathbb{Z}^d_{\geq 0}, x_1^{a_1/p^n}\cdots x_d^{a_d/p^n} \notin J_n\right\} = \mathrm{fpt}(\mathfrak{a};\Delta).\end{aligned}$$

On the other hand, by Remark 7.12, we may compute $\mathrm{ppt}(\mathfrak{a};\Delta) = \lim_{e\in S} \mathrm{Pt}(\mathfrak{a}R_{\mathrm{perf}}, J_e)$, so it is immediate from Theorem 7.5 that

$$\mathrm{ppt}(\mathfrak{a};\Delta) = \lim_{e\in S} \sup\left\{\frac{|a|}{p^n} \mid a \in \mathbb{Z}^d_{\geq 0}, x_1^{a_1/p^n}\cdots x_d^{a_d/p^n} \notin J_e\right\} \geq \mathrm{fpt}(\mathfrak{a};\Delta).$$

□

Remark 7.23. The proof of Proposition 7.22 shows that the perfectoid threshold generalizes the Frobenius closure version of F-threshold: if R has characteristic $p > 0$ and $\mathfrak{a}, J$ are ideals, then

$$\mathrm{Pt}(\mathfrak{a}R_{\mathrm{perf}}, JR_{\mathrm{perf}}) = \lim_{e\to\infty} \frac{1}{p^e}\sup\{m \mid \mathfrak{a}^m \not\subseteq (J^{[p^e]})^F\}.$$

In the same spirit [RS25, Corollary 3.9] shows that a tight closure version of F-threshold can be recovered as $\mathrm{Pt}(\mathfrak{a}B, JB)$ for certain BCM algebras B.

As we mentioned, we do not know a good analog of Proposition 7.22 in mixed characteristic.

Question 7.24. *Let $(R,\mathfrak{m})$ be a Noetherian local ring of mixed characteristic $(0, p > 0)$ that is perfectoid pure. Then is $\mathrm{ppt}(\mathfrak{a}) = \mathrm{ppt}(\mathfrak{a}; B)$ whenever $R \to B$ is pure and B is perfectoid?*

An affirmative answer to this question will be a generalization of the well-known fact that in a strongly F-regular ring the strongly F-regular threshold coincides with F-pure threshold. We record the following partial result towards this question.

Proposition 7.25. *If $(R,\mathfrak{m},k)$ is a complete regular local ring of residual characteristic $p > 0$ then $\mathrm{ppt}(\mathfrak{a}) = \mathrm{ppt}(\mathfrak{a}; \widehat{R^+})$ for any ideal $\mathfrak{a} \subset R$.*

Proof. Let C_k be a coefficient ring of R. Set $R' = R\widehat{\otimes}_{C_k} W(k^{1/p^\infty})$. Note that the map $C_k \to W(k^{1/p^\infty})$ is the p-adic completion of an integral map and thus $W(k^{1/p^\infty})$ and hence R' admit a map to $\widehat{R^+}$. Let $x_1,\ldots,x_d$ be a regular system of parameters of R and fix a compatible system of p-power roots of them in R^+. Set $R_n = R'[x_1^{1/p^n},\ldots,x_d^{1/p^n}]$ and $R_\infty = \widehat{\varinjlim} R_n$. Then R_∞ is perfectoid and each R_n is regular local. Let R'' be the $\mathfrak{m}$-adic completion of R'; then by [Bha20, Theorem 5.16], the natural map $R_n \to \widehat{R''^+}$ is faithfully flat and hence $R_\infty \to \widehat{R''^+}$ is p-completely faithfully flat. Set $\mathfrak{a} = (f_1,\ldots,f_n)$ and let $S = R[y_1,\ldots,y_n]$. Consider the natural map $S \to R$ sending y_i to f_i and define $R^{S_\infty}_{\mathrm{perfd}}$ as in Notation 2.7. The ring $B := \widehat{R''^+}\widehat{\otimes}^{\mathbb{L}}_{R_\infty} R^{S_\infty}_{\mathrm{perfd}}$ is then perfectoid and p-completely faithfully flat over $R^{S_\infty}_{\mathrm{perfd}}$. Then we have

$$\mathrm{ppt}(\mathfrak{a}) \geq \mathrm{ppt}(\mathfrak{a};\widehat{R^+}) \geq \mathrm{ppt}(\mathfrak{a};\widehat{R''^+}) \geq \mathrm{ppt}(\mathfrak{a};B) = \mathrm{ppt}(\mathfrak{a};R^{S_\infty}_{\mathrm{perfd}}) = \mathrm{ppt}(\mathfrak{a})$$

by Corollary 7.13 and Proposition 7.17. Thus we must have equalities throughout which completes the proof. □

7.3. **Localization and regular extensions.** In this subsection we will strengthen and generalize [BMP+24, Corollary 4.21 (a)] by proving a faithfully flat ascent result for pairs, Theorem 7.28. In particular, we will show that perfectoid purity is preserved under faithfully flat extensions with regular closed fiber (thus answering [BMP+24, Conjecture 4.22 (a)]).

We start with a lemma treating a partial case.

Lemma 7.26. *Let $(R, \mathfrak{m})$ be a complete local ring of residual characteristic $p > 0$ and P be a perfectoid R-algebra. Set $T = P[x^{1/p^\infty}]^{\wedge(\mathfrak{m},x)}$. If M is a pointed P-module, then*

$$\mathcal{I}^{R[\![x]\!]}(T, M \otimes_P T) = \mathcal{I}^R(P, M)T + xT.$$

Proof. Set $M_T = M \otimes_P T$ and note that x is a regular element on $M_T/(N \otimes_P T)$ for any submodule $N \subset M$. By Lemma 2.27 and Lemma 2.24, it suffices to show that $\mathcal{I}^{R[\![x]\!]}(T, M_T) \subseteq \mathcal{I}^R(P, M)T + xT$. We may further assume that $R \to P$ is pure, so R is reduced.

Let I_t be an approximately Gorenstein sequence in R and u_t be elements such that u_tR/I_t is the socle ideal. By Lemma 2.25 we may compute

$$\mathcal{I}^R(T, M_T) = \bigcup I_tM_T :_T u_t\mathbf{1}_{M_T} \text{ and } \mathcal{I}^{R[\![x]\!]}(T, M_T) = \bigcup (I_t + x^t)M_T :_T u_tx^{t-1}\mathbf{1}_{M_T}.$$

Let $a \in (I_n + x^n)M_T :_T u_nx^{n-1}\mathbf{1}_{M_T}$. By definition, there is $f \in M_T$ such that

$$x^{n-1}(au_n\mathbf{1}_{M_T} + fx) \in I_nM_T.$$

Note x is a regular element here since $I_nM_T \subseteq M_T$ is obtained by base change from P, and so $(au_n\mathbf{1}_{M_T} + fx) \in I_nM_T$, i.e.,

$$fx \in (I_nM_T, Tu_n\mathbf{1}_{M_T}) = (I_nM, Pu_n\mathbf{1}_M) \otimes_P T.$$

Hence, $f \in (I_nM_T, Tu_n\mathbf{1}_{M_T})$, so there is $b \in T$ such that $f - bu_n\mathbf{1}_{M_T} \in I_nM_T$. This implies that $(a+bx)u_n\mathbf{1}_{M_T} \in I_nM_T$, thus $a+bx \in I_nM_T :_T u_n\mathbf{1}_{M_T} \subseteq \mathcal{I}^R(T, M_T)$. It remains to note that $\mathcal{I}^R(T, M_T) \subseteq \mathcal{I}^R(P, M)T + xT$, which follows from Lemma 2.27 (c) and (f) applied to the $\mathfrak{m}$-completely faithfully flat map $P \to T/xT$. □

Corollary 7.27. *Let $(R, \mathfrak{m})$ be a complete local perfectoid pure ring of residue characteristic p, $\mathfrak{a}$ be an ideal of R, and Δ be an effective Mumford $\mathbb{Q}$-divisor on R. If $\Delta \neq 0$, then further assume that R is an excellent reduced $\mathbf{S}_2$ ring. Let $\Delta[\![x]\!]$ be the flat pullback of Δ to $R[\![x]\!]$. Then*

$$\operatorname{ppt}((\mathfrak{a} + x)R[\![x]\!]; \Delta[\![x]\!]) = \operatorname{ppt}(\mathfrak{a}; \Delta) + 1.$$

Proof. Fix a surjection $A \twoheadrightarrow R$ such that $\operatorname{ppt}(\mathfrak{a}; \Delta; R^{A_\infty}_{\text{perfd}}) = \operatorname{ppt}(\mathfrak{a}; \Delta)$ (Proposition 7.17). It follows from the universal property that there is a natural map

$$R[\![x]\!]^{(A[\![x]\!])_\infty}_{\text{perfd}} \to (R^{A_\infty}_{\text{perfd}}[x^{1/p^\infty}])^{\wedge(\mathfrak{m},x)} \cong \left(R[\![x]\!]^{(A[\![x]\!])_\infty}_{\text{perfd}}\right)^{\wedge(\mathfrak{m},x)} =: T.$$

By Corollary 7.13 and Proposition 7.17 we know that

$$\operatorname{ppt}((\mathfrak{a} + x)R[\![x]\!]; \Delta[\![x]\!]) = \operatorname{ppt}((\mathfrak{a} + x)R[\![x]\!]; \Delta[\![x]\!]; T).$$

Applying Lemma 7.26 to the sequence $(R^{A_\infty}_{\text{perfd}}, M_{R^{A_\infty}_{\text{perfd}}}(\mathfrak{d}_e))$ from Definition 4.15 as in Proposition 4.24, we obtain that

$$\mathcal{I}^{R[\![x]\!]}(T, <\Delta[\![x]\!]) = \mathcal{I}^R(R^{A_\infty}_{\text{perfd}}, <\Delta)T + xT. \tag{7.27.1}$$

By Theorem 7.5 it remains to show that, writing $\mathfrak{a} = (y_1, \dots, y_n)$ so that each y_i admits a compatible system of p-power roots in $R^{A_\infty}_{\text{perfd}}$, if a monomial $m = y_1^{a_1/p^e} \cdots y_n^{a_n/p^e} \in R^{A_\infty}_{\text{perfd}}$ is not contained in $\mathcal{I}^R(R^{A_\infty}_{\text{perfd}}, <\Delta)$ then $x^{1-1/p^e} m \notin \mathcal{I}^{R[\![x]\!]}(T, <\Delta[\![x]\!])$. If not, then

$$x^{1-1/p^e} m - xs \in \mathcal{I}^R(R^{A_\infty}_{\text{perfd}}, <\Delta)T$$

for some $s \in T$. Since x is a regular element modulo IT for every $I \subseteq R^{A_\infty}_{\text{perfd}}$, we must have

$$m - sx^{1/p^e} \in \mathcal{I}^R(R^{A_\infty}_{\text{perfd}}, <\Delta)T, \tag{7.27.2}$$

i.e., $sx^{1/p^e} \in (m, \mathcal{I}^R(R^{A_\infty}_{\text{perfd}}, <\Delta))T$. Thus, there is $t \in T$ such that $s - mt \in \mathcal{I}^R(R^{A_\infty}_{\text{perfd}}, <\Delta)T$. Plugging this into (7.27.2) we obtain that

$$m(1 - tx^{1/p^e}) \in \mathcal{I}^R(R^{A_\infty}_{\text{perfd}}, <\Delta)T.$$

Since T is (x)-complete, $(1 - tx^{1/p^e})$ is invertible, so by Lemma 2.27 and (7.27.1)

$$m \in \mathcal{I}^R(R^{A_\infty}_{\text{perfd}}, <\Delta)T \cap R^{A_\infty}_{\text{perfd}} \subseteq \mathcal{I}^R(T, <\Delta) \cap R^{A_\infty}_{\text{perfd}} = \mathcal{I}^R(R^{A_\infty}_{\text{perfd}}, <\Delta),$$

a contradiction. □

Theorem 7.28. *Let $(R, \mathfrak{m}) \to (S, \mathfrak{n})$ be a faithfully flat local map of Noetherian local rings of residue characteristic p such that $S/\mathfrak{m}S$ is regular. Let $f_1, \dots, f_n \in S$ be elements whose images form a regular system of parameters on $S/\mathfrak{m}S$. Let $\mathfrak{a}$ be an ideal of R and Δ be an effective Mumford $\mathbb{Q}$-divisor on R. If $\Delta \neq 0$, further assume that R and S are excellent reduced $\mathbf{S}_2$ rings and that the flat pullback Δ_S of Δ is a Mumford $\mathbb{Q}$-divisor (Remark 2.17). Then*

$$\operatorname{ppt}((\mathfrak{a}S + (f_1, \dots, f_n))S; \Delta_S) = \operatorname{ppt}(\mathfrak{a}; \Delta) + \dim(S/\mathfrak{m}S).$$

In particular, if (R, Δ) is perfectoid pure, then so is (S, Δ_S).

Proof. We may assume that R is perfectoid pure, because if it is not then S is also not perfectoid pure (since an S-algebra is an R-algebra) so all the thresholds involved are $-\infty$. We may also assume that both rings are complete by Corollary 7.16. Using Corollary 7.27, by adjoining power series variables $x_1, \dots, x_n$ to R and mapping them to $f_1, \dots, f_n$ in S, we reduce the theorem to the case where $\mathfrak{n} = \mathfrak{m}S$, in which case we need to show that $\operatorname{ppt}(\mathfrak{a}; \Delta) = \operatorname{ppt}(\mathfrak{a}S; \Delta_S)$.

Let us choose coefficient rings $V_R \subseteq R$ and $V_S \subseteq S$. Fix a surjection $B' := V_S[\![y_1, \dots, y_h]\!] \twoheadrightarrow S$ such that $\operatorname{ppt}(\mathfrak{a}S; \Delta_S; S^{B'_\infty}_{\text{perfd}}) = \operatorname{ppt}(\mathfrak{a}S; \Delta_S)$ (Proposition 7.17). By [Gro64, IV, 19.8.6.i], the composition $V_R \to S$ can be lifted to a homomorphism $V_R \to B' = V_S[\![y_1, \dots, y_h]\!]$. We now take $A = V_R[\![x_1, \dots, x_g]\!] \twoheadrightarrow R$ to be any surjection such that $\operatorname{ppt}(\mathfrak{a}; \Delta; R^{A_\infty}_{\text{perfd}}) = \operatorname{ppt}(\mathfrak{a}; \Delta)$ (Proposition 7.17) and define $B = V_S[\![x_1, \dots, x_g, y_1, \dots, y_h]\!]$. We have a faithfully flat map $A \to B$. Also note that $\operatorname{ppt}(\mathfrak{a}S; \Delta_S; S^{B'_\infty}_{\text{perfd}}) = \operatorname{ppt}(\mathfrak{a}S; \Delta_S; S^{B_\infty}_{\text{perfd}})$ by Corollary 7.13 since $S^{B'_\infty}_{\text{perfd}} \to S^{B_\infty}_{\text{perfd}}$ is p-completely faithfully flat by André's lemma [BS22, Proposition 7.11].

Consider the perfectoid ring $P = (R^{A_\infty}_{\text{perfd}} \widehat{\otimes}^{\mathbb{L}}_R S \widehat{\otimes}^{\mathbb{L}}_B B_\infty)_{\text{perfd}}$. By [BS22, Proposition 8.13], for any diagram $T_1 \leftarrow T_3 \to T_2$ we know that $(T_1 \widehat{\otimes}^{\mathbb{L}}_{T_3} T_2)_{\text{perfd}} = (T_{1,\text{perfd}} \widehat{\otimes}^{\mathbb{L}}_{T_3} T_2)_{\text{perfd}}$, thus we have

$$P = (A_\infty \widehat{\otimes}^{\mathbb{L}}_A R \widehat{\otimes}^{\mathbb{L}}_R S \widehat{\otimes}^{\mathbb{L}}_B B_\infty)_{\text{perfd}} = (A_\infty \widehat{\otimes}^{\mathbb{L}}_A S \widehat{\otimes}^{\mathbb{L}}_B B_\infty)_{\text{perfd}} = (A_\infty \widehat{\otimes}^{\mathbb{L}}_A S^{B_\infty}_{\text{perfd}})_{\text{perfd}}.$$

By André's lemma ([BS22, Proposition 7.11]), we find that $S^{B_\infty}_{\mathrm{perfd}} \to P$ is p-completely faithfully flat. Therefore, by Corollary 7.13 we have

$$\mathrm{ppt}(\mathfrak{a};\Delta) = \mathrm{ppt}(\mathfrak{a};\Delta;R^{A_\infty}_{\mathrm{perfd}}) \quad \text{and} \quad \mathrm{ppt}(\mathfrak{a}S;\Delta_S;P) = \mathrm{ppt}(\mathfrak{a}S;\Delta_S;S^{B_\infty}_{\mathrm{perfd}}) = \mathrm{ppt}(\mathfrak{a}S;\Delta_S).$$

By Corollary 7.15 we also have $\mathrm{ppt}(\mathfrak{a};\Delta;P) = \mathrm{ppt}(\mathfrak{a}S;\Delta_S;P)$, so it follows immediately that

$$\mathrm{ppt}(\mathfrak{a};\Delta) = \mathrm{ppt}(\mathfrak{a};\Delta;R^{A_\infty}_{\mathrm{perfd}}) \geq \mathrm{ppt}(\mathfrak{a};\Delta;P) = \mathrm{ppt}(\mathfrak{a}S;\Delta_S).$$

Thus, it suffices to show that $\mathrm{ppt}(\mathfrak{a};\Delta;R^{A_\infty}_{\mathrm{perfd}}) \leq \mathrm{ppt}(\mathfrak{a};\Delta;P)$.

We claim that the p-completed base change $R^{A_\infty}_{\mathrm{perfd}} \to R^{A_\infty}_{\mathrm{perfd}}\widehat{\otimes}^{\mathbb{L}}_R S\widehat{\otimes}^{\mathbb{L}}_B B_\infty$ is quasisyntomic. First, since $R \to S$ is a complete intersection morphism, [Avr99, Theorem 1.2] shows that the Tor dimension of the cotangent complex $L_{S/R}$ is at most one. This shows that the p-completely faithfully flat base change $R^{A_\infty}_{\mathrm{perfd}} \to R^{A_\infty}_{\mathrm{perfd}}\widehat{\otimes}^{\mathbb{L}}_R S$ is quasisyntomic. Second, $B \to B_\infty$ is ind-syntomic, so a further base change $R^{A_\infty}_{\mathrm{perfd}}\widehat{\otimes}^{\mathbb{L}}_R S \to R^{A_\infty}_{\mathrm{perfd}}\widehat{\otimes}^{\mathbb{L}}_R S\widehat{\otimes}^{\mathbb{L}}_B B_\infty$ is quasisyntomic.

Since $R^{A_\infty}_{\mathrm{perfd}} \to R^{A_\infty}_{\mathrm{perfd}}\widehat{\otimes}^{\mathbb{L}}_R S\widehat{\otimes}^{\mathbb{L}}_B B_\infty$ is quasisyntomic, by [BS22, Proposition 7.11] there exists a perfectoid P-algebra T such that the composition $R^{A_\infty}_{\mathrm{perfd}} \to P \to T$ is p-completely faithfully flat. Thus, Corollary 7.13 yields the desired $\mathrm{ppt}(\mathfrak{a};\Delta;P) \geq \mathrm{ppt}(\mathfrak{a};\Delta;T) = \mathrm{ppt}(\mathfrak{a};\Delta;R^{A_\infty}_{\mathrm{perfd}})$. □

We end this subsection by describing the behavior of perfectoid pure threshold of the maximal ideal under localizations.

Proposition 7.29. *Let $(R,\mathfrak{m})$ be an excellent local ring of residual characteristic $p>0$, Δ be an effective Mumford $\mathbb{Q}$-divisor on R. If $\Delta \neq 0$, we further assume that R is reduced and $\mathbf{S}_2$. If $p \in \mathfrak{q}$ is a prime ideal such that $\dim(R_\mathfrak{q}) + \dim(R/\mathfrak{q}) = \dim(R)$, then*

$$\mathrm{ppt}(\mathfrak{m};\Delta) \leq \dim(R/\mathfrak{q}) + \mathrm{ppt}(\mathfrak{q}R_\mathfrak{q};\Delta_\mathfrak{q}).$$

Proof. By Corollary 7.16 and Theorem 7.28, we may assume that R is complete. By a straightforward induction we may assume that $\dim(R/\mathfrak{q}) = 1$.

Since $R/\mathfrak{q}$ is a complete local domain of dimension one, its normalization V is a DVR module-finite over $R/\mathfrak{q}$. Let $1, f_1, \ldots, f_n$, where $f_i \in V$, generate V as a module over $R/\mathfrak{q}$. Consider the surjective map

$$(S,\mathfrak{n}) := R[x_1,\ldots,x_n]_\mathfrak{n} \twoheadrightarrow V$$

sending x_i to f_i where $\mathfrak{n}$ is the inverse image of $\mathfrak{m}_V$. Let Q be the kernel of this map; then $Q \in \mathrm{Spec}(S)$ lies over $\mathfrak{q} \in \mathrm{Spec}(R)$. In particular, we have an induced commutative diagram:

$$\begin{array}{ccc} R & \longrightarrow & S \\ \downarrow & & \downarrow \\ R_\mathfrak{q} & \longrightarrow & S_Q \end{array}$$

where both horizontal maps are faithfully flat with regular closed fibers of dimension n: this is obvious for $R \to S$, and for $R_\mathfrak{q} \to S_Q$, note that $\dim(S_Q/\mathfrak{q}S_Q) = n$ while $Q(S_Q/\mathfrak{q}S_Q)$ can be generated by $(x_1 - f_1, \ldots, x_n - f_n)$ since $f_i \in V \subseteq R_\mathfrak{q}/\mathfrak{q}R_\mathfrak{q} \subseteq S_Q/\mathfrak{q}S_Q$.

Let Δ_S and Δ_{S_Q} be the obvious pullbacks of Δ to S and S_Q respectively. By Theorem 7.28, we have that

$$\operatorname{ppt}(\mathfrak{m};\Delta) = \operatorname{ppt}(\mathfrak{n};\Delta_S) - n \text{ and } \operatorname{ppt}(\mathfrak{q}R_{\mathfrak{q}};\Delta_{\mathfrak{q}}) = \operatorname{ppt}(QS_Q;\Delta_{S_Q}) - n.$$

Therefore, in order to prove the proposition, we may further replace R by S and $\mathfrak{q}$ by Q to assume that $R/\mathfrak{q}$ is a DVR, i.e., $\mathfrak{m} = (\mathfrak{q}, x)$ for some $x \in \mathfrak{m}$. Now by Proposition 7.19 we have

$$\operatorname{ppt}(\mathfrak{m};\Delta) = \operatorname{ppt}((\mathfrak{q}, x);\Delta) \leq \operatorname{ppt}(\mathfrak{q};\Delta) + \operatorname{ppt}(x;\Delta) \leq \operatorname{ppt}(\mathfrak{q}R_{\mathfrak{q}};\Delta_{\mathfrak{q}}) + 1.$$

□

7.4. **Measuring singularities.** In this subsection we connect both $\operatorname{ppt}(\mathfrak{a})$, respectively $\operatorname{ppt}(\mathfrak{a};\Delta)$, with the singularities of R, respectively (R,Δ). We start by proving some bounds on the perfectoid pure threshold using analytic spread.

Corollary 7.30. *Let $(R,\mathfrak{m})$ be a Noetherian local ring of residual characteristic $p>0$ and $\mathfrak{a}\subseteq\mathfrak{m}$ be an ideal. Then $\operatorname{ppt}(\mathfrak{a}) \leq \ell(\mathfrak{a})$, where $\ell(\mathfrak{a})$ denotes the analytic spread of $\mathfrak{a}$. Moreover,*

$$\operatorname{ppt}(\mathfrak{a}) \leq \min\{\dim(R_Q) \mid (p,\mathfrak{a}) \subseteq Q \in \operatorname{Spec}(R)\}.$$

Proof. We may assume that R is perfectoid pure as otherwise $\operatorname{ppt}(\mathfrak{a}) = -\infty$. First, if $\mathfrak{a} = xR$ is principal, then $\operatorname{ppt}(\mathfrak{a}) \leq 1$ by Theorem 7.5 since $xB \subseteq \mathcal{I}(B)$ for any R-algebra B.

We may replace R with $R(T)$ by Theorem 7.28 to assume that the residue field is infinite. Hence, if $\lambda = \ell(\mathfrak{a})$ then there are elements $x_1,\ldots,x_\lambda \in R$ such that $\overline{(x_1,\ldots,x_\lambda)} = \overline{\mathfrak{a}}$. It now follows from Proposition 7.19 that

$$\operatorname{ppt}(\mathfrak{a}) \leq \operatorname{ppt}(x_1R) + \cdots + \operatorname{ppt}(x_\lambda R) \leq \lambda = \ell(\mathfrak{a}).$$

For the second bound, since $\operatorname{ppt}(\mathfrak{a}) \leq \operatorname{ppt}((p,\mathfrak{a})) \leq \operatorname{ppt}((p,\mathfrak{a})R_Q)$, we may assume that $p \in \mathfrak{a}$ and $\mathfrak{a}$ is $\mathfrak{m}$-primary. But then $\ell(\mathfrak{a}) = \dim(R)$ and the conclusion follows from the first claim. □

We now extend an argument of Takagi–Watanabe [TW04, Proposition 2.6] to our setting.

Proposition 7.31. *Let $(R,\mathfrak{m},k)$ be a perfectoid pure Noetherian local ring with infinite residue field k. Let $\mathfrak{a}$ be an ideal of analytic spread λ and $x_1,\ldots,x_\lambda \in R$ form a minimal reduction of $\mathfrak{a}$. If $\operatorname{ppt}(\mathfrak{a}) > \lambda - r$ for some integer $\lambda \geq r \geq 1$, then $\mathfrak{a}^r \subseteq (x_1,\ldots,x_\lambda)$.*

Proof. We may assume that R is complete by Corollary 7.16. Let B be an absolutely integrally closed perfectoid R-algebra such that $\operatorname{ppt}(\mathfrak{a}) = \operatorname{ppt}(\mathfrak{a};B)$. By the assumption on $\operatorname{ppt}(\mathfrak{a})$ we may find an integer n and an R-linear map $\phi\colon B \to R$ such that $\phi(\overline{\mathfrak{a}^{1/n}B}\cdot\overline{\mathfrak{a}^{\lambda-r}B}) = R$. By Corollary 2.4, $\overline{\mathfrak{a}^{1/n}B} \subseteq (\overline{\mathfrak{a}^{1/n}B})_{\operatorname{perfd}} = (\mathfrak{a}B)_{\operatorname{perfd}}$. Thus by Proposition 7.4 applied to $I = (x_1,\ldots,x_\lambda)$, we have

$$\mathfrak{a}^r = \mathfrak{a}^r\phi((\mathfrak{a}B)_{\operatorname{perfd}}\overline{\mathfrak{a}^{\lambda-r}B}) \subseteq \phi(\mathfrak{a}_{\operatorname{perfd}}\overline{\mathfrak{a}^{\lambda}}B) \subseteq \phi((x_1,\ldots,x_\lambda)B) \subseteq (x_1,\ldots,x_\lambda).$$

□

As a consequence, we show that the perfectoid pure threshold of the maximal ideal detects regularity, generalizing the analogous characteristic $p>0$ result for F-pure thresholds, [TW04, Theorem 2.7(1)].

Corollary 7.32. *Let $(R, \mathfrak{m}, k)$ be a perfectoid pure Noetherian local ring of residual characteristic $p > 0$. Then $\operatorname{ppt}(\mathfrak{m}) > \dim(R) - 1$ if and only if R is regular (in which case $\operatorname{ppt}(\mathfrak{m}) = \dim(R)$).*

Proof. We may assume that R is complete by Corollary 7.16. By Theorem 7.28 we may assume that k is infinite. Taking $r = 1$ in Proposition 7.31 shows that $\mathfrak{m} \subseteq (x_1, \ldots, x_{\dim R}) \subseteq \mathfrak{m}$ and so R is regular. The converse is Corollary 7.21. □

We next prove some behavior of the perfectoid pure threshold under adjunction for *compatible* ideals (generalizing [DSNnBS26, Lemma 2.19]).

Proposition 7.33. *Let $(R, \mathfrak{m})$ be a Noetherian complete local ring of residue characteristic $p > 0$ and let B be a perfectoid R-algebra. If $I \subseteq R$ is ϕ-compatible for all $\phi \in \operatorname{Hom}_R(B, R)$, then for all $\mathfrak{a} \subseteq R$ such that $\mathfrak{a}B$ can be generated by elements admitting compatible systems of p-power roots we have the inequality*

$$\operatorname{ppt}(\mathfrak{a}(R/I); (B/IB)_{\text{perfd}}) \geq \operatorname{ppt}(\mathfrak{a}; B).$$

Proof. By Proposition 2.30, any $\phi \in \operatorname{Hom}_R(B, R)$ descends to $\overline{\phi} \in \operatorname{Hom}_R((B/IB)_{\text{perfd}}, R/I)$. Since a surjection would descend to a surjection, it follows immediately that

$$\mathcal{I}^R(B)(B/IB)_{\text{perfd}} \supseteq \mathcal{I}^{R/I}((B/IB)_{\text{perfd}})$$

and the proposition follows from Theorem 7.5. □

This result does not have an obvious extension to divisor pairs, since, in general, there is no appropriate divisor $\Delta_{\mathfrak{a}}$ on $R/\mathfrak{a}$ even in positive characteristic. In characteristic $p > 0$ though, there is a correct Cartier algebra-type statement and that is what we will mimic below. We will provide two results: one generalizing the notion of pairs and one putting an extra assumption ensuring that $\Delta_{\mathfrak{a}}$ is defined.

Definition 7.34. Let $(R, \mathfrak{m})$ be a Noetherian complete local ring of residual characteristic $p > 0$ that is reduced and $\mathbf{S}_2$, and let Δ be an effective Mumford $\mathbb{Q}$-divisor on R. We say that $\mathfrak{a}$ *is* (R, Δ)*-compatible* if for any perfectoid R-algebra B on which $M_B(<\Delta)$ can be defined and every $\phi \in \operatorname{Hom}_R(M_B(<\Delta), R)$, $(\phi \circ \iota_{B,\Delta})((\mathfrak{a}B)_{\text{perfd}}) \subset \mathfrak{a}$.

Remark 7.35. With notation as above, $\phi \circ \iota_{B,\Delta} \in \operatorname{Hom}_R(B, R)$. The set of all maps obtained this way, *i.e.* the image of $\operatorname{Hom}_R(M_B(<\Delta), R)$ in $\operatorname{Hom}_R(B, R)$ via precomposition with $\iota_{B,\Delta}$ forms a B-submodule, which we call $\mathscr{C}(B, <\Delta)$. The B-module structure comes from precomposition by multiplication by an element of B. Thus $\mathfrak{a}$ is (R, Δ)-compatible if and only if it is $\mathscr{C}(B, <\Delta)$-compatible. In particular, if $\mathfrak{a}$ is uniformly compatible then it is (R, Δ)-compatible for all Δ.

Furthermore, if $\mathfrak{a}$ is Δ-compatible, then by Proposition 2.30, $\mathscr{C}(B, <\Delta)$ descends to a $(B/\mathfrak{a})_{\text{perfd}}$-submodule $\overline{\mathscr{C}(B, <\Delta)} \subseteq \operatorname{Hom}_{R/\mathfrak{a}}((B/\mathfrak{a})_{\text{perfd}}, R/\mathfrak{a})$.

Definition 7.36. Let $(R, \mathfrak{m})$ be a Noetherian complete local ring, B a perfectoid R-algebra, and $\mathscr{C} \subseteq \operatorname{Hom}_R(B, R)$ be a B-submodule (i.e., $\mathscr{C}$ is stable under precomposition with elements in B). Then we call

$$\mathcal{I}(B, \mathscr{C}) \coloneqq \{x \in B \mid \phi(x) \in \mathfrak{m} \ \forall \phi \in \mathscr{C}\}$$

the *non-pure ideal* of $\mathscr{C}$.

The verification that $\mathcal{I}(B,\mathscr{C})$ is an ideal is straightforward. First, if $x \in \mathcal{I}(B,\mathscr{C})$ and $b \in B$, then for any $\phi \in \mathscr{C}$, the composition $\phi \circ \mathrm{mult}_b \in \mathscr{C}$, so $\phi(bx) \in \mathfrak{m}$. Second, if $x, y \in \mathcal{I}(B,\mathscr{C})$ then $\phi(x+y) = \phi(x) + \phi(y) \in \mathfrak{m}$ for all $\phi \in \mathscr{C}$.

Lemma 7.37. *Let $(R,\mathfrak{m})$ be a Noetherian complete local ring of residual characteristic $p > 0$ that is reduced and $\mathbf{S}_2$, and let Δ be an effective Mumford $\mathbb{Q}$-divisor on R. Let B be a perfectoid R-algebra such that $M_B(<\Delta)$ is defined. If $\mathfrak{a}$ is an (R,Δ)-compatible ideal, then*

$$\mathrm{ppt}(\mathfrak{b};\Delta;B) \leq \mathrm{Pt}(\mathfrak{b}(B/\mathfrak{a}B)_{\mathrm{perfd}}, \mathcal{I}((B/\mathfrak{a}B)_{\mathrm{perfd}}, \overline{\mathscr{C}(B,<\Delta)}))$$

for all $\mathfrak{b} \subseteq R$ (such that $\mathfrak{b}B$ can be generated by elements admitting compatible systems of p-power roots).

Proof. By Remark 7.35, $\mathcal{I}(B,<\Delta) = \mathcal{I}(B,\mathscr{C}(B,<\Delta))$. Since $\mathfrak{a}$ is (R,Δ)-compatible, for any $\phi \in \mathscr{C}(B,<\Delta)$, $\overline{\phi}(\overline{x}) \in \mathfrak{m}(R/\mathfrak{a})$ if and only if $\phi(x) \in \mathfrak{m}$. It follows immediately that

$$\mathcal{I}(B,<\Delta)(B/\mathfrak{a}B)_{\mathrm{perfd}} = \mathcal{I}((B/\mathfrak{a}B)_{\mathrm{perfd}}, \overline{\mathscr{C}(B,<\Delta)}).$$

The conclusion follows from Theorem 7.5. □

Proposition 7.38. *Let $(R,\mathfrak{m})$ be a Noetherian complete local normal domain of residual characteristic $p > 0$ and Δ be an integral effective Weil divisor on R such that (R,Δ) is perfectoid pure. Let $\mathfrak{a}$ be an (R,Δ)-compatible ideal of R with $R/\mathfrak{a}$ normal and define $\Delta_{\mathfrak{a}}$ to be the divisor corresponding to $I_{\Delta_{\mathfrak{a}}} \coloneqq (((I_\Delta \colon \mathfrak{a}) + \mathfrak{a})/\mathfrak{a})^{**}$ (which we assume to be nonzero) on $R/\mathfrak{a}$. If $\mathfrak{b}$ is an ideal of R, then*

$$\mathrm{ppt}(\mathfrak{b};\Delta) \leq \mathrm{ppt}(\mathfrak{b}(R/\mathfrak{a});\Delta_{\mathfrak{a}}).$$

Proof. By Proposition 7.17, we may find an absolutely integrally closed perfectoid R-algebra B such that $\mathrm{ppt}(\mathfrak{b};\Delta) = \mathrm{ppt}(\mathfrak{b};\Delta;B)$. In particular, $R \to M_B(<\Delta)$ splits, say by $\tilde{\phi}$. It follows that $\phi \coloneqq \widetilde{\phi} \circ \iota_{B,\Delta} \in \mathscr{C}(B,<\Delta)$ is surjective. Since I_Δ is $\mathscr{C}(B,<\Delta)$-compatible by Corollary 4.27, so is $(I_\Delta \colon \mathfrak{a})$ by Proposition 2.30. It follows from [Fay25, Proposition 2.34] that $J = (I_\Delta \colon \mathfrak{a}) + \mathfrak{a}$ is $\mathscr{C}(B,<\Delta)$-compatible and in particular radical by Proposition 2.30 (we assumed that (R,Δ) is perfectoid pure). Since every element of $(B/\mathfrak{a}B)_{\mathrm{perfd}}$ has a compatible system of p-th power roots, $M_{(B/\mathfrak{a}B)_{\mathrm{perfd}}}(<\Delta_{\mathfrak{a}})$ exists.

We now claim that

$$\overline{\mathscr{C}(B,<\Delta)} \subseteq \mathscr{C}((B/\mathfrak{a}B)_{\mathrm{perfd}},<\Delta_{\mathfrak{a}}).$$

Indeed, the image $J(R/\mathfrak{a})$ is radical and by [Fay25, Proposition 2.20], it is $\overline{\mathscr{C}(B,<\Delta)}$-compatible. Then, Proposition 2.30 gives that $(J(R/\mathfrak{a}))^{**} = I_{\Delta_{\mathfrak{a}}}$ is $\overline{\mathscr{C}(B,<\Delta)}$-compatible. The claim follows from Corollary 4.27.

The claim then implies that $\mathcal{I}((B/\mathfrak{a}B)_{\mathrm{perfd}},<\Delta_{\mathfrak{a}}) \subseteq \mathcal{I}((B/\mathfrak{a}B)_{\mathrm{perfd}}, \overline{\mathscr{C}(B,<\Delta)})$. Thus by Lemma 7.37 and Lemma 7.3 we obtain that

$$\begin{aligned}\mathrm{ppt}(\mathfrak{b};\Delta) = \mathrm{ppt}(\mathfrak{b};\Delta;B) &\leq \mathrm{Pt}(\mathfrak{b}(B/\mathfrak{a}B)_{\mathrm{perfd}}, \mathcal{I}((B/\mathfrak{a}B)_{\mathrm{perfd}}, \overline{\mathscr{C}(B,<\Delta)}))\\ &\leq \mathrm{Pt}(\mathfrak{b}(B/\mathfrak{a}B)_{\mathrm{perfd}}, \mathcal{I}((B/\mathfrak{a}B)_{\mathrm{perfd}},<\Delta_{\mathfrak{a}}))\\ &= \mathrm{ppt}(\mathfrak{b}(R/\mathfrak{a});\Delta_{\mathfrak{a}};(B/\mathfrak{a}B)_{\mathrm{perfd}}) \leq \mathrm{ppt}(\mathfrak{b}(R/\mathfrak{a});\Delta_{\mathfrak{a}}).\end{aligned}$$

□

We now present a quantification of the inversion of adjunction in [BMP+24, Theorem 6.6].

Proposition 7.39. *Let $(R, \mathfrak{m}, k)$ be a perfectoid pure complete intersection of residual characteristic $p > 0$ and $\mathfrak{a}$ be an ideal. If $f \in \mathfrak{a}$ is a regular element, then* $\operatorname{ppt}(\mathfrak{a}) \geq \operatorname{ppt}(\mathfrak{a}(R/fR)) + 1$.

Proof. Take a surjection $S := \mathrm{C}(k)[\![x, x_1, \ldots, x_d]\!] \to R$ such that $x \mapsto f$ and $\mathfrak{a}R^{S_\infty}_{\mathrm{perfd}}$ is generated by elements $(y_1, \ldots, y_n)$ that admit compatible systems of p-power roots (e.g., we could arrange that $\{y_1, \ldots, y_n\}$ is a subset of the images of $\{x_1, \ldots, x_d\}$).

If the image of $y_1^{a_1/p^m} \cdots y_n^{a_n/p^m}$ is not in $\mathcal{I}((R/fR)^{S_\infty}_{\mathrm{perfd}})$, then as in the proof of Theorem 3.14, we know that $R \xrightarrow{\cdot f y_1^{a_1/p^m} \cdots y_n^{a_n/p^m}} ((f)^{1/p^\infty} R^{S_\infty}_{\mathrm{perfd}})^-$ is pure. Thus by Lemma 3.2, we have

$$f^{(p^e-1)/p^e} y_1^{a_1/p^m} \cdots y_n^{a_n/p^m} \notin \mathcal{I}(R^{S_\infty}_{\mathrm{perfd}})$$

for all $e \gg 0$. The desired inequality then follows from Theorem 7.5. □

We end this subsection by comparing the perfectoid pure threshold with the log canonical threshold.

Proposition 7.40. *Let $(R, \mathfrak{m})$ be an excellent normal local ring of residual characteristic $p > 0$ admitting a dualizing complex and let $\mathfrak{a} \subseteq R$ be an ideal. Let Δ be an effective $\mathbb{Q}$-divisor on R such that $K_R + \Delta$ is $\mathbb{Q}$-Cartier. Then* $\operatorname{ppt}(\mathfrak{a}; \Delta) \leq \operatorname{lct}(\mathfrak{a}; \Delta)$.

Proof. We can choose a generating set $\mathfrak{a} = (x_1, \ldots, x_m)$ that satisfies Corollary 7.18. By Remark 4.26 and Theorem 5.6, we have

$$\begin{aligned}
\operatorname{ppt}(\mathfrak{a}; \Delta) &= \sup\left\{ \frac{|a|}{p^e} \mid a \in \mathbb{Z}^m_{\geq 0}, (R, \Delta + \frac{1}{p^e}\operatorname{div}(x_1^{a_1} \cdots x_m^{a_m})) \text{ is perfectoid pure} \right\} \\
&\leq \sup\left\{ \frac{|a|}{p^e} \mid a \in \mathbb{Z}^m_{\geq 0}, (R, \Delta + \frac{1}{p^e}\operatorname{div}(x_1^{a_1} \cdots x_m^{a_m})) \text{ is log canonical} \right\} \\
&\leq \sup\left\{ \frac{n}{p^e} \mid n \in \mathbb{Z}_{\geq 0}, (R, \mathfrak{a}^{n/p^e}, \Delta) \text{ is log canonical} \right\} = \operatorname{lct}(\mathfrak{a}; \Delta).
\end{aligned}$$

□

7.5. **Examples.** In this subsection we collect some computations on the perfectoid pure threshold, using various properties we developed earlier this section.

Example 7.41 (A_n singularities)**.** Let $R = \mathbb{Z}_p[\![a, b]\!]$ and $f = xy + z^{n+1}$ where x, y, z coincide with p, a, b up to a permutation. Then $1 \geq \operatorname{ppt}(\mathfrak{m}(R/fR))$ by Corollary 7.32 and $\operatorname{ppt}(\mathfrak{m}(R/fR)) \geq 1$ by Proposition 7.39 because $R/(f, z)R$ is perfectoid pure. Therefore $\operatorname{ppt}(\mathfrak{m}(R/fR)) = 1$ and thus coincides with the characteristic p value computed in [TW04].

Example 7.42. Consider the ring $R = \mathbb{Z}_p[\![x, y, z]\!]/(x^3 + y^3 + z^3)$. First, if $p \equiv 1 \pmod 3$, then R/pR is F-pure. From this we obtain that $\operatorname{ppt}(\mathfrak{m}) = 1$: the lower bound holds by Proposition 7.39 and the upper bound by Proposition 7.31.

Second, we claim that $\operatorname{ppt}(\mathfrak{m}) = 1 - p/(p^2 - 1)$ if $p \equiv 2 \pmod 3$. To see this, first note that by [Yos25, Proposition 6.10], we have $\operatorname{ppt}(p) = 1 - p/(p^2 - 1)$. However, (x, y, z) is the multiplier ideal of R and hence is uniformly compatible by [Fay25, Corollary 4.14]. In particular, $(x, y, z)_{\mathrm{perfd}} \subseteq \mathcal{I}(R^{A_\infty}_{\mathrm{perfd}})$ and thus by Proposition 7.8, we have $\operatorname{ppt}(\mathfrak{m}) = \operatorname{ppt}(p)$ since only monomials in p have nonzero thresholds.

Example 7.43. In a similar way, $R = \mathbb{Z}_p[\![x,y]\!]/(p^3+x^3+y^3)$ is not perfectoid pure if $p \equiv 2 \pmod 3$ (one can see this via Theorem 3.14: let $A = \mathbb{Z}_p[\![x,y]\!]$; we have $\operatorname{ppt}((p^3+x^3+y^3)A) < 1$ by [BJP+26, Theorem 4.5]). On the other hand, if $p \equiv 1 \pmod 3$ then R is perfectoid pure by [BMP+24, Example 7.4] and in fact we have $\operatorname{ppt}(\mathfrak{m}) = 0$ in this case: this is because the multiplier ideal is the entire $\mathfrak{m}$ and thus only units can split.

Example 7.44. More generally, consider $R = \mathbb{Z}_p[\![x_1,\dots,x_n]\!]/F_d(x_1,\dots,x_n)$, where F_d is a degree d form so that $F_d \neq 0 \bmod p$ and such that R/pR is F-pure. Then by Proposition 7.39 we have that $\operatorname{ppt}(\mathfrak{m}) \geq 1 + \operatorname{fpt}(\mathfrak{m}(R/pR)) = n - d + 1$, where the equality is due to [DSNB18, Theorem B]. On the other hand, if J is a homogeneous minimal reduction, then the socle degree of R/J is $\mathrm{e}(R) - 1 = d - 1$. Thus $\mathfrak{m}^{d-1} \not\subset J$ and by Proposition 7.31 we obtain that $n - d + 1 = \dim(R) - (d-1) \geq \operatorname{ppt}(\mathfrak{m})$. It follows that $\operatorname{ppt}(\mathfrak{m}) = n - d + 1$.

8. Perfectoid signature for pairs

Let $(A, \mathfrak{m}_A, k)$ be a complete regular local ring of dimension d with perfect residue field k of characteristic $p > 0$. Consider the following two cases:

- If $\operatorname{char} A = p > 0$, the *equicharacteristic $p > 0$ case*, then we set $A = \Lambda[\![x_1,\dots,x_d]\!]$, where $\Lambda = k$.
- If $\operatorname{char} A = 0$, the *unramified mixed characteristic case*, then we define $A = \Lambda[\![x_2,\dots,x_d]\!]$, where $\Lambda = W(k)$. In this case we set $x_1 = p$.

In either case, $x_1,\dots,x_d$ form a regular system of parameters, and we define

$$A_n := A[x_1^{1/p^n}, x_2^{1/p^n}, \dots, x_d^{1/p^n}].$$

These rings form a filtered system, and we set

$$A_\infty^{nc} := \varinjlim_n A_n.$$

Faltings's normalized length $\lambda_\infty(-)$ is then defined on $\mathfrak{m}_A$-power torsion A_∞^{nc}-modules and has the following properties:

- It is additive on short exact sequences;
- It is defined on almost categories;
- It commutes with filtered colimits with injective transition maps.

Our primary use of normalized length will be in the following setting. Let $(R, \mathfrak{m}, k)$ be a complete Noetherian normal local domain with perfect residue field of positive characteristic $p > 0$, and choose a Cohen–Gabber normalization $A \subset R$. With A_n and A_∞^{nc} as above, the *perfectoid signature* of R is then defined as

$$s^{\underline{x}}_{\mathrm{perfd}}(R) = \lambda_\infty(R^{A_\infty}_{\mathrm{perfd}}/I_\infty),$$

where

$$I_\infty := \mathcal{I}^R(R^{A_\infty}_{\mathrm{perfd}}) = \{z \in R^{A_\infty}_{\mathrm{perfd}} \mid R \xrightarrow{1\mapsto z} R^{A_\infty}_{\mathrm{perfd}} \text{ is not pure}\}$$

is the non-splitting ideal over R, in the notation of Definition 2.23. We retain here $\underline{x}$ in the notation because independence of the choice of parameters is not known in mixed characteristic. For more properties of the normalized length and perfectoid signature we refer the reader to [CLM+26] and the references within.

In equicharacteristic $p > 0$, the perfectoid signature agrees with the F-signature, giving a new interpretation of the F-signature via perfection [CLM+26, Theorem 3.7]. Analogously, to motivate our definition of the perfectoid signature of divisor pairs, we first give below a similar interpretation of the F-signature of divisor pairs from [BST12] via perfection. Throughout this section, we will make use of the following notation. For a finite extension $R \subseteq S$ of normal domains, let $\pi\colon \operatorname{Spec}(S) \to \operatorname{Spec}(R)$ be the induced morphism. If Δ_R is an effective $\mathbb{Q}$-divisor on $\operatorname{Spec}(R)$ whose pullback $\pi^*\Delta_R$ has integral coefficients, by writing $S(\Delta_R)$ we mean $S(\pi^*\Delta_R)$. As we are working with finite extensions of normal rings, such abuses of notation are harmless. Moreover, we also write $R^+(\Delta_R)$ to mean the union over all $S(\Delta_R)$ where $R \subseteq S \subseteq R^+$ are finite normal extensions such that the pullback of Δ_R to S has integral coefficients.

Theorem 8.1. *Let $(R, \mathfrak{m}, k)$ be a complete Noetherian normal local domain of characteristic $p > 0$ with perfect residue field. Fix a Noether-Cohen-Gabber normalization $A \subseteq R$ and a regular system of parameters $\underline{x}$ for A as above. Let $\Delta_R \geq 0$ be a $\mathbb{Q}$-divisor on $\operatorname{Spec}(R)$, and let $(R, \mathfrak{m}) \subseteq (S, \mathfrak{n})$ be a finite generically étale extension of complete normal local domains such that the pullback of Δ_R to S has integral coefficients. If*

$$\mathcal{I}^R(S_{\mathrm{perf}}, \Delta_R) = \{z \in S_{\mathrm{perf}} \mid R \xrightarrow{1 \mapsto z \otimes 1} S_{\mathrm{perf}} \otimes_S S(\Delta_R) \text{ is not pure}\}$$

is the non-splitting ideal over R, then

$$s(R, \Delta_R) = \frac{1}{r} \cdot \lambda_\infty(S_{\mathrm{perf}}/\mathcal{I}^R(S_{\mathrm{perf}}, \Delta_R))$$

where $r = \operatorname{rank}_R S$.

Proof. Write $d = \dim R$, and for $q = p^e$ let

$$\begin{aligned} J_e &= \{z \in S \mid R \xrightarrow{1 \mapsto z^{1/q} \otimes 1} S^{1/q} \otimes_S S(\Delta_R) \text{ is not pure}\} \\ &= \{z \in S \mid \varphi(z^{1/q}) \in \mathfrak{m} \text{ for all } \varphi \in \operatorname{Hom}_R(S(q\Delta_R)^{1/q}, R)\}. \end{aligned}$$

We first claim that

$$\lambda_\infty(S_{\mathrm{perf}}/\mathcal{I}^R(S_{\mathrm{perf}}, \Delta_R)) = \lim_{e \to \infty} \ell_R(S/J_e)/q^d. \tag{8.1.1}$$

To see this, choose $0 \neq g \in A$ such that $g^{1/q} S_{\mathrm{perf}} \subseteq S^{1/q}[A_{\mathrm{perf}}] = S^{1/q} \otimes_{A^{1/q}} A_{\mathrm{perf}}$ for all e (see [PT18, Lemma 2.3] for further details). We have

$$J_e^{1/q} \cdot S^{1/q}[A_{\mathrm{perf}}] \subseteq \mathcal{I}^R(S_{\mathrm{perf}}, \Delta_R) \cap S^{1/q}[A_{\mathrm{perf}}] \subseteq (J_e : g)^{1/q} \cdot S^{1/q}[A_{\mathrm{perf}}]$$

where the first containment follows from the string of inclusion $S^{1/q} \subseteq S^{1/q}[A_{\mathrm{perf}}] \subseteq S_{\mathrm{perf}}$ and the second from the argument in [CLM+26, Lemma 3.6], both after applying $_ \otimes_S S(\Delta_R)$. Consequently,

$$0 \leq \frac{\ell_R(S/J_e)}{q^d} - \lambda_\infty\left(\frac{S^{1/q}[A_{\mathrm{perf}}]}{\mathcal{I}^R(S_{\mathrm{perf}}, \Delta_R) \cap S^{1/q}[A_{\mathrm{perf}}]}\right) \leq \frac{\ell_R(S/(J_e, g))}{q^d}$$

which limits to 0 as $e \to \infty$ as $\mathfrak{m}^{[q]} \cdot S \subseteq J_e$ and $\dim S/(g) < d$. As $S_{\mathrm{perf}} = \bigcup_e S^{1/q}[A_{\mathrm{perf}}]$, (8.1.1) then follows from the definition of the normalized length as in [CLM+26, proof of Theorem 3.7].

Let Ram be the ramification divisor of the generically étale extension $R \subseteq S$, consider now the sequence of ideals $J'_e = \{z \in S \mid \varphi(z^{1/q}) \in \mathfrak{m} \text{ for all } \varphi \in \operatorname{Hom}_R(S(q\Delta_R + \mathrm{Ram})^{1/q}, R)\}$. To

conclude, it suffices to observe

$$\lim_{e\to\infty} \frac{\ell_R(S/J_e)}{q^d} = \lim_{e\to\infty} \frac{\ell_R(S/J'_e)}{q^d} = r\cdot s(R,\Delta_R).$$

The first equality follows from [CRST18, proof of Lemma 2.8] after noting that $J_e \subseteq J'_e \subseteq (J_e : x)$ for any $x \in R$ with $\operatorname{div}_S(x) \geq \operatorname{Ram}$. The second equality is precisely [CRST18, Lemma 4.5], whose argument notably does not make use of the additional hypothesis $\pi^*\Delta_R - \operatorname{Ram} \geq 0$. □

8.1. **Definition and first properties.**

Definition 8.2. Let $(R,\mathfrak{m},k)$ be a complete Noetherian normal local domain of mixed characteristic $(0,p)$ with perfect residue field. Fix a Noether–Cohen–Gabber normalization $A \subseteq R$ and a regular system of parameters $\underline{x}$ for A as above. Let $\Delta_R \geq 0$ be a $\mathbb{Q}$-divisor on $\operatorname{Spec}(R)$, and let $f\colon (R,\mathfrak{m}) \to (S,\mathfrak{n})$ be a finite generically étale extension of complete normal local domains such that the pullback of Δ_R to S has integral coefficients. Following the notation from Definition 2.23 with some abbreviation, we set

(8.2.1)

$$\mathcal{I}^R(S^{A_\infty}_{\mathrm{perfd}},\Delta_R) = \{x \in S^{A_\infty}_{\mathrm{perfd}} \mid R \xrightarrow{1\mapsto x\otimes 1} S^{A_\infty}_{\mathrm{perfd}} \otimes_S S(\Delta_R) \text{ is not pure}\}$$

(8.2.2)

$$= \{x \in S^{A_\infty}_{\mathrm{perfd}} \mid \psi(x\otimes 1) \in \mathfrak{m}_R \text{ for any } \psi \in \operatorname{Hom}_R(S^{A_\infty}_{\mathrm{perfd}} \otimes_S S(\Delta_R), R)\}.$$

The *perfectoid signature* of the pair (R,Δ_R) is then defined by

$$s^{\underline{x}}_{\mathrm{perfd}}(R,\Delta_R) = \frac{1}{r}\cdot \lambda_\infty(S^{A_\infty}_{\mathrm{perfd}}/\mathcal{I}^R(S^{A_\infty}_{\mathrm{perfd}},\Delta_R))$$

where $r = \operatorname{rank}_R S$.

We now prove that the definition is independent of the choice of S.

Lemma 8.3. *In the setting of Definition 8.2, let $f'\colon (S,\mathfrak{n}) \to (S',\mathfrak{n}')$ be another generically étale finite extension of complete normal local domains. Choose $0 \neq g \in A$ such that $f[1/g]$ and $f'[1/g]$ are étale. Then the canonical map*

$$S'^{A_\infty}_{\mathrm{perfd}}/\mathcal{I}^R(S^{A_\infty}_{\mathrm{perfd}},\Delta_R)S'^{A_\infty}_{\mathrm{perfd}} \to S'^{A_\infty}_{\mathrm{perfd}}/\mathcal{I}^R(S'^{A_\infty}_{\mathrm{perfd}},\Delta_R)$$

is a g-almost isomorphism, i.e., the ideal $\mathcal{I}^R(S^{A_\infty}_{\mathrm{perfd}},\Delta_R)$ is independent (in the sense of extensions) of the choice of S up to almost isomorphism.

Proof. By Lemma 2.25, we have

$$\mathcal{I}^R(S^{A_\infty}_{\mathrm{perfd}},\Delta_R) = \bigcup_t (I_t S^{A_\infty}_{\mathrm{perfd}} \otimes_S S(\Delta_R) :_{S^{A_\infty}_{\mathrm{perfd}}} u_t\otimes 1).$$

By [BS22, Theorem 10.9], $S^{A_\infty}_{\mathrm{perfd}} \to S'^{A_\infty}_{\mathrm{perfd}}$ is g-almost étale. Hence by [CLM+26, Proposition 2.25], it is p-complete g-almost faithfully flat. Since one can write $I_t S^{A_\infty}_{\mathrm{perfd}} \otimes_S S(\Delta_R) :_{S^{A_\infty}_{\mathrm{perfd}}} u_t\otimes 1$ as

$$\ker(S^{A_\infty}_{\mathrm{perfd}} \xrightarrow{1\mapsto u_t\otimes 1} S^{A_\infty}_{\mathrm{perfd}} \otimes_S S(\Delta_R)/I_t S^{A_\infty}_{\mathrm{perfd}} \otimes_S S(\Delta_R))$$

we have

$$\mathcal{I}^R(S^{A_\infty}_{\mathrm{perfd}},\Delta_R)S'^{A_\infty}_{\mathrm{perfd}} \overset{a}{\simeq} \bigcup_t (I_t S'^{A_\infty}_{\mathrm{perfd}} \otimes_S S(\Delta_R) :_{S'^{A_\infty}_{\mathrm{perfd}}} u_t\otimes 1)$$

by p-complete g-almost faithfully flatness of ${S'}^{A_\infty}_{\text{perfd}}$. Now it suffices to show that

$$\bigcup_t (I_t {S'}^{A_\infty}_{\text{perfd}} \otimes_S S(\Delta_R) :_{{S'}^{A_\infty}_{\text{perfd}}} u_t \otimes 1) = \bigcup_t (I_t {S'}^{A_\infty}_{\text{perfd}} \otimes_{S'} S'(\Delta_R) :_{{S'}^{A_\infty}_{\text{perfd}}} u_t \otimes 1)$$

since the latter is $\mathcal{I}^R({S'}^{A_\infty}_{\text{perfd}}, \Delta_R)$ by Lemma 2.25 again. In other words, we need to show that $R \to {S'}^{A_\infty}_{\text{perfd}} \otimes_S S(\Delta_R)$ sending 1 to $x \otimes 1$ is pure if and only if $R \to {S'}^{A_\infty}_{\text{perfd}} \otimes_{S'} S'(\Delta_R)$ sending 1 to $x \otimes 1$ is pure. By local duality it suffices to show that $H^d_{\mathfrak{m}}(R(K_R)) \to H^d_{\mathfrak{m}}(R(K_R) \otimes_R {S'}^{A_\infty}_{\text{perfd}} \otimes_S S(\Delta_R))$ is injective if and only if $H^d_{\mathfrak{m}}(R(K_R)) \to H^d_{\mathfrak{m}}(R(K_R) \otimes_R {S'}^{A_\infty}_{\text{perfd}} \otimes_{S'} S'(\Delta_R))$ is injective. We claim that $H^d_{\mathfrak{m}}(R(K_R) \otimes_R {S'}^{A_\infty}_{\text{perfd}} \otimes_S S(\Delta_R)) \simeq H^d_{\mathfrak{m}}(R(K_R) \otimes_R {S'}^{A_\infty}_{\text{perfd}} \otimes_{S'} S'(\Delta_R))$. Note that top degree local cohomology of two modules are the same if the difference is supported on a codimension ≥ 2 subset of $\operatorname{Spec}(S)$. Now if we restrict to the regular locus U of S, then ${S'}^{A_\infty}_{\text{perfd}} \otimes_{S'} S'(\Delta_R) \cong {S'}^{A_\infty}_{\text{perfd}} \otimes_S S(\Delta_R)$ because $S(\Delta_R)$ is a Cartier divisor when restricting to U. Since the complement of U has codimension at least two on $\operatorname{Spec}(S)$ as S is normal, we obtain the isomorphism on top local cohomology as wanted. □

Lemma 8.4. *Let R be a perfectoid algebra and $R^\flat$ be its tilt. Denote by $\sharp : R^\flat \to R$ the untilt map. Then for any $x, y \in R^\flat$ we have*

$$(x+y)^\sharp = x^\sharp + y^\sharp + p \cdot (x^{1/p} y^{1/p})^\sharp \cdot u + p^2 \cdot v$$

for some $u, v \in R$.

Proof. Factor the untilt map as

$$\sharp \colon R^\flat \xrightarrow{[\cdot]} A_{\text{inf}}(R) \xrightarrow{\theta} R$$

where $[\cdot]$ is the Teichmüller lift and θ is Fontaine's map. Hence it suffices to show that $[x+y] = [x] + [y] + p \cdot [x^{1/p} y^{1/p}] \cdot u' + p^2 \cdot v'$ for some $u', v' \in A_{\text{inf}}(R)$. This follows from a simple Witt vector computation. □

Recall from the beginning of this section that A_∞ is the p-completion of the union of $A_n = A[x_1^{1/p^n}, x_2^{1/p^n}, \ldots, x_d^{1/p^n}]$ where $A = W(k)[\![x_2, \ldots, x_d]\!]$.

Lemma 8.5. *Let $t = p^\flat, x_2^\flat, \ldots, x_d^\flat \in A_\infty^\flat$ satisfy $t^\sharp = p, (x_2^\flat)^\sharp = x_2, \ldots, (x_d^\flat)^\sharp = x_d$, respectively, and put $(A_\infty^\flat)^{\text{un}} = k[\![p^\flat, x_2^\flat, \ldots, x_d^\flat]\!]_{\text{perf}}$ so that $k[\![p^\flat, x_2^\flat, \ldots, x_d^\flat]\!] \subseteq (A_\infty^\flat)^{\text{un}} \subseteq A_\infty^\flat$. For any $0 \neq f \in \mathfrak{m}_A$, we can find elements $f_2, \ldots, f_d$ inside the maximal ideal of $k[\![p^\flat, x_2^\flat, \ldots, x_d^\flat]\!]$ (so also $f_2^\sharp, \ldots, f_d^\sharp \in \mathfrak{m}_A A_\infty$) such that $\sqrt{(f, f_2^\sharp, \ldots, f_d^\sharp)} = \mathfrak{m}_{A_\infty}$.*

Proof. We first recall the following fact which we shall use repeatedly throughout the proof: for any element $z \in A_\infty$, there is some $y \in (A_\infty^\flat)^{\text{un}}$ and $y' \in A_\infty$ so that $z = y^\sharp + p \cdot y'$ in A_∞. Indeed, to see this, begin by writing $z = y_1^\sharp + p \cdot y_1'$ for some $y_1 \in A_\infty^\flat$ and $y_1' \in A_\infty$. Since $(A_\infty^\flat)^{\text{un}}/(t) = A_\infty^\flat/(t)$, we may write $y_1 = y + t \cdot e$ for some $y \in (A_\infty^\flat)^{\text{un}}$ and $e \in A_\infty^\flat$. This gives $y_1^\sharp = y^\sharp + p \cdot y_2'$ (e.g., by Lemma 8.4) for some $y_2' \in A_\infty$, and putting $y' = y_1' + y_2'$ we have $z = y^\sharp + p \cdot y'$ for $y \in (A_\infty^\flat)^{\text{un}}$ and $y' \in A_\infty$.

Proceeding with the proof, write $f = p^n a$ in A such that $p \nmid a$ with $n \in \mathbb{Z}_{\geq 0}$ and $a \in A$ non-zero. If a is a unit, then we may take $f_2 := x_2^\flat, \ldots, f_d := x_d^\flat$. Thus, we may assume a is not a unit, so that the height of (p, a) in A is two. Choose $\tilde{f}_3, \ldots, \tilde{f}_d$ in A so that $p, a, \tilde{f}_3, \ldots, \tilde{f}_d$ is a system of parameters of A. Write $a = x^\sharp + p \cdot b$ for $x \in (A_\infty^\flat)^{\text{un}}$ and $b \in A_\infty$.

Assume first that b is not a unit. In this case, we first argue that $a, p \in \sqrt{(f, (x+t)^\sharp)}$. By Lemma 8.4 we have $(x+t)^\sharp = x^\sharp + p + p^{(p+1)/p} \cdot c$ for some $c \in A_\infty$. This gives

$$p^n \cdot (x+t)^\sharp - f = p^{n+1} \cdot (1 + p^{1/p} \cdot c - b).$$

Since $1 + p^{1/p} \cdot c - b$ is a unit, we see $p \in \sqrt{(f, (x+t)^\sharp)}$. As $(x+t)^\sharp = x^\sharp + p + p^{(p+1)/p} \cdot c$, this gives $x^\sharp \in \sqrt{(f, (x+t)^\sharp)}$ and hence also $a = x^\sharp + p \cdot b \in \sqrt{(f, (x+t)^\sharp)}$ as claimed. For $j > 2$, we can then write $\tilde{f}_j = g_j^\sharp + pv_j$ for $g_j \in (A_\infty^\flat)^{\mathrm{un}}$ and $v_j \in A_\infty$. Since $A_\infty/(p) = A_\infty^\flat/(t)$, note also that $g_j \in \sqrt{(t, x_2^\flat, \ldots, x_d^\flat)A_\infty^\flat}$. Putting $g_2 = x + t$, again noting that $g_2 \in \sqrt{(t, x_2^\flat, \ldots, x_d^\flat)A_\infty^\flat}$ as $f \in \mathfrak{m}_A$, we have that $\mathfrak{m}_{A_\infty} = \sqrt{(p, a, \tilde{f}_3, \ldots, \tilde{f}_d)} = \sqrt{(f, g_2^\sharp, \ldots, g_d^\sharp)}$. Taking $N = p^e$ for $e \gg 0$ so that $f_2 = g_2^N, \ldots, f_d = g_d^N$ are all in $(t, x_2^\flat, \ldots, x_d^\flat) \subseteq k[\![t, x_2^\flat, \ldots, x_d^\flat]\!]$, this gives $\sqrt{(f, f_2^\sharp, \ldots, f_d^\sharp)} = \sqrt{(f, g_2^\sharp, \ldots, g_d^\sharp)} = \mathfrak{m}_{A_\infty}$ as desired.

It remains to treat the case where b is a unit, where we begin by arguing that $a, p \in \sqrt{(f, (x+t^p)^\sharp)}$. Again by Lemma 8.4 we have $(x+t^p)^\sharp = x^\sharp + p^p + p^2 \cdot d$ for some $d \in A_\infty$. This gives

$$p^n \cdot (x+t^p)^\sharp - f = p^{n+1} \cdot (p^{p-1} + p \cdot d - b).$$

Since $p^{p-1} + p \cdot d - b$ is a unit, we see $p \in \sqrt{(f, (x+t^p)^\sharp)}$. As $(x+t^p)^\sharp = x^\sharp + p^p + p^2 \cdot d$, this gives $x^\sharp \in \sqrt{(f, (x+t^p)^\sharp)}$ and hence also $a = x^\sharp + p \cdot b \in \sqrt{(f, (x+t^p)^\sharp)}$ as claimed. For $j > 2$, we can then write $\tilde{f}_j = g_j^\sharp + pv_j$ for $g_j \in (A_\infty^\flat)^{\mathrm{un}}$ and $v_j \in A_\infty$. Since $A_\infty/(p) = A_\infty^\flat/(t)$, note also that $g_j \in \sqrt{(t, x_2^\flat, \ldots, x_d^\flat)A_\infty^\flat}$. Putting $g_2 = x + t^p$, again noting that $g_2 \in \sqrt{(t, x_2^\flat, \ldots, x_d^\flat)A_\infty^\flat}$ as $f \in \mathfrak{m}_A$, we have that $\mathfrak{m}_{A_\infty} = \sqrt{(p, a, \tilde{f}_3, \ldots, \tilde{f}_d)} = \sqrt{(f, g_2^\sharp, \ldots, g_d^\sharp)}$. Taking $N = p^e$ for $e \gg 0$ so that $f_2 = g_2^N, \ldots, f_d = g_d^N$ are all in $(t, x_2^\flat, \ldots, x_d^\flat) \subseteq k[\![t, x_2^\flat, \ldots, x_d^\flat]\!]$, this gives $\sqrt{(f, f_2^\sharp, \ldots, f_d^\sharp)} = \sqrt{(f, g_2^\sharp, \ldots, g_d^\sharp)} = \mathfrak{m}_{A_\infty}$ as desired. □

Proposition 8.6. *With notations as in Lemma 8.3. Let $r' = \mathrm{rank}_S S'$, then we have*

$$\lambda_\infty(S'^{A_\infty}_{\mathrm{perfd}}/\mathcal{I}^R(S'^{A_\infty}_{\mathrm{perfd}}, \Delta_R)) = r' \cdot \lambda_\infty(S^{A_\infty}_{\mathrm{perfd}}/\mathcal{I}^R(S^{A_\infty}_{\mathrm{perfd}}, \Delta_R)).$$

Proof. We follow the proof of [CLM+26, Proposition 6.12]. Choose $0 \neq g \in \mathfrak{m}_A$ such that $A[1/g] \to S[1/g] \to S'[1/g]$ are finite étale maps and $S'[1/g]$ is a finite free $S[1/g]$-module. By [BS22, Theorem 10.9], we have that $A_\infty/p \to S^{A_\infty}_{\mathrm{perfd}}/p \to S'^{A_\infty}_{\mathrm{perfd}}/p$ are g-almost finite étale maps. By Lemma 8.5, we can pick $f_2^\sharp, \ldots, f_d^\sharp \in \mathfrak{m}_A A_\infty$ such that $\sqrt{(pg, f_2^\sharp, \ldots, f_d^\sharp)} = \mathfrak{m}_{A_\infty}$. Thus, modulo $(f_2^\sharp, \ldots, f_d^\sharp)_{\mathrm{perfd}} = ((f_2^\sharp)^{1/p^\infty}, \ldots, (f_d^\sharp)^{1/p^\infty})^- \subseteq A_\infty$, we have

$$A_\infty/(p, (f_2^\sharp, \ldots, f_d^\sharp)_{\mathrm{perfd}}) \to S^{A_\infty}_{\mathrm{perfd}}/(p, (f_2^\sharp, \ldots, f_d^\sharp)_{\mathrm{perfd}}) \to S'^{A_\infty}_{\mathrm{perfd}}/(p, (f_2^\sharp, \ldots, f_d^\sharp)_{\mathrm{perfd}})$$

are p-almost finite étale maps since $\sqrt{p} = (g)_{\mathrm{perfd}} = \mathfrak{m}_{A_\infty}$ modulo $(f_2^\sharp, \ldots, f_d^\sharp)_{\mathrm{perfd}}$. Note that for any perfectoid ring, its p-torsion is exactly $\sqrt{p}$-torsion which is p-almost zero. Hence by modding out p-torsion if necessary, we may assume that $A_\infty/(f_2^\sharp, \ldots, f_d^\sharp)_{\mathrm{perfd}}$, $S^{A_\infty}_{\mathrm{perfd}}/(f_2^\sharp, \ldots, f_d^\sharp)_{\mathrm{perfd}}$ and $S'^{A_\infty}_{\mathrm{perfd}}/(f_2^\sharp, \ldots, f_d^\sharp)_{\mathrm{perfd}}$ are all p-torsion free. Removing p-torsion gives p-almost isomorphisms, so it does not change normalized length by [CLM+26, Proposition 4.10]. The reductions modulo

p are also p-almost isomorphic, so it suffices to prove uniform p-almost freeness of rank r' after removing p-torsion.

Now by the tilting equivalence (see [Sch12, Theorem 5.25]), we know that

$$(A_\infty^\flat/(f_2,\dots,f_d)_{\mathrm{perf}})^\wedge \to (S^{A_\infty}_{\mathrm{perfd}}/(f_2^\sharp,\dots,f_d^\sharp)_{\mathrm{perfd}})^\flat \to (S'^{A_\infty}_{\mathrm{perfd}}/(f_2^\sharp,\dots,f_d^\sharp)_{\mathrm{perfd}})^\flat$$

is t-almost finite étale, where the completion is t-adic. We then claim that the natural map

$$k[\![t]\!]^\wedge_{\mathrm{perf}} \to (A_\infty^\flat/(f_2,\dots,f_d)_{\mathrm{perf}})^\wedge \simeq (k[\![t,x_2^\flat,\dots,x_d^\flat]\!]/(f_2,\dots,f_d))^\wedge_{\mathrm{perf}}$$

is t-almost finite étale. It suffices to prove without the completion. By the almost purity theorem in characteristic p, it suffices to show that $k[\![t]\!]_{\mathrm{perf}}[1/t] \to (k[\![t,x_2^\flat,\dots,x_d^\flat]\!]/(f_2,\dots,f_d))_{\mathrm{perf}}[1/t]$ is finite étale. Since perfection commutes with localization, we may proceed the localization first. We claim that

$$k[\![t]\!][1/t]_{\mathrm{perf}} \to (k[\![t,x_2^\flat,\dots,x_d^\flat]\!]/(f_2,\dots,f_d))[1/t]_{\mathrm{perf}}$$

is a finite étale extension: before perfection, $(k[\![t,x_2^\flat,\dots,x_d^\flat]\!]/(f_2,\dots,f_d))[1/t]$ is a finite extension of a field, so $(k[\![t,x_2^\flat,\dots,x_d^\flat]\!]/(f_2,\dots,f_d))[1/t]_{\mathrm{red}}$ is a finite product of finite field extension of $k[\![t]\!][1/t]$, thus after perfection, it is a finite product of finite (étale) extension of the perfect field $k[\![t]\!][1/t]_{\mathrm{perf}}$. This proves the claim.

It follows that, in particular, $(A_\infty^\flat/(f_2,\dots,f_d)_{\mathrm{perf}})^\wedge[1/p^\flat]$ and thus $(S^{A_\infty}_{\mathrm{perfd}}/(f_2^\sharp,\dots,f_d^\sharp)_{\mathrm{perfd}})^\flat[1/p^\flat]$ are product of perfect fields, hence $S^{A_\infty}_{\mathrm{perfd}}/(f_2^\sharp,\dots,f_d^\sharp)_{\mathrm{perfd}}[1/p]$ is a finite product of perfectoid fields. But we also know that $(S^{A_\infty}_{\mathrm{perfd}}/(f_2^\sharp,\dots,f_d^\sharp)_{\mathrm{perfd}})[1/p] \to (S'^{A_\infty}_{\mathrm{perfd}}/(f_2^\sharp,\dots,f_d^\sharp)_{\mathrm{perfd}})[1/p]$ is finite étale and finite free (of constant rank r'). Hence the latter is also a finite product of perfectoid fields. By tilting again, we know that

$$(S^{A_\infty}_{\mathrm{perfd}}/(f_2^\sharp,\dots,f_d^\sharp)_{\mathrm{perfd}})^\flat[1/p^\flat] \to (S'^{A_\infty}_{\mathrm{perfd}}/(f_2^\sharp,\dots,f_d^\sharp)_{\mathrm{perfd}})^\flat[1/p^\flat].$$

is a finite free extension of constant rank r' (and both are finite product of perfect fields).

By [CLM+26, Lemma 6.8], we know that $(S'^{A_\infty}_{\mathrm{perfd}}/(f_2^\sharp,\dots,f_d^\sharp)_{\mathrm{perfd}})^\flat$ is uniformly $p^\flat$-almost finite free of rank r' over $(S^{A_\infty}_{\mathrm{perfd}}/(f_2^\sharp,\dots,f_d^\sharp)_{\mathrm{perfd}})^\flat$, and thus $S'^{A_\infty}_{\mathrm{perfd}}/(p,(f_2^\sharp,\dots,f_d^\sharp)_{\mathrm{perfd}})$ is uniformly p-almost finite free of rank r' over $S^{A_\infty}_{\mathrm{perfd}}/(p,(f_2^\sharp,\dots,f_d^\sharp)_{\mathrm{perfd}})$. Also note that $(g)_{\mathrm{perfd}} = (p^{1/p^\infty})$ in $A_\infty/(f_2^\sharp,\dots,f_d^\sharp)_{\mathrm{perfd}}$. We choose a sequence of elements $\varepsilon_n \in (g)_{\mathrm{perfd}}$ so that $\varepsilon_n = p^{1/p^n} \pmod{(f_2^\sharp,\dots,f_d^\sharp)_{\mathrm{perfd}}}$. By uniformly p-almost finite freeness (see [CLM+26, Definition 6.7]),

$$\begin{aligned} S'^{A_\infty}_{\mathrm{perfd}}/(p,(f_2^\sharp,\dots,f_d^\sharp)_{\mathrm{perfd}}) &\xrightarrow{\phi_{\varepsilon_n}} \left(S^{A_\infty}_{\mathrm{perfd}}/(p,(f_2^\sharp,\dots,f_d^\sharp)_{\mathrm{perfd}})\right)^{r'} \\ &\xrightarrow{\psi_{\varepsilon_n}} S'^{A_\infty}_{\mathrm{perfd}}/(p,(f_2^\sharp,\dots,f_d^\sharp)_{\mathrm{perfd}}) \end{aligned}$$

where the two-way compositions are both multiplication by p^{1/p^n}. Since $S'^{A_\infty}_{\mathrm{perfd}}/p$ is g-almost finite projective over $S^{A_\infty}_{\mathrm{perfd}}/p$ by [BS22, Theorem 10.9], we can apply [CLM+26, Lemma 6.1.1]

to obtain the following commutative diagram:

$$\begin{array}{ccccc} {S'}^{A_\infty}_{\mathrm{perfd}}/p & \xrightarrow{\widetilde{\phi}_{\varepsilon_n}} & (S^{A_\infty}_{\mathrm{perfd}}/p)^{r'} & \xrightarrow{\widetilde{\psi}_{\varepsilon_n}} & {S'}^{A_\infty}_{\mathrm{perfd}}/p \\ \downarrow{\scriptstyle can} & & \downarrow{\scriptstyle can} & & \downarrow{\scriptstyle can} \\ {S'}^{A_\infty}_{\mathrm{perfd}}/(p,(f_2^\sharp,\dots,f_d^\sharp)_{\mathrm{perfd}}) & \xrightarrow{\varepsilon_n\cdot\phi_{\varepsilon_n}} & (S^{A_\infty}_{\mathrm{perfd}}/(p,(f_2^\sharp,\dots,f_d^\sharp)_{\mathrm{perfd}}))^{r'} & \xrightarrow{\psi_{\varepsilon_n}} & {S'}^{A_\infty}_{\mathrm{perfd}}/(p,(f_2^\sharp,\dots,f_d^\sharp)_{\mathrm{perfd}}) \end{array}$$

Below we will only show

$$\lambda_\infty({S'}^{A_\infty}_{\mathrm{perfd}}/\mathcal{I}^R({S'}^{A_\infty}_{\mathrm{perfd}},\Delta_R)) \le r'\cdot\lambda_\infty(S^{A_\infty}_{\mathrm{perfd}}/\mathcal{I}^R(S^{A_\infty}_{\mathrm{perfd}},\Delta_R))$$

as the other direction is similar (and slightly easier).

We set $C_{\varepsilon_n}=\mathrm{coker}(\widetilde{\psi}_{\varepsilon_n})$. Since $(p,(f_2^\sharp,\dots,f_d^\sharp)_{\mathrm{perfd}})=(p,(f_2^\sharp)^{1/p^\infty},\dots,(f_d^\sharp)^{1/p^\infty})$ and C_{ε_n} is almost finitely generated, we can find $m\gg n$ such that $\varepsilon_n^3\cdot C_{\varepsilon_n}\subset((f_2^\sharp)^{1/p^m},\dots,(f_d^\sharp)^{1/p^m})C_{\varepsilon_n}$, i.e.,

$$\varepsilon_n^3\cdot({S'}^{A_\infty}_{\mathrm{perfd}}/p)\subset\mathrm{im}(\widetilde{\psi}_{\varepsilon_n})+((f_2^\sharp)^{1/p^m},\dots,(f_d^\sharp)^{1/p^m})({S'}^{A_\infty}_{\mathrm{perfd}}/p). \tag{8.6.1}$$

Increasing m if necessary, we may assume that $\varepsilon_n=p^{1/p^n}$ in $A_\infty/((f_2^\sharp)^{1/p^m},\dots,(f_d^\sharp)^{1/p^m})$ up to units. By Lemma 8.3, we have

$$\begin{aligned}\lambda_\infty({S'}^{A_\infty}_{\mathrm{perfd}}/\mathcal{I}^R({S'}^{A_\infty}_{\mathrm{perfd}},\Delta_R)) &= \lambda_\infty({S'}^{A_\infty}_{\mathrm{perfd}}/\mathcal{I}^R(S^{A_\infty}_{\mathrm{perfd}},\Delta_R){S'}^{A_\infty}_{\mathrm{perfd}})\\ &\le r'\cdot\lambda_\infty(S^{A_\infty}_{\mathrm{perfd}}/\mathcal{I}^R(S^{A_\infty}_{\mathrm{perfd}},\Delta_R))+\lambda_\infty(C_{\varepsilon_n}/\mathcal{I}^R(S^{A_\infty}_{\mathrm{perfd}},\Delta_R)C_{\varepsilon_n})\end{aligned}$$

Therefore it suffices to show that for any $\delta>0$, there exists some $n>0$ such that

$$\lambda_\infty(C_{\varepsilon_n}/\mathcal{I}^R(S^{A_\infty}_{\mathrm{perfd}},\Delta_R)C_{\varepsilon_n})<\delta.$$

Now we estimate the normalized length:

$$\begin{aligned}&\lambda_\infty(C_{\varepsilon_n}/\mathcal{I}^R(S^{A_\infty}_{\mathrm{perfd}},\Delta_R)C_{\varepsilon_n})\\ &\le\lambda_\infty({S'}^{A_\infty}_{\mathrm{perfd}}/((p,f_2^\sharp,\dots,f_d^\sharp){S'}^{A_\infty}_{\mathrm{perfd}}+\mathrm{im}(\widetilde{\psi}_{\varepsilon_n})))\\ &\le p^{m(d-1)}\cdot\lambda_\infty({S'}^{A_\infty}_{\mathrm{perfd}}/((p,(f_2^\sharp)^{1/p^m},\dots,(f_d^\sharp)^{1/p^m}){S'}^{A_\infty}_{\mathrm{perfd}}+\mathrm{im}(\widetilde{\psi}_{\varepsilon_n})))\\ &\le p^{m(d-1)}\cdot\lambda_\infty({S'}^{A_\infty}_{\mathrm{perfd}}/(\varepsilon_n^3,(f_2^\sharp)^{1/p^m},\dots,(f_d^\sharp)^{1/p^m}))\\ &= p^{m(d-1)}\cdot(\mathrm{rank}_A\,S')\cdot\lambda_\infty(A_\infty/(\varepsilon_n^3,(f_2^\sharp)^{1/p^m},\dots,(f_d^\sharp)^{1/p^m}))\\ &= p^{m(d-1)}\cdot(\mathrm{rank}_A\,S')\cdot\lambda_\infty(A_\infty/(p^{3/p^n},(f_2^\sharp)^{1/p^m},\dots,(f_d^\sharp)^{1/p^m}))\\ &= (3/p^n)\cdot(\mathrm{rank}_A\,S')\cdot\lambda_\infty(A_\infty/(p,(f_2^\sharp),\dots,(f_d^\sharp)))\end{aligned}$$

where the inequality on the third line follows from (8.6.1), the equality on the fourth line follows from [CLM+26, Proposition 4.16]; the equality in the fifth line follows from our choice that $\varepsilon_n=p^{1/p^n}$ up to units in $A_\infty/((f_2^\sharp)^{1/p^m},\dots,(f_d^\sharp)^{1/p^m})$. The equality in the sixth line follows from taking a filtration and noting that $p,f_2^\sharp,\dots,f_d^\sharp$ is a regular sequence on A_∞: after modulo p, $(f_2^\sharp,\dots,f_d^\sharp)(A_\infty/p)$ has the same radical with $\mathfrak{m}_A(A_\infty/p)$, so the images of $f_2^\sharp,\dots,f_d^\sharp$ in A_∞/p, which we may assume comes from A_n for $n\gg0$, form a system of parameters and thus a regular sequence on A_n/p, and hence they form a regular sequence on A_∞/p. □

Remark 8.7. Using the method of Proposition 8.6, one can generalize [CLM+26][Proposition 6.12] to get rid of the issue that one needs to change A_∞ whenever considering a new extension $R \to S$. Explicitly, one can run through the proof of [CLM+26][Proposition 6.12] while replacing the canonical choice of system of parameters in A_∞ by the system of parameters given by Lemma 8.5.

Corollary 8.8. *Definition 8.2 is well-defined, i.e., independent of the choice of S.*

Proof. Suppose $f_1 \colon R \to S_1$ and $f_2 \colon R \to S_2$ are two extensions of normal domains such that $\pi_1^* \Delta_R$ and $\pi_2^* \Delta_R$ are integral. Then we can find a third one S_3 that dominates S_1 and S_2. Now we are done by Proposition 8.6. □

Now we establish a transformation rule for pairs using the relative canonical divisor.

Theorem 8.9. *Let $\phi \colon (R, \mathfrak{m}, k) \to (R', \mathfrak{n}, l)$ be a finite generically étale extension of complete normal local domains of mixed characteristic $(0, p)$ with perfect residue fields and with fraction fields $K = K(R)$ and $L = K(R')$. Let $\tau \colon \operatorname{Spec}(R') \to \operatorname{Spec}(R)$ be the corresponding morphism of schemes. Suppose $\Delta_R \geq 0$ is a $\mathbb{Q}$-divisor such that $\Delta_{R'} := \tau^* \Delta_R - K_{R'/R} \geq 0$. Let t be the maximal number of direct Tr-summands[9] of R' and $r = [L : K]$. Then we have*

$$t \cdot s^{\underline{x}}_{\mathrm{perfd}}(R', \Delta_{R'}) = r \cdot s^{\underline{x}}_{\mathrm{perfd}}(R, \Delta_R).$$

Proof. Let $f \colon R' \to T$ be a finite generically étale extension of complete normal local domains such that Δ_R and $\Delta_{R'}$ are integral after pullback to $\operatorname{Spec}(T)$. Set $M = K(T)$ and $\pi = \operatorname{Spec}(f)$.

Suppose first that $t = 0$, so that the trace map is not surjective and $\operatorname{Tr}(R') \subseteq \mathfrak{m}$. By Lemma 8.10 below, every $\varphi \in \operatorname{Hom}_R(T^{A_\infty}_{\mathrm{perfd}} \otimes_T T(\Delta_R), R)$ can be expressed as $\varphi = \operatorname{Tr} \circ \psi$ for some $\psi \in \operatorname{Hom}_{R'}(T^{A_\infty}_{\mathrm{perfd}} \otimes_T T(\Delta_{R'}), R')$. In particular,

$$\varphi(z \otimes 1) = \operatorname{Tr}(\psi(z \otimes 1)) \in \operatorname{Tr}(R') \subseteq \mathfrak{m}$$

for all $z \in T^{A_\infty}_{\mathrm{perfd}}$. Thus, $\mathcal{I}^R(T^{A_\infty}_{\mathrm{perfd}}, \Delta_R) = T^{A_\infty}_{\mathrm{perfd}}$ by (8.2.2), and hence $s^{\underline{x}}_{\mathrm{perfd}}(R, \Delta_R) = 0$. The desired equality therefore holds in this case.

Now assume that the trace map is surjective. By [CRST18, Lemma 2.13], we know that $t = [l : k]$. Notice that

$$\lambda^{A_\infty}_\infty(T^{A_\infty}_{\mathrm{perfd}} / \mathcal{I}^R(T^{A_\infty}_{\mathrm{perfd}}, \Delta_R)) = r \cdot [M : L] \cdot s^{\underline{x}}_{\mathrm{perfd}}(R, \Delta_R)$$

by the independence of our definition, where the normalized length is computed with respect to the Noether-Cohen-Gabber normalization $A = W(k)[\![x_2, \ldots, x_d]\!] \to R$. Similarly, we see that

$$\lambda^{A'_\infty}_\infty(T^{A'_\infty}_{\mathrm{perfd}} / \mathcal{I}^{R'}(T^{A'_\infty}_{\mathrm{perfd}}, \Delta_{R'})) = [M : L] \cdot s^{\underline{x}}_{\mathrm{perfd}}(R', \Delta_{R'}).$$

where the normalized length is computed with respect to the Noether-Cohen-Gabber normalization $A' = W(l)[\![x_2, \ldots, x_d]\!] \to R'$. Using the argument behind [CLM+26, Equation (6.2)], we have

$$t \cdot \lambda^{A'_\infty}_\infty(T^{A'_\infty}_{\mathrm{perfd}} / \mathcal{I}^{R'}(T^{A'_\infty}_{\mathrm{perfd}}, \Delta_{R'})) = \lambda^{A_\infty}_\infty(T^{A_\infty}_{\mathrm{perfd}} / \mathcal{I}^{R'}(T^{A_\infty}_{\mathrm{perfd}}, \Delta_{R'}))$$

and keeping in mind that $t = [l : k]$. Therefore, it suffices to show that

$$\mathcal{I}^R(T^{A_\infty}_{\mathrm{perfd}}, \Delta_R) = \mathcal{I}^{R'}(T^{A_\infty}_{\mathrm{perfd}}, \Delta_{R'}).$$

[9] That is, a rank-1 free R-module summand of R' such that if $\rho \colon R' \to R$ is projection on that summand, then $\rho = \operatorname{Tr} \cdot s$ for some $s \in R'$, see also [CRST18, Definition 2.12].

This follows by using Lemma 8.10 below and [CRST18, Lemma 2.10], applied to the descriptions of these ideals in (8.2.2). Thus the proof is complete. □

Lemma 8.10. *With notations as in the proof of Theorem 8.9, we have*

$$\mathrm{Hom}_R(T^{A_\infty}_{\mathrm{perfd}} \otimes_T T(\Delta_R), R) \simeq \mathrm{Hom}_{R'}(T^{A_\infty}_{\mathrm{perfd}} \otimes_T T(\Delta_{R'}), R')$$

via the composition by the trace map $\mathrm{Tr}\colon R'(K_{R'/R}) \to R$.

Proof. Note that we have

$$\begin{aligned}\mathrm{Hom}_R(T^{A_\infty}_{\mathrm{perfd}} \otimes_T T(\Delta_R), R) &\simeq \mathrm{Hom}_R(T^{A_\infty}_{\mathrm{perfd}} \otimes_T T(\Delta_R) \otimes_{R'} R', R)\\ &\simeq \mathrm{Hom}_{R'}(T^{A_\infty}_{\mathrm{perfd}} \otimes_T T(\Delta_R), \mathrm{Hom}_R(R', R))\\ &\simeq \mathrm{Hom}_{R'}(T^{A_\infty}_{\mathrm{perfd}} \otimes_T T(\Delta_R), R'(K_{R'/R}))\end{aligned}$$

where the last isomorphism follows by identifying $\mathrm{Hom}_R(R', R)$ with $R'(K_{R'/R})$. Note that under this identification, the evaluation at 1 map $\mathrm{Hom}_R(R', R) \to R$ is identified with the trace map $\mathrm{Tr}\colon R'(K_{R'/R}) \to R$. Then we are done after we have observed the following isomorphism

$$\begin{aligned}\mathrm{Hom}_{R'}(T^{A_\infty}_{\mathrm{perfd}} \otimes_T T(\Delta_R), R'(K_{R'/R})) &\simeq \mathrm{Hom}_T(T^{A_\infty}_{\mathrm{perfd}}, \mathrm{Hom}_{R'}(T(\Delta_R), R'(K_{R'/R})))\\ &\simeq \mathrm{Hom}_T(T^{A_\infty}_{\mathrm{perfd}}, \mathrm{Hom}_{R'}(T(\Delta_R - \pi^* K_{R'/R}), R'))\\ &= \mathrm{Hom}_T(T^{A_\infty}_{\mathrm{perfd}}, \mathrm{Hom}_{R'}(T(\Delta_{R'}), R'))\\ &\simeq \mathrm{Hom}_{R'}(T^{A_\infty}_{\mathrm{perfd}} \otimes_T T(\Delta_{R'}), R').\end{aligned}$$

The second isomorphism is true by using the fact that any morphism between two reflexive sheaves is determined by any open subset whose complement has codimension ≥ 2. Hence we may restrict ourselves to the regular locus which clearly has the desired isomorphism. □

8.2. **BCM-regularity.** In this subsection we will show that a pair (R, Δ) is BCM-regular in the sense of [MS21] if and only if the perfectoid signature of pair $s^{\underline{x}}_{\mathrm{perfd}}(R, \Delta) > 0$, generalizing the result in [CLM+26, Section 5] to the pair version. Throughout this subsection we work in the following setting.

Setting 8.11. Suppose $(R, \mathfrak{m})$ is a Noetherian complete local normal domain of mixed characteristic $(0, p)$ with perfect residue field. Fix $\Delta \geq 0$ a $\mathbb{Q}$-divisor on $\mathrm{Spec}\, R$ such that $K_R + \Delta$ is $\mathbb{Q}$-Cartier. Choose $R \subseteq S \subseteq R^+$ a finite extension of normal domains such that $S(K_R + \Delta) \cong S$ (that is, the pullback of $K_R + \Delta$ is Cartier).

We also fix a Noether–Cohen normalization $A \subseteq R$ and $0 \neq g \in A$ such that $A[g^{-1}] \subseteq R[g^{-1}]$ and $A[g^{-1}] \subseteq S[g^{-1}]$ are étale.

Definition 8.12. With notation as in Setting 8.11 we say that (R, Δ) is *perfectoid BCM-regular* if for each perfectoid big Cohen-Macaulay R^+-algebra B, we have that

$$R \to B \otimes_S S(\Delta)$$

is pure where S is a normal domain finite over R so that the pullback of Δ to S is a Weil divisor.

Remark 8.13. Set E to be the injective hull of the residue field. Notice that (R, Δ) is perfectoid BCM-regular if and only if for all perfectoid big Cohen-Macaulay B, we have that

$$E \to B \otimes_S S(\Delta) \otimes_R E$$

is injective. Since $E = H^d_{\mathfrak{m}}(R(K_R))$, we see that $S(\Delta) \otimes_R E \cong H^d_{\mathfrak{m}}(S(K_R + \Delta)) \cong H^d_{\mathfrak{m}}(S)$. Hence, we have that (R, Δ) is perfectoid BCM-regular if and only if

$$H^d_{\mathfrak{m}}(R(K_R)) \to H^d_m(B \otimes_S S(K_R + \Delta)) \cong H^d_m(B) \tag{8.13.1}$$

is injective.

Our goal in this section is to prove that positivity of perfectoid signature is equivalent to BCM-regularity of pairs (R, Δ). We recall the following result from [CLM+26], also see [BMP+25, Claim 7.11].

Lemma 8.14 ([CLM+26, Lemma 5.6]). *Suppose $(S, \mathfrak{n}, L)$ is a complete Noetherian local domain of mixed characteristic and fix $W(k)[\![x_2, \ldots, x_d]\!] = A \subseteq S$ a regular subring such that S is a finite A-module (we do not assume that the induced $k \subseteq L$ is an equality). Let $0 \neq g \in A$ with $A[1/g] \to S[1/g]$ finite étale. Suppose M is a p^∞-torsion S-module. Consider the following submodules of M:*

$$\begin{aligned}
M_1 &:= \{\eta \in M \mid (g)_{\mathrm{perfd}} \otimes \eta = 0 \text{ in } S^{A_\infty}_{\mathrm{perfd}} \otimes_S M\}.\\
M_2 &:= \{\eta \in M \mid (g^{1/p^\infty}) \otimes \eta = 0 \text{ in } S^+ \otimes_S M\}.\\
M_3 &:= \{\eta \in M \mid (g^{1/p^\infty}) \otimes \eta = 0 \text{ in } B \otimes_S M \text{ for some perfectoid big Cohen-Macaulay } S^+\text{-algebra } B\}.\\
M_4 &:= \{\eta \in M \mid 1 \otimes \eta = 0 \text{ in } B \otimes_S M \text{ for some perfectoid big Cohen-Macaulay } S^+\text{-algebra } B\}
\end{aligned}$$

Then we have $M_1 = M_2 = M_3 \supseteq M_4$. Moreover, if M is finitely generated, then we have $M_3 = M_4$.

The purpose of this lemma is seen in the next result.

Proposition 8.15. *With notation as in Setting 8.11, the following are equivalent.*

(a) (R, Δ) is perfectoid BCM-regular.

(b) For some $c \in (g)_{\mathrm{perfd}} \subseteq S^{A_\infty}_{\mathrm{perfd}}$ we have that the following map is pure:

$$R \xrightarrow{1 \mapsto c} S^{A_\infty}_{\mathrm{perfd}} \otimes_S S(\Delta).$$

(c) For some $e \gg 0$, the following map is pure:

$$R \xrightarrow{1 \mapsto g^{1/p^e}} \widehat{R^+}(\Delta).$$

(d) For some $e \gg 0$, and for every perfectoid big Cohen-Macaulay R^+-algebra B, the following map is pure:

$$R \xrightarrow{1 \mapsto g^{1/p^e} \otimes 1} B \otimes_S S(\Delta).$$

Proof. Let E be the injective hull of the residue field of R. Then by Remark 8.13, we know that (R, Δ) is BCM-regular if and only if for each perfectoid big Cohen-Macaulay R^+-algebra B, the map

$$E \to E \otimes_R B \otimes_S S(\Delta)$$

is injective, namely the following is zero:

$$\{\eta \in E \mid \eta \otimes 1 = 0 \text{ in } E \otimes_R B \otimes_S S(\Delta) \text{ for some perfectoid big Cohen-Macaulay } R^+\text{-algebra } B\}.$$

Similarly, we also note that (b) holds if and only if

$$\{\eta \in E \mid \eta \otimes (g)_{\text{perfd}} = 0 \text{ in } E \otimes_R S^{A_\infty}_{\text{perfd}} \otimes_S S(\Delta)\} = 0$$

and (c) holds if and only if

$$\{\eta \in E \mid \eta \otimes (g^{1/p^\infty}) = 0 \text{ in } E \otimes_R R^+ \otimes_S S(\Delta)\} = 0.$$

Finally (d) holds if and only if

$$\left\{\eta \in E \;\middle|\; \begin{array}{c} \eta \otimes (g^{1/p^\infty}) = 0 \text{ in } E \otimes_R B \otimes_S S(\Delta) \\ \text{for some perfectoid big Cohen–Macaulay } R^+\text{-algebra } B \end{array}\right\} = 0.$$

With the above observation the equivalence between (a), (b), (c) and (d) follows from Lemma 8.14. Note that the equality of M_3 and M_4 requires M to be finitely generated in Lemma 8.14. However, since $K_R + \Delta$ is $\mathbb{Q}$-Cartier, by Remark 8.13 we know that

$$E \otimes_R B \otimes_S S(\Delta) \simeq H^d_{\mathfrak{m}}(B) \simeq H^d_{\mathfrak{m}}(R) \otimes_R B.$$

The proof of [CLM+26, Lemma 5.6] holds verbatim for the top local cohomology module by the Čech-complex description. Alternatively, choose a system of parameters $y_1, \ldots, y_d$ of R. Using $S(K_R + \Delta) \cong S$, represent the image of $\eta \in E$ in $H^d_{\mathfrak{m}}(S)$ by $u \in S/(y_1^N, \ldots, y_d^N)$ for some N. We may assume that B is balanced, so $B/(y_1^N, \ldots, y_d^N) \longrightarrow H^d_{\mathfrak{m}}(B)$ is injective. Thus, if every g^{1/p^e} annihilates the image of η in $H^d_{\mathfrak{m}}(B)$, they all annihilate the image of u in $B/(y_1^N, \ldots, y_d^N)$. Since $S/(y_1^N, \ldots, y_d^N)$ is a finite p^∞-torsion S-module, Lemma 8.14 gives a perfectoid big Cohen–Macaulay algebra B' such that u maps to zero in $B'/(y_1^N, \ldots, y_d^N)$. Hence η maps to zero in $H^d_{\mathfrak{m}}(B')$. □

We can now state and prove the main result of this subsection.

Theorem 8.16. *With the notation as in Setting 8.11, we have (R, Δ) is BCM-regular if and only if*

$$s^{\underline{x}}_{\text{perfd}}(R, \Delta) > 0.$$

Proof. The proof closely follows the one in [CLM+26, Proposition 5.8 and 5.9]. We first prove the "if" direction. We will prove that for every nonzero $g \in R^+$, there exists $e \gg 0$ such that for any perfectoid big Cohen-Macaulay R^+-algebra B, the map $R \to B \otimes_S S(\Delta)$ sending $1 \mapsto g^{1/p^e}$ is pure. Replacing g with a multiple if necessary, we may assume that $g \in A$ and that $A[g^{-1}] \subseteq S[g^{-1}]$ is étale. By Proposition 8.15, it suffices to show that there exists $g_1 \in (g)_{\text{perfd}}$ such that $R \to S^{A_\infty}_{\text{perfd}} \otimes_S S(\Delta)$ sending 1 to g_1 is pure. If this is not the case, then $(g)_{\text{perfd}} \subseteq \mathcal{I}^R(S^{A_\infty}_{\text{perfd}}, \Delta)$ and hence $s^{\underline{x}}_{\text{perfd}}(R, \Delta) = \frac{1}{[K(S):K(R)]} \lambda_\infty\big(S^{A_\infty}_{\text{perfd}}/\mathcal{I}^R(S^{A_\infty}_{\text{perfd}}, \Delta)\big) = 0$ by [CLM+26, Proposition 4.10] which is a contradiction.

We next prove the "only if" direction. We prove this by contradiction. So suppose $s^{\underline{x}}_{\text{perfd}}(R, \Delta) = (1/\operatorname{rank}_R S)\lambda_\infty(S^{A_\infty}_{\text{perfd}}/\mathcal{I}^R(S^{A_\infty}_{\text{perfd}}, \Delta)) = 0$. Set v to be the $\mathbb{Z}[1/p]$-valued valuation that sends $f \in A_\infty$ to the lowest degree (in terms of the rational-power monomials in $p, x_2, \ldots, x_d$) of a term (see [CLM+26, Definition 4.11]). By [Shi07, Proposition 2.15] or [CLM+26, Lemma 4.13], there is a sequence of elements $\{c_i\} \subseteq A_\infty$ such that $v(c_i) \to 0$ and with $c_i \in \mathcal{I}^R(S^{A_\infty}_{\text{perfd}}, \Delta)$. In particular, the map $R \to S^{A_\infty}_{\text{perfd}} \otimes_S S(\Delta)$ sending $1 \mapsto c_i \otimes 1$ is not pure.

Suppose $\eta \in H^d_{\mathfrak{m}}(R(K_R)) = E$ generates the socle. Then $c_i \otimes 1 \otimes \eta = 0$ inside $S^{A_\infty}_{\text{perfd}} \otimes_S S(\Delta) \otimes_R E$. Since $R \to \widehat{R^+} \otimes_S S(\Delta)$ factors through $S^{A_\infty}_{\text{perfd}} \otimes_S S(\Delta)$, we have that $c_i \otimes 1 \otimes \eta = 0$ inside $\widehat{R^+} \otimes_S S(\Delta) \otimes_R E \cong R^+ \otimes_S S(\Delta) \otimes_R E$. It follows that $c_i \otimes 1 \otimes \eta = 0$ inside

$$R^+ \otimes_S S(\Delta) \otimes_R E \cong R^+ \otimes_S S(\Delta) \otimes_R H^d_{\mathfrak{m}}(R(K_R)) \cong H^d_{\mathfrak{m}}(R^+(K_R + \Delta)).$$

To see the second isomorphism, note that while the canonical map $R^+ \otimes_R R(K_R) \otimes_S S(\Delta) \to R^+(K_R+\Delta)$ need not be an isomorphism, it is an isomorphism over the complement of a codimension ≥ 2 subset of $\operatorname{Spec} R$ (where all the divisors considered are Cartier), and hence the induced map

$$H^d_{\mathfrak{m}}(R(K_R)) \otimes_R R^+ \otimes_S S(\Delta) \cong H^d_{\mathfrak{m}}\big(R^+ \otimes_R R(K_R) \otimes_S S(\Delta)\big) \to H^d_{\mathfrak{m}}\big(R^+(K_R + \Delta)\big)$$

is an isomorphism.

Since $K_R + \Delta$ is $\mathbb{Q}$-Cartier and R is local, we see that $R^+(K_R + \Delta) \cong R^+$. It follows from an argument of Gabber [Gab18] in the form of [CLM+26, Corollary 2.28] that there exists a perfectoid big Cohen-Macaulay R^+-algebra B such that the image of η is zero under the map

$$H^d_{\mathfrak{m}}(R^+(K_R + \Delta)) \cong H^d_{\mathfrak{m}}(R^+) \to H^d_{\mathfrak{m}}(B).$$

Hence (R, Δ) is not BCM-regular which is a contradiction. □

8.3. **Continuity and Convexity.**

Theorem 8.17. *Suppose $(R, \mathfrak{m}, k)$ is a complete Noetherian normal local domain of mixed characteristic $(0, p)$ with perfect residue field and $f \in \mathfrak{m}$ is non-zero. Let $D = \operatorname{div}(f)$ on $\operatorname{Spec}(R)$. The function $\mathbb{Q}_{\geq 0} \to \mathbb{R}$ given by*

$$t \mapsto s^{\underline{x}}_{\text{perfd}}(R, tD)$$

is continuous. In particular, it extends naturally to a function $\mathbb{R}_{\geq 0} \to \mathbb{R}$.

Proof. Let $y_1 := f, y_2, \ldots, y_d$ be a fixed system of parameters of R beginning with f. Suppose $t \in \mathbb{Q}_{\geq 0}$ and $\varepsilon \in \mathbb{Q}_{>0}$. Choose $(S, \mathfrak{n})$ a finite complete normal local extension domain of $(R, \mathfrak{m})$ so that $f^t, f^{t+\varepsilon} \in S$. Observe that, after tracing through the definitions, we have

$$\mathcal{I}^R(S^{A_\infty}_{\text{perfd}}, tD) = \mathcal{I}^R(S^{A_\infty}_{\text{perfd}}, 0) :_{S^{A_\infty}_{\text{perfd}}} f^t \quad \text{and} \quad \mathcal{I}^R(S^{A_\infty}_{\text{perfd}}, (t+\varepsilon)D) = \mathcal{I}^R(S^{A_\infty}_{\text{perfd}}, 0) :_{S^{A_\infty}_{\text{perfd}}} f^{t+\varepsilon}$$

so that in particular $\mathcal{I}^R(S^{A_\infty}_{\text{perfd}}, (t+\varepsilon)D) = \mathcal{I}^R(S^{A_\infty}_{\text{perfd}}, tD) :_{S^{A_\infty}_{\text{perfd}}} f^\varepsilon$. If $r = \operatorname{rank}_R(S)$, then we have

$$\begin{aligned}
&s^{\underline{x}}_{\text{perfd}}(R, tD) - s^{\underline{x}}_{\text{perfd}}(R, (t+\varepsilon)D) \\
&= \frac{1}{r}\left(\lambda_\infty\left(S^{A_\infty}_{\text{perfd}}/\mathcal{I}^R(S^{A_\infty}_{\text{perfd}}, tD)\right) - \lambda_\infty\left(S^{A_\infty}_{\text{perfd}}/\mathcal{I}^R(S^{A_\infty}_{\text{perfd}}, (t+\varepsilon)D)\right)\right) \\
&= \frac{1}{r}\lambda_\infty\left(\frac{S^{A_\infty}_{\text{perfd}}}{(\mathcal{I}^R(S^{A_\infty}_{\text{perfd}}, tD), f^\varepsilon)S^{A_\infty}_{\text{perfd}}}\right) \\
&\leq \frac{1}{r}\lambda_\infty\left(\frac{S^{A_\infty}_{\text{perfd}}}{(y_1^\varepsilon, y_2, \ldots, y_d)S^{A_\infty}_{\text{perfd}}}\right).
\end{aligned}$$

Set $\ell = [S/\mathfrak{n} : k]$. Now $y_1^\varepsilon, y_2, \ldots, y_d$ is also a system of parameters for S, and so by [CLM+26, Proposition 4.14] it follows that

$$\begin{aligned}\lambda_\infty\left(\frac{S^{A_\infty}_{\mathrm{perfd}}}{(y_1^\varepsilon, y_2, \ldots, y_d)S^{A_\infty}_{\mathrm{perfd}}}\right) &= \ell \cdot e^{\underline{x}}_{\mathrm{perfd}}(S, (y_1^\varepsilon, y_2, \ldots, y_d)) \\ &= \ell \cdot e(S, (y_1^\varepsilon, y_2, \ldots, y_d)) \\ &= \varepsilon \cdot r \cdot e(R, (y_1, y_2, \ldots, y_d)).\end{aligned}$$

Putting these together gives that

$$0 \leq s^{\underline{x}}_{\mathrm{perfd}}(R, tD) - s^{\underline{x}}_{\mathrm{perfd}}(R, (t+\varepsilon)D) \leq \varepsilon \cdot e(R, (y_1, y_2, \ldots, y_d)).$$

Thus, we have that the function $\mathbb{Q}_{\geq 0} \to \mathbb{R}$ given by $t \mapsto s^{\underline{x}}_{\mathrm{perfd}}(R, tD)$ is Lipschitz with constant $e(R, (y_1, y_2, \ldots, y_d))$, and in particular continuous as claimed. □

Theorem 8.18. *Suppose $(R, \mathfrak{m}, k)$ is a complete Noetherian normal local domain of mixed characteristic $(0, p)$ with perfect residue field and $f \in \mathfrak{m}$ is non-zero with $D = \operatorname{div}(f)$ a divisor on $\operatorname{Spec} R$. The function $\mathbb{R}_{\geq 0} \to \mathbb{R}$ given by*

$$t \mapsto s^{\underline{x}}_{\mathrm{perfd}}(R, tD)$$

is convex.

Proof. If $0 \leq t < t_1 < t_2$, we need to verify that

(8.18.1)
$$D(t, t_1) = \frac{s^{\underline{x}}_{\mathrm{perfd}}(R, t_1 D) - s^{\underline{x}}_{\mathrm{perfd}}(R, tD)}{t_1 - t} \leq \frac{s^{\underline{x}}_{\mathrm{perfd}}(R, t_2 D) - s^{\underline{x}}_{\mathrm{perfd}}(R, t_1 D)}{t_2 - t_1} = D(t_1, t_2).$$

By continuity, it suffices to check this inequality on the dense subset of rational numbers. Taking a common denominator $n \in \mathbb{Z}_{>0}$, we may assume that

$$t = \frac{a}{n} \qquad t_1 = \frac{a_1}{n} \qquad t_2 = \frac{a_2}{n}$$

where $a, a_1, a_2 \in \mathbb{Z}$ with $0 \leq a < a_1 < a_2$. Moreover, it suffices to treat the case $a_1 = a + 1$ and $a_2 = a + 2$.

Choose $(S, \mathfrak{n})$ a complete normal local finite extension domain of $(R, \mathfrak{m})$ so that $f^{1/n} \in S$. If $r = \operatorname{rank}_R(S)$, the inequality (8.18.1) reduces to the following

$$\begin{aligned}&\frac{n}{r}\left(\lambda_\infty\left(\frac{S^{A_\infty}_{\mathrm{perfd}}}{\mathcal{I}^R(S^{A_\infty}_{\mathrm{perfd}}, tD) :_{S^{A_\infty}_{\mathrm{perfd}}} f^{1/n}}\right) - \lambda_\infty\left(\frac{S^{A_\infty}_{\mathrm{perfd}}}{\mathcal{I}^R(S^{A_\infty}_{\mathrm{perfd}}, tD)}\right)\right) \\ &\qquad \leq \frac{n}{r}\left(\lambda_\infty\left(\frac{S^{A_\infty}_{\mathrm{perfd}}}{\mathcal{I}^R(S^{A_\infty}_{\mathrm{perfd}}, t_1 D) :_{S^{A_\infty}_{\mathrm{perfd}}} f^{1/n}}\right) - \lambda_\infty\left(\frac{S^{A_\infty}_{\mathrm{perfd}}}{\mathcal{I}^R(S^{A_\infty}_{\mathrm{perfd}}, t_1 D)}\right)\right)\end{aligned}$$

With similar reductions as in the proof of Theorem 8.17 the above is equivalent to

$$-\frac{n}{r}\left(\lambda_\infty\left(\frac{S^{A_\infty}_{\mathrm{perfd}}}{(\mathcal{I}^R(S^{A_\infty}_{\mathrm{perfd}}, tD), f^{1/n})S^{A_\infty}_{\mathrm{perfd}}}\right)\right) \leq -\frac{n}{r}\left(\lambda_\infty\left(\frac{S^{A_\infty}_{\mathrm{perfd}}}{(\mathcal{I}^R(S^{A_\infty}_{\mathrm{perfd}}, t_1 D), f^{1/n})S^{A_\infty}_{\mathrm{perfd}}}\right)\right)$$

or simply

$$\lambda_\infty\left(\frac{S^{A_\infty}_{\mathrm{perfd}}}{(\mathcal{I}^R(S^{A_\infty}_{\mathrm{perfd}},tD),f^{1/n})S^{A_\infty}_{\mathrm{perfd}}}\right)\geq\lambda_\infty\left(\frac{S^{A_\infty}_{\mathrm{perfd}}}{(\mathcal{I}^R(S^{A_\infty}_{\mathrm{perfd}},t_1D),f^{1/n})S^{A_\infty}_{\mathrm{perfd}}}\right)$$

which follows from the inclusion $\mathcal{I}^R(S^{A_\infty}_{\mathrm{perfd}},tD)\subseteq\mathcal{I}^R(S^{A_\infty}_{\mathrm{perfd}},t_1D)=\mathcal{I}^R(S^{A_\infty}_{\mathrm{perfd}},tD):_{S^{A_\infty}_{\mathrm{perfd}}}f^{1/n}$. □

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